

SMITHIAN FOLD THEORY - BIOLOGY AND LIFE SCIENCES PAPER 001

# From Fold to Life

An Exact, Zero-Parameter and Machine-Closed Foundational Reconstruction of  
Biology and Life Sciences from Smithian Fold Theory

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Ernos Labs

Open Source Science Platform and Knowledge Tree

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Version 1.0.0 - current-evidence closed, extension-open foundation

75 required laws - 19,200 exact candidates

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Paper: CC BY 4.0 - Code: Apache-2.0

LOCAL PREPUBLICATION MANUSCRIPT - PUBLICATION NOT YET AUTHORIZED

# Abstract

This paper reports the foundational Biology and Life Sciences branch of the third clean-room Smithian Fold Theory reconstruction at its current-evidence-closed, extension-open boundary. Seventy-five obligations in twelve ordered families generate 19,200 exact candidates and decisions, seventy-five unique survivors, seventy-five depth-independent closure certificates, three hundred passing adverse controls, seventy-five implementation-distinct reconstructions and seventy-five post-seal external comparisons. Every dependency graph reaches the single premise-free theorem, There Is No Nothing. The branch uses zero axioms, zero free or fitted parameters, no negative proof quantities, no irrational or imaginary proof values, no target-selected rule and no opaque predictor.

The exact headline construction generates four DNA coding labels from two held distinctions, three ordered codon positions, sixty-four complete codon words, sixteen first-two-position boxes and four third-position words per box. The paper also closes explicit laws for living organization, compartments, metabolism, inheritance, evolution, genes and genomes, protein folding and assemblies, cells, development, organisms, physiology, populations, ecosystems and biological evidence. A complete audit of 763 prior V1/V2 entries identifies thirty Biology-owned atoms; all thirty now map to immutable V3 receipts, with twenty-one corrected at a stricter evidence boundary. Those corrections are scientific results: beneficial variants do not inevitably fix; protein state cannot be reduced to one rigid shape; biological homochirality is observed while its origin remains unresolved; a universal three-quarter allometric exponent is not supported; strongly scale-free networks are empirically rare; and denominator parity, cycling or one periodic orbit cannot by themselves identify senescence, cancer or ecosystem stability.

## Results first: the biological findings

| Headline result              | Exact or bounded result                                                                                          | Scientific meaning                                                                     |
|------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Nucleic-acid coding alphabet | $2 \times 2 = 4$ held symbol labels                                                                              | Four is generated before the NCBI genetic-code table is opened.                        |
| Codon word space             | $4 \times 4 \times 4 = 64$ ordered triplets                                                                      | All sixty-four words are explicitly enumerated; no codon table selects the count.      |
| Codon prefix boxes           | $4 \times 4 = 16$ boxes, each holding 4 third-position words                                                     | The complete partition covers each of the sixty-four codons exactly once.              |
| Living/nonliving boundary    | Compartment + maintenance + inheritance + regulated resource exchange                                            | Autocatalysis or dissipation alone is an eliminated incomplete form.                   |
| Evolution                    | Heritable variation + differential continuation + controlled environment                                         | Selection is distinguished from drift; benefit alone does not force fixation.          |
| Protein structure            | Directed condition-bounded fold plus retained conformational ensemble                                            | Fast reduction is admitted; one rigid universal native shape is not.                   |
| Homochirality                | L-amino-acid/D-sugar predominance observed; origin mechanism unresolved                                          | The old parity-selection overclaim is removed instead of hidden.                       |
| Networks and scaling         | Whole-graph hostile alternatives; 3/4, 2/3, unity and nonuniversal allometry retained                            | Scale-free and three-quarter universality are not imported as laws.                    |
| Complete evidence            | 75 laws; 19,200 candidates; 75 survivors; 300 controls; 75 independent reconstructions; 75 post-seal comparisons | Every result has a root trace, falsifier, custody record and immutable engine receipt. |

## Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown

questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Biology, a database label, selected organism, favorable specimen, opaque structure predictor or consensus classification cannot stand for a law of life. Organism, lineage, compartment, environment, condition, method, missingness and adverse observations remain in the evidence. Exact code counts are stated as exact; variable biological magnitudes remain bounded to their measured populations and protocols.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com), submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

## 1. Publication, authorship and open-science boundary

**LOCAL PREPUBLICATION MANUSCRIPT. Publication is not yet authorized.** Building this paper performs no push, release, upload, DOI creation or Zenodo action.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com). Reproducibility reports and submissions: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

Copyright preserves Maria Smith's authorship. The paper and documentation are prepared under CC BY 4.0 and code under Apache-2.0. The Ernos Labs name is a separate, revocable standards designation: reuse is open, but the designation requires continued adherence to the public constitution, unchanged admission engine, complete adverse evidence and open critical review.

## 2. Exact scope and closure language

Foundational closure means every obligation in the frozen 75-law question surface has an engine-admitted theorem inside its declared exact grammar, a complete dependency path, adverse controls, an independent reconstruction and a post-seal external comparison. It does not mean Biology is permanently closed, that every species or future observation has been enumerated, or that the full field-wide Layer Two roadmap is complete. This foundation is current-evidence closed and extension-open: any lawful discovery, correction or invalidation can be admitted through the same public process.

## 3. Constitutional mathematical domain

Structural absence is Empty One and may be displayed as  $\emptyset$ ; it is not conventional numerical nothing. Counts are generated positive wholes and exact parts are positive held fractions. Opposition, side and orientation are labels rather than negative scalars. Negative proof quantities, irrational or imaginary proof values, floating equality, completed infinity, ungenerated continua, axioms, free parameters, fitted coefficients, imported biological equations, pretrained predictors, consensus-selected survivors and application-selected laws are prohibited. Conventional notation appears only in downstream correspondence prose where it cannot act as proof input.

## 4. Dependency spine

Every Biology registration names the single premise-free root and zero axioms. The first law depends only on already admitted Foundation, Mathematics, Information Science, Physics, Chemistry and Materials receipts. Each later law also depends on its immediate Biology predecessor, producing an ordered 75-law chain. The engine accepts a dependency only if its immutable model-admission receipt already exists. Thus every law is traceable to the root through actual claim identities, not through a narrative assertion that a branch exists.

## 5. Complete target-blind seal

Before any external source identity was selected, the whole 75-law inventory, exact counting module and target-blind blueprint set were frozen. The seal is

sha256:4b3e1ba191d363a1b67e1a02853f071cdf2c9d3d86081fced05ab3c5d079e639. It binds seventy-five predictions, 19,200 candidate identities and the exact source files while recording `external_source_identities_selected=false` and `external_target_content_opened=false`. No later source discovery can change those candidates or survivors without breaking the seal.

## 6. Exact codon certificate

The independent finite certificate is `{'held_distinctions': 2, 'alphabet_count': 4, 'word_length': 3, 'codon_count': 64, 'box_count': 16, 'box_widths': (4,), 'alphabet_complete': True, 'codon_census_complete': True, 'partition_complete': True, 'each_word_once': True}`. Two held distinctions generate four ordered labels. The complete length-three product contains sixty-four distinct words. Prefix equivalence on the first two positions forms sixteen boxes. Every box holds four third-position alternatives; their union contains all sixty-four words exactly once. NCBI is then used as an external genetic-code correspondence, not as the generator.

## 7. External evidence and preserved adversity

Seventeen byte-frozen usable sources cover the twelve families: US National Academies/NCBI organization records, NCBI cell and molecular-biology chapters, Gene Ontology evidence rules, Sequence Ontology, NCBI's genetic-code table, the primary RCSB 1COP deposition, GBIF survey standards, evolution/genetics, protein folding, development, homeostasis, aging, homochirality, and full-text network/allometry studies. Each claim binds at least two purpose-matched fragments. The prediction process cannot read filesystem, network, clock, environment, dynamic import, subprocess or external target. A separate custodian opens sources after sealing and releases a target only for a matching prediction seal.

The evidence ledger also preserves four failures: the Sequence Ontology web endpoint failed the local TLS handshake; one GBIF page returned 403; and two PMC printable endpoints returned challenge pages rather than article content. Registered official release or Europe PMC endpoints supplied the same purposes in new addenda without deleting the failed rows. Scientific adverse results are likewise retained: nonuniversal allometry, rarity of strong scale-freeness and unresolved homochirality mechanism are part of the accepted evidence rather than inconvenient exceptions.

## 8. V1/V2 categorical audit and corrected legacy boundaries

All 356 V1 rows and 407 V2 steps were reviewed. Nine V1 rows and nine V2 steps contain Biology-owned atoms after mixed statements are decomposed at categorical boundaries. Thirty atomic obligations are identified; thirty are mapped to current Biology receipts; none remains open. Twenty-one are explicitly corrected rather than cosmetically called historical. Physics retains irreversibility, Chemistry retains molecular chirality, computation retains general complexity, and Biology owns living organization, inheritance, protein behavior, evolution, organismal and ecological consequences.

The corrected boundaries include: no inevitable selection-to-fixation law; no single rigid protein state; no parity-derived biological hand; no universal numerical autocatalytic ignition threshold; no denominator-parity aging law; no half-One membrane-voltage claim; no recurrence-only cancer diagnosis; no one-orbit ecosystem stability; no universal scale-free/small-world topology; no universal three-quarter allometry; and no claim that the inaccurate prior Lambda Cro docking output was a validated predictor.

## 9. Reading the exhaustive derivation ledger

Each of the seventy-five sections below states the theorem, dependencies, full eight-axis grammar, all 256 candidate decisions, unique survivor, depth-independent base/successor certificate, operational witnesses, scientific meaning, four adverse controls, independent implementation, post-seal sources, falsification boundary and immutable identities. The machine-readable files remain authoritative; this prose makes their meaning inspectable.

## 10. Life Boundary

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

### 1. Living organization

Claim identity: SFT-BIO-LIFE-ORGANIZATION-001

**Question and exact theorem.** Living organization is a bounded process network that retains its identity by regulated material and energy turnover within a declared viability window.

bounded-process-carrier\_\_self-maintaining-transition-relation\_\_identity-through-turnover\_\_viability-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bounded-process-carrier                       | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | self-maintaining-transition-relation          | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | identity-through-turnover                     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | viability-window-declared                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: bounded-process-carrier\_\_self-maintaining-transition-relation\_\_identity-through-turnover\_\_viability-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces living organization is a bounded process network that retains its identity by regulated material and energy turnover within a declared viability window. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 7f8401dbad0de0173b17aeb86d4767451a27b3d6ead06ae49bfff58c0cfd7a83.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 229cb278b021705235de343c3410bf41a8ad0905ea9f021f203157126628ba7c.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 44bd25efd4d5d0b63e46f2c6a025ce10212fe1ba01d57b1acc62143c0ed0d7b1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 1f094b9a0efad2fbfb92d1898bf06fbf7481c66cdd970401b3d9deef7ec88b8e.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 71d1b8c495ef217f138b07250c0fe1824b091a459e6f8b92c49fdcfca548d2e5c; certificate  
sha256 : 8b21df6f71702810feee6f44c058033dbab72f2cde31b60cfc5389ed7f07e882; external-validation  
sha256 : 2c825984088115cf9e12793a84bfaadfd872d61efdc513da4a4d06030b1a21df.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256 : 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256 : ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-life-organization-001-authority-correspondence: predicted  
biology:bounded-process-carrier\_\_self-maintaining-transition-relation\_\_identity-through-turnover\_\_viability-window-declared;  
source-derived  
biology:bounded-process-carrier\_\_self-maintaining-transition-relation\_\_identity-through-turnover\_\_viability-window-declared;  
exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value



- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:39d6894670b5f8d32e79bab74dcf3db34675ae9078f74b0ed94070d3b006c06e; derivation seal sha256:1a6fe3a05037607c2a49884c06748c0865e143f524ed6730035d17b41332967b; engine receipt sha256:c81ed1467e76985489eb75341f8444d9b7dd70806158c065f76276adb038e5ff at receipts/engine/model\_admitted/SFT-BIO-LIFE-ORGANIZATION-001-c81ed1467e769854.json; empirical validation sha256:a090ae092d9cba93ecf9ade559164769dbb16ba36636a8b75b1a2408fff3c858; measurement receipt sha256:7a021ff265a0247600733ac54b5a7ba587d279fe15c832eaddf609c981e9082b; isolation sha256:d5402c9744920b45328881f4aeee7d6cf447f9b3f9cb00c7157c48050b7e9481; custody sha256:1975c36d2c3c73793a18420df38bebaa5a249bc57405fff918f7233a1ccf76c7.

## 2. Living and nonliving chemical-network boundary

Claim identity: SFT-BIO-LIFE-CHEMICAL-BOUNDARY-001

**Question and exact theorem.** A chemical network is biologically living only when compartment, maintenance, heritable continuation and regulated resource exchange are jointly retained; chemistry alone does not supply that classification.

complete-reaction-network-carrier\_\_network-to-maintenance-comparison\_\_closure-and-inheritance-retained\_\_condition-bounded-life-class\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001



- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-ORGANIZATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-reaction-network-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | network-to-maintenance-comparison             | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | closure-and-inheritance-retained              | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | condition-bounded-life-class                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-reaction-network-carrier\_\_network-to-maintenance-comparison\_\_closure-and-inheritance-retained\_\_condition-bounded-life-class\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- carrier: the biological carrier is explicit and nonempty; passed true.

- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The survivor requires compartment, maintenance, hereditary continuation and regulated resource exchange together. Autocatalysis or dissipation alone remains an eliminated incomplete form. This is a finite operational boundary, not a claim that every disputed biological entity has already been classified.

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 5bfc61fb3d7abdae8a4d32e515cb6d3c1fdc87f1e62f331dac47a47a358a2ba5.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 43dd61e298af31a325908ab1d613676b01b67ec4a5abaf8da2f0a196fe5e67c9.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: d564960a6d32d676b86679cdf8f13620e59b97dc86419b3091d42dc258d0e984.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 4f2e203d73164b45a6a9257e9cbe6d14ddf8e630cc263d40de1d82522a8533a7.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 5efb3410a944b475f5eb6be4579a1b9f9df8e8760f29906489502ef741c8bc73; certificate  
sha256: 3bd2996b764240aa2514713d78c7e97acce74f5f2fdebcbadaad2e96bd1692c7e; external-validation  
sha256: 1aedf58a998e7587f1c2038896b75816ea2bc1712d552b87fa7b6a6f12586705.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

#### Sources:

- **SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010** - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

#### Comparison record:

- sft-bio-life-chemical-boundary-001-authority-correspondence: predicted biology:complete-reaction-network-carrier\_\_network-to-maintenance-comparison\_\_closure-and-inheritance-retained\_\_condition-bounded-life-class; source-derived biology:complete-reaction-network-carrier\_\_network-to-maintenance-comparison\_\_closure-and-inheritance-retained\_\_condition-bounded-life-class; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:13fefc9e71d71033fab18115d7775d8c82d2ce6519fb0ae5fa8212798cbf12b; derivation seal sha256:e9a3d4212e573a8a05ec21572512b5d96a70824e9e098039636982794d2ac587; engine receipt sha256:ea8f6776c77a0aa597926fc3919a4fed35c47ec6f0cf0b63189b296c953efca0 at receipts/engine/model\_admitted/SFT-BIO-LIFE-CHEMICAL-BOUNDARY-001-ea8f6776c77a0aa5.json; empirical validation sha256:b0e986124faa31e3ee747b9932dc39ef01d8be7c4f20562904e6c6dcb2311506; measurement receipt sha256:8ebbbb951b0d4ddb127607904707f84f0c12eb8fb43faec0a878bf3b988b46bd; isolation sha256:cfdd7636989b1d3b6b3dfea1783274ea417fedd38302586570622553916450d7; custody sha256:7a59707d87e65910ed925f45b374988c0ec7ff60f530326c610898c2e7f61366.

### 3. Autocatalytic closure and origin-of-life boundary

Claim identity: SFT-BIO-LIFE-AUTOCATALYTIC-CLOSURE-001

**Question and exact theorem.** Autocatalytic closure is the least complete finite reaction support in which every required catalytic role is regenerated by the network while resources and boundary conditions remain explicit.

complete-catalytic-network\_\_product-to-required-catalyst-recurrence\_\_boundary-and-resource-input-retained\_\_finite-network-closure-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-CHEMICAL-BOUNDARY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-catalytic-network                    | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | product-to-required-catalyst-recurrence       | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | boundary-and-resource-input-retained          | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-network-closure-test                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-catalytic-network\_\_product-to-required-catalyst-recurrence\_\_boundary-and-resource-input-retained\_\_finite-network-closure-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite

-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The generated closure test asks whether every required catalytic role is regenerated inside the declared finite reaction support while external resources and the compartment are still recorded. The earlier numerical 1/4 to 1/2 to 1 story is not imported as a chemical concentration or universal ignition threshold.

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:1f293441bcd8513014e4c13ac1bb2bbaec8b813f0f493a6949f255f886698cd5.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:91bb2db52fed69f18c66729b39295995647410a9195dfef2220c391e1de9134d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:5564768635fdad2d74279deb13cbe1096ebe7c5631c95a72aa274a1191e4b38e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:e46d554c0f3ae731d8dc8ed7eb4b931e4d4862f122d137848dce89eb323126fe.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:c2ae56170be0b056d3443181dcfc295090f984517f778a289efa48958e4190fa; certificate  
sha256:a9bf3a6b0fb89418101b4be39b5b7155162c674f6ed5f48a1d6c621a6bfe242c; external-validation  
sha256:889a0d8e3a737153f5c0a9df85f2f1646e00f60223dcb795b65fd63af1275b49.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

**Comparison record:**

- sft-bio-life-autocatalytic-closure-001-authority-correspondence: predicted biology:complete-catalytic-network\_\_product-to-required-catalyst-recurrence\_\_boundary-and-resource-input-retained\_\_finite-network-closure-test; source-derived biology:complete-catalytic-network\_\_product-to-required-catalyst-recurrence\_\_boundary-and-resource-input-retained\_\_finite-network-closure-test; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:585a7b17f052bc5791e59a7e8241dff2be1ad85d43bfd68796909b644ad99936; derivation seal sha256:32275b20b50e2ea9fa3c0784cf4c863bac4986b9496fb301e24680ab99bd6ff8; engine receipt sha256:2ba082e116d52b085704fa2b521e74749aca028326a9a3714e7a7f3e343115ff at receipts/engine/model\_admitted/SFT-BIO-LIFE-AUTOCATALYTIC-CLOSURE-001-2ba082e116d52b08.json; empirical validation sha256:38141f6d68c395e72074e85ea71dfd2e3b6267543a2ee0462271c30aeaa2c404; measurement receipt sha256:e5ff9493b80f06e3f9a258889dc82d8aeb44ba00112974f72adcc0c2c9f0ea59; isolation sha256:ed16406bcefd22ffe11ed65a4511d1ecbf573e2d592bbdd97da32164bea0c1af; custody sha256:38f932d5c5b3442ce76672adc008b14416ee6b916829c2212ef80eb4ba3f4a91.

**4. Biological maintenance**

Claim identity: SFT-BIO-LIFE-MAINTENANCE-001

**Question and exact theorem.** Maintenance is the generated set of transitions that restores identity-defining biological relations faster than incompatible distinctions accumulate at the declared boundary.

living-state-and-repair-carrier\_\_damage-to-restoration-transition\_\_identity-defining-relations-retained\_\_time-and-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-AUTOCATALYTIC-CLOSURE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|----------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | living-state-and-repair-carrier          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased     | damage-to-restoration-transition         | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed                 | identity-defining-relations-retained     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded        | time-and-condition-boundary              | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record       | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing               | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.            |



| Axis       | Eliminated form                    | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-favorable-instance | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule   | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `living-state-and-repair-carrier__damage-to-restoration-transition__identity-defining-relations-retained__time-and-condition-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces maintenance is the generated set of transitions that restores identity-defining biological relations faster than incompatible distinctions accumulate at the declared boundary. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 28ec14a33e711a06a408189d20df2a6b577c0e11c76f4292b04389e3d0c23097.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f8086752b3781e9615839b764966f5cd7769b1319537201c52a02b38c5b2686b.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 6863d7fe7e6c368e6435f543bd801364131f04b6eac71d191b35f043be9deda2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 9b9dda3f85497a84c10e9cd97f0671527d93029673da9b2aef24b60d5baf1dae.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: bc018094c727cf5871ad1cf893c0536926cc9dc70cc64675510b79258f850b39; certificate  
sha256: f24154a79f3ff73bd663596edc4b3c2d67fa516bcb46da05e3ab2094ae7e6118; external-validation  
sha256: 16dfa440c7d403d0192283a8ed198f60531d1c9b9bb41c7b70b4a28ed6339c7e.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
 sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
 sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-life-maintenance-001-authority-correspondence: predicted biology:living-state-and-repair-carrier\_\_damage-to-restoration-transition\_\_identity-defining-relations-retained\_\_time-and-condition-bounded; source-derived biology:living-state-and-repair-carrier\_\_damage-to-restoration-transition\_\_identity-defining-relations-retained\_\_time-and-condition-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256: 8675fa5142326b46a7d349eaa943e7dc4dc8a38b26eb3b5f590e9e97c52deb8b; derivation seal sha256: 8d7e9e27f29e060635e6aa96bede6877f4d4efb027dc36a171b2d74727720540; engine receipt sha256: b90688b59f2dcb1654c668f5068c08772bcea2c46ff7762f7db78d0b3588f60d at receipts/engine/model\_admitted/SFT-BIO-LIFE-MAINTENANCE-001-b90688b59f2dcb16.json; empirical validation sha256: ff95deb575324da3052d5c9863fd0439217e26da756ac4c961a30fa779d7da; measurement receipt sha256: 25a45b75f33b130b03066b7ca1247fd203791df69520d90d01e18870f810242a; isolation sha256: b95843082b56d7175b85794d50cf3b192fcb75eaabab830becc0905e4e041a85; custody sha256: bcef070f64c0993d4d579f40baff463194baf60a618469adc8c5095d629bf29b.

## 5. Identity through constituent turnover

Claim identity: SFT-BIO-LIFE-IDENTITY-TURNOVER-001

**Question and exact theorem.** Biological identity persists through constituent replacement exactly when the lineage-bound organization and its required transition relations remain reconstructible.

lineage-bound-organization-carrier\_\_replacement-with-relation-preservation\_\_organization-and-provenance-retained\_\_scale-and-time-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-MAINTENANCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                            | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | lineage-bound-organization-carrier     | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | replacement-with-relation-preservation | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | organization-and-provenance-retained          | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | scale-and-time-declared                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: lineage-bound-organization-carrier\_\_replacement-with-relation-preservation\_\_organization-and-provenance-retained\_\_scale-and-time-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces biological identity persists through constituent replacement exactly when the lineage-bound organization and its required transition relations remain reconstructible. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:1e3f73e57619daf8cfb5816a3b111091380b935f4bcd1c5f030d24c83986fbc2.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4a824b2141efac6901125947800cb635b2eb843ad080fd57759633863787ddf1.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:587d61f3be6832408e103c3eccdb22ab7392ed0c3c5c5b09dbcc3214de81d178.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:85b4fc11c432a567517c160be1596de09382f4d244e34ab289de2e73d99f0ef1.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:bb5f84f6a89b47426f4a21ce6a4a6f40c7ae864edc3ea901c38d6284a4d2849a; certificate  
sha256:efeb05db503292d0544ccac3d9cad414bc1a35fa9927d92ec5ea610354e4740c; external-validation  
sha256:d32841cedf29348c99ed95f7b586d3f2b656deb7499e2ceff3eb582ebaa28d92.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-life-identity-turnover-001-authority-correspondence: predicted biology:lineage-bound-organization-carrier\_\_replacement-with-relation-preservation\_\_organization-and-provenance-retained\_\_scale-and-time-declared; source-derived biology:lineage-bound-organization-carrier\_\_replacement-with-relation-preservation\_\_organization-and-provenance-retained\_\_scale-and-time-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:ae380075058f1ed2efe9b2db4345a4c969b6e1d308961baef23a38f5e880f818; derivation seal sha256:019c8df470182fb87abbfc7f8b55386fd3c8b21426b48c31452aeb51f4b3207a; engine receipt sha256:a95e5dc8151d49e4dddab3a193e7b5a66f63d04a231d72e1a4064cc72c2755ab at receipts/engine/model\_admitted/SFT-BIO-LIFE-IDENTITY-TURNOVER-001-a95e5dc8151d49e4.json; empirical validation sha256:dbcf121608d8d1573e202f572e4923b16117db517a309abef4f6af41a2f4061f; measurement receipt sha256:08d8910e65023f96849337e8406da42db7c8cf90255f4ed6d5016f0919ab0c5b; isolation sha256:0892964b200e509ba58ade8ae844ce34df74471cd32d4497d79a60d49fd76325; custody sha256:81339d5b92a48c43246d106e479aa24fa988c45ba962adf98a1d4f1948302c2e.

## 6. Loss of living closure

Claim identity: SFT-BIO-LIFE-DEATH-001

**Question and exact theorem.** Biological death is the condition-bounded loss of the jointly required maintenance, regulation and integrated continuation relations, not numerical zero or disappearance of material.

formerly-living-process-carrier\_\_irrecoverable-closure-loss\_\_material-remains-process-identity-not-retained\_\_recovery-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                    | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | formerly-living-process-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | irrecoverable-closure-loss                     | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | material-remains-process-identity-not-retained | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | recovery-boundary-declared                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record       | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                     | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure  | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                  | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `formerly-living-process-carrier__irrecoverable-closure-loss__material-remains-process-identity-not-retained__recovery-boundary-declared__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces biological death is the condition-bounded loss of the jointly required maintenance, regulation and integrated continuation relations, not numerical zero or disappearance of material. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.



The branch begins by separating a merely reactive chemical network from a bounded organization that jointly maintains identity, resources, compartment and hereditary continuation. No single familiar property is allowed to stand in for life.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:3657b3403fd47eda9f2f261dc3399814a1b50c1dc251eb49c131fb72d2d6d864.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:194d952e39f170067fbd663c10110189829633385c92df14851d4477d40f6e75.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:754da5cbda31ebba7e172361efb8975f10ebf4331ff4058002b480229343bd29.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:0ced547b2ba7f44bc09660f98248222a59c2b6159a64c51e22038ad3a883d475.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:ebf9bf1aac6a13707d548d273754b72dd61a2fbd515c113e581add9714dea040; certificate  
sha256:a5d54077dfc125bc4c791af74b14d2607dddc73a085069c228c8dd7d498f22da; external-validation  
sha256:30f5812b0f37ceb8da2646703c72ce36812db58620cbb5aaaa6118ee0f742b1e.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:563bc675734c215823ebffd0174ed0c21d53e19d5426e54326ad3f4e3bcb6960. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-life-death-001-authority-correspondence: predicted biology:formerly-living-process-carrier\_\_irrecoverable-closure-loss\_\_material-remains-process-identity-not-retained\_\_recovery-boundary-declared; source-derived biology:formerly-living-process-carrier\_\_irrecoverable-closure-loss\_\_material-remains-process-identity-not-retained\_\_recovery-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:675b7e8e628e6c58a479c140605a371319fb5d5996a3a2fb1b66ddd45192a6d7; derivation seal sha256:01e6645ba83b833f757418e11c749c5b6d814b57a996eccbc5647ef91d09bb05; engine receipt sha256:12c472570e01cb1d43d7d5f9446284d9c3c459dbbb4f7b756db26728c9e7d750 at receipts/engine/model\_admitted/SFT-BIO-LIFE-DEATH-001-12c472570e01cb1d.json; empirical validation sha256:1ba2445b0afbc253d13ede2df0bda79f43b75d4ec4efc091bd0d03508045a4b3; measurement receipt sha256:d4d0d747416e3dd7f81210be24bcc3b0ebcd5cfffbed028ca427975367258e9ef; isolation sha256:18d4277d50fab384ab6ba14c60f76750422bf2d92f9a4d48d6ed8de283be77ec; custody sha256:d7545ef6da5d9ea9d6bb9e212e32d4d9b39e4410cf95340681ab963110c4bc34.

## 11. Compartment Identity

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

### 7. Biological compartment

Claim identity: SFT-BIO-COMPARTMENT-001

**Question and exact theorem.** A biological compartment is a connected boundary that retains distinguishable internal and external content classes together with every permitted transfer path.

boundary-and-content-carrier\_\_boundary-separates-content-classes\_\_inside-outside  
-distinction-held\_\_resolution-declared\_\_complete-state-transition-control-record  
\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-  
extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-DEATH-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | boundary-and-content-carrier                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | boundary-separates-content-classes            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | inside-outside-distinction-held               | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | resolution-declared                           | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: boundary-and-content-carrier\_\_boundary-separates-content-classes\_\_inside-outside-distinction-held\_\_resolution-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

**Base:** The least positive finite carrier retains every required biological distinction and its complete control record.

**Successor:** Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

**Closure scope:** `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological compartment is a connected boundary that retains distinguishable internal and external content classes together with every permitted transfer path. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 6564fae19c41fa0931d32060ab8bb004f74d45f8deca0c13b177092fef9a8b3f.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c6b0ed2d6f06d4d904c4b0eb719b237dd36bb5ee3feb637cc208f38dc8872439.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 9f9d5be82da591069b4c698480f32a2d91ea57dff6094f1e14e84d495cc5ea41.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 6e56035ddf8ea23160e821ad946f693380836333c57398441fbac9bbba4434f1.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 71bdd6b72e70a08a56e37302e432037e7c330e1d09b1aa1b0b3a6634bc0303cf; certificate  
sha256: 0d851745d6a7b265c1ea357dc22ea1e30fec79d6ecea55571f14d9620ec5c48b; external-validation  
sha256: d7d62db44fe33283820327a0551e651a28fe013d7408ed855bc24f1472d91398.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

**Comparison record:**

- sft-bio-compartment-001-authority-correspondence: predicted  
biology:boundary-and-content-carrier\_\_boundary-separates-content-classes\_\_inside-outside-distinction-held\_\_resolution-declared;  
source-derived biology:boundary-and-content-carrier\_\_boundary-separates-content-classes\_\_inside-outside-distinction-held\_\_resolution-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256: dd324f3358f584884be050c764712eb23a22667eff8bd1f42cb1c39cc4b1d538; derivation seal sha256: befda2d7ac993da93de23d188f2e3c6e91359eaf860df054d49b7ecc87e6621b; engine receipt sha256: 7803f76dac841eb4ca77f8f93b867ec6dab24c521c368692d2fe176de0bc2cef at receipts/engine/model\_admitted/SFT-BIO-COMPARTMENT-001-7803f76dac841eb4.json; empirical validation sha256: 473963374b5755d38bb9d27d07060705fa4921f5bf06e2c8f3f48a10b7fe00a9; measurement receipt sha256: d0634d5ccca45e4a79ff9fd074bb525c2736cff02068ff5c84624d4230780591; isolation sha256: fc1070cf19652be6241069562ee4136a53e066354578206dabc3e907a60d6ade; custody sha256: b8c2af977699d9d4b501ad729e7b92c184cd9bb368be38f05a974406cd94a1c0.

**8. Inside/outside orientation**

Claim identity: SFT-BIO-INSIDE-OUTSIDE-001

**Question and exact theorem.** A retained compartment boundary forces two held side labels, inside and outside, whose distinction cannot be erased by a lawful transfer description.

oriented-boundary-carrier\_\_opposed-side-label-relation\_\_two-sides-never-merged\_\_observer-orientation-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-COMPARTMENT-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|----------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | oriented-boundary-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased     | opposed-side-label-relation              | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed                 | two-sides-never-merged                   | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded        | observer-orientation-declared            | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record       | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing               | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.            |

| Axis       | Eliminated form                    | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-favorable-instance | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule   | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: oriented-boundary-carrier\_\_opposed-side-label-relation\_\_two-sides-never-merged\_\_observer-orientation-declared\_\_complete-state-transition-on-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a retained compartment boundary forces two held side labels, inside and outside, whose distinction cannot be erased by a lawful transfer description. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 22801cfb2f7e3fdb065edbcd0b7efd70120ad5021c717eebdef38bc986802e73.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3d8afa04fc8206c6058520b113f070378abd9765375d30f88ef54aa32935ce66.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 568b821f5d3072c769edd66594d4cc30735512b3ca269e748ef615b183e3b693.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 5077badde2ae1aad6f284fc0f25a3c780f94b7b38b3e99fba5ceca3412fd9965.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: b9efa8970fd11067f19829fff8660a2f8597ddb3afd96a65bf572ebfcbf4f208; certificate  
sha256: 638c98bc796250545946fbe6bc8d398588cbb5864eb73533020b61688087b881; external-validation  
sha256: a8a2d150817de9b29e7ab8f1fada442fdb5a9369c0986d3d6879c8a706ce08c5.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after



derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
 sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells;  
 membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
 sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular  
 function; cellular component; biological process and relations.

Comparison record:

- sft-bio-inside-outside-001-authority-correspondence: predicted  
 biology:oriented-boundary-carrier\_\_opposed-side-label-relation\_\_two-sides-never-merged\_\_observer-orientation-declared;  
 source-derived  
 biology:oriented-boundary-carrier\_\_opposed-side-label-relation\_\_two-sides-never-merged\_\_observer-orientation-declared;  
 exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
 sha256:628df53dfa8dfe0ced0b20643da80003d621fbb94a553a98da96a7a9c2572064; derivation seal  
 sha256:f5e8891205268749215983b8a2f16e2b2d74a11cab2e05bf3a981807c57cb22c; engine receipt  
 sha256:b40c7c3a9150268343bf0170331f0dd0ad6ad2e97e0a415d5a3c5ab03892b60d at  
 receipts/engine/model\_admitted/SFT-BIO-INSIDE-OUTSIDE-001-b40c7c3a91502683.json; empirical  
 validation sha256:d503f4298341aecea9964cc6be94dab62ff23f0e293d0a3c67a12efa9659b025;  
 measurement receipt sha256:a4a306a2a8254b93d8f6b5ecc5ef1fa22b38ace2d0fc4cc24efb14cdabe6958a;  
 isolation sha256:baec94d98735d742c574526d44e088a4efab85517480efd849f7724fb0643fa2; custody  
 sha256:df7e9c2261f4d81258147968e2032c9a908717447ef5f2c62bc1a2012dbd05f7.

## 9. Membrane organization

Claim identity: SFT-BIO-MEMBRANE-001

**Question and exact theorem.** A membrane is a continuous oriented component organization that maintains a compartment while retaining composition, adjacency, state and interruption records.

complete-membrane-component-carrier\_\_component-adjacency-and-orientation\_\_continuous-boundary-support\_\_state-and-scale-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-INSIDE-OUTSIDE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                         | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | complete-membrane-component-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | component-adjacency-and-orientation | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | continuous-boundary-support                   | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | state-and-scale-bounded                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `complete-membrane-component-carrier__component-adjacency-and-orientation__continuous-boundary-support__state-and-scale-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a membrane is a continuous oriented component organization that maintains a compartment while retaining composition, adjacency, state and interruption records. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6910b041bde4b8edb788ac9cdc8cf1f23a74c6c9d66cc2cbb8d13e384e36ba14`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5cf028049ac91052e22dc51926a89236ee6fc8d7e8e622ef516b9493a438c68b`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:a6b610f067b17c90dee702aedee6d483e34406b2940447dcd1252aaa67ccba42.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:b139b7dc6f7d14703813dbbbce9f2670f88c7e50c6669daa5381dd3e8e589a73.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:df8d2ad329a1bc683f4c5ea5e6cc5a3cca222f502213604338828b6e644ce55e; certificate  
sha256:c6435be36b03b80c514ef7258f684c99500821010a15953f17f2dd4d400241b9; external-validation  
sha256:412eb8d94b4a3d02b8e0c59563f12c9a04d00c68367586e64302c10760dd68be.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-membrane-001-authority-correspondence: predicted biology:complete-membrane-component-carrier\_\_component-adjacency-and-orientation\_\_continuous-boundary-support\_\_state-and-scale-bounded; source-derived biology:complete-membrane-component-carrier\_\_component-adjacency-and-orientation\_\_continuous-boundary-support\_\_state-and-scale-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:b0639060306bce02f7f981be5611f657256a6c1121384c7de4820ea5953e5fb0; derivation seal sha256:0fc5bb9e125f5803dd99493fd1f9ca06f033b3366ca7dbdd325b9196acc4773d; engine receipt sha256:0f6b2cec6d4396ab957fc2f3ed3f4204b677170ecca50b9707631a77c2e12 at receipts/engine/model\_admitted/SFT-BIO-MEMBRANE-001-0f6b2cec6d4396ab.json; empirical validation sha256:9c2c0e94e34676558465b898f003683ff41a76d8e87642fa1968d7336ccbe1b4; measurement receipt sha256:345fc078c2c3f88b63b5281c0d7cea0348a51fbe63761420b71cd8739a7a2a42; isolation sha256:90909cb590d8eb01abb866a23a7d9191153a3776af833d3552bf8c9dd528ebc5; custody sha256:f29bb22478aaf7d197b323e9ef77e23e28c1d1912151ea723f314b01ab9e8595.

## 10. Selective permeability

Claim identity: SFT-BIO-SELECTIVE-PERMEABILITY-001

**Question and exact theorem.** Selective permeability is the complete partition of candidate boundary crossings into permitted and rejected species-state paths under declared conditions.

boundary-species-state-carrier\_\_species-specific-transition-permission\_\_allowed-and-rejected-paths-retained\_\_condition-bounded-transport-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-MEMBRANE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | boundary-species-state-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | species-specific-transition-permission        | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | allowed-and-rejected-paths-retained           | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | condition-bounded-transport-test              | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: boundary-species-state-carrier\_\_species-specific-transition-permission\_\_allowed-and-rejected-paths-retained\_\_condition-bounded-transport-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces selective permeability is the complete partition of candidate boundary crossings into permitted and rejected species-state paths under declared conditions. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 45c6fd5ecccfc493555bdd1a38e346b97f031707df94ea5fd06403bea0e15d6.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 99e574d9d4b1c5ae83afc484d271686e00be28c40108a6bab79976b3474eb0a9.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 47dcf21b945737e2e110b04a6fea0a4813ea4d2c9ef6febb688d4ed170fb3caa.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 02678b4a512925504a3d760d61b3a88e0c6a254aea634d9188fa0baef1bbf9dd.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 6c66f4dbd5d5b3ea5b2f362baaf3f7d7d25bd192ba956ef84c18443379b68550; certificate  
sha256: 517e5b1fef0b176c35cac2de506f025a3456466522246dab2da43ed8eb474f1d; external-validation  
sha256: f7a60d1344c8d6ce621df1eb835b220b0806b6c8a70a8a3023a0410f78eb4b43.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- **sft-bio-selective-permeability-001-authority-correspondence**: predicted biology:boundary-species-state-carrier\_\_species-specific-transition-permission\_\_allowed-and-rejected-paths-retained\_\_condition-bounded-transport-test; source-derived biology:boundary-species-state-carrier\_\_species-specific-transition-permission\_\_allowed-and-rejected-paths-retained\_\_condition-bounded-transport-test; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology



- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:cc52c4d469a0275b46121b2ec4117c93a0e84cd22d4b9c5617a91b1518138da0; derivation seal sha256:5ef5f9aecefeaeaf6a74eb40f368755ba5a47bd41d96b812cf14f4efb3696ed4a; engine receipt sha256:2ef57b37bde96ec4bee92de6174564ed8f70b4c72530d367c7316d628b2074c7 at receipts/engine/model\_admitted/SFT-BIO-SELECTIVE-PERMEABILITY-001-2ef57b37bde96ec4.json; empirical validation sha256:25d729118f0aa5aea3f9d01bde54ffdede06eaf43dc4b90f9ff10952b6246e35; measurement receipt sha256:9e3c903853cda72e89b6fb9fb4061219e81a663e6e55321573bfb8965326d38; isolation sha256:993887839a42728b4fe490d09101cf0a682ad305967757193b9f4041254e53dc; custody sha256:92e714801243c1206528c6a01b310d06ed2fb755d3c6872ab4b6ed2a1c766460.

## 11. Nested biological compartments

Claim identity: SFT-BIO-NESTED-COMPARTMENT-001

**Question and exact theorem.** Nested biological organization is a finite containment tree in which every compartment boundary, side label and cross-boundary transfer remains separately reconstructible.

finite-compartment-tree-carrier\_\_parent-child-containment\_\_each-boundary-and-transfer-retained\_\_nesting-depth-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-SELECTIVE-PERMEABILITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | finite-compartment-tree-carrier               | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | parent-child-containment                      | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | each-boundary-and-transfer-retained           | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | nesting-depth-declared                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: finite-compartment-tree-carrier\_\_parent-child-containment\_\_each-boundary-and-transfer-retained\_\_nesting-depth-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces nested biological organization is a finite containment tree in which every compartment boundary, side label and cross-boundary transfer remains separately reconstructible. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 8d20ce57ce44a6f887d93acd44473241b40d2242abf087b78bd0cfe337b2d8.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8085ae8f3b177785c0e1d76717706aff698e7032d9ab159d7a5f5f51ef7516fb.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: e93483cd85549c39d813a85557b2abbbfe1b7db780fce9c78ff3dc54eeade902.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: cdaed20a0935bfdb25ffe6be06083b3d68680bb0f72b5c8342d7e7e06e97fc2.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 6e7ced9b65febebd65af211e2b27c7d0e85f72b3ed4c2e6adda6da360ea8f818; certificate  
sha256: 79838475941a0bd7ad476361d289579aa0ebf31e67126fe444ea039c1881468e; external-validation  
sha256: 675506d58ba5ef1eb1f2ff0bf88946c04d17dd582f4b0f2a2079a75b09fa28e4.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- **sft-bio-nested-compartment-001-authority-correspondence:** predicted biology:finite-compartment-tree-carrier\_\_parent-child-containment\_\_each-boundary-and-transfer-retained\_\_nesting-depth-declared; source-derived biology:finite-compartment-tree-carrier\_\_parent-child-containment\_\_each-boundary-and-transfer-retained\_\_nesting-depth-declared; exact match **True**
- every required authoritative fragment and snapshot hash reproduced

- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:213fd7e96daad382da7524f126c4c22c48ca3fcd58a835b0fd151afb4caa3c21; derivation seal sha256:f5cd1f2a6ccc7a1f886d1f022603f397e5f53a0b571a74ad9fe09f371c3f5bc0; engine receipt sha256:4a92735e1f74f82ff0058013c136ef9ce2753437f6b822fcd61ac1ff6fe81111 at receipts/engine/model\_admitted/SFT-BIO-NESTED-COMPARTMENT-001-4a92735e1f74f82f.json; empirical validation sha256:2f78ee03adc4226ddc0a2b72c1d34ddfd45ff2170a257520ec33b28d87a3c5b3; measurement receipt sha256:cebc4175b41c07f12f5abff74699d11406b8901345ac70ba5b99e1ef19677c37; isolation sha256:778369ffb074f3aeff663f8d3a2adb52c8cdf2fc266722781a20066f9c2e2190; custody sha256:8593769dcc8b0df8745200a447174684cac2e209d550b281a8cf9a8ab9ad1599.

## 12. Compartment lineage provenance

Claim identity: SFT-BIO-COMPARTMENT-PROVENANCE-001

**Question and exact theorem.** Compartment identity claims require an unbroken predecessor-to-successor record across division, fusion, growth and component turnover.

source-bound-compartment-carrier\_\_predecessor-to-successor-boundary-map\_\_division-fusion-and-transfer-recorded\_\_lineage-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-NESTED-COMPARTMENT-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-bound-compartment-carrier              | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | predecessor-to-successor-boundary-map         | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | division-fusion-and-transfer-recorded         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | lineage-window-declared                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: source-bound-compartment-carrier\_\_predecessor-to-successor-boundary-map\_\_division-fusion-and-transfer-recorded\_\_lineage-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces compartment identity claims require an unbroken predecessor-to-successor record across division, fusion, growth and component turnover. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological identity requires a held boundary. Inside and outside, membrane organization, selective transitions, nested compartments and lineage provenance remain separate reconstructible relations.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : a7049b4a3821bac84602f091ff89910c588fb5af2631396c3efb4366621d9736.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 09ff66ac1e2895fdfe10f841216623ec1a7992869fb8ac2430dbb35f7bd364cb.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 5dca4d9bc98e9a71bfbbbfa2566b30d239c11d2be40276f88a781346f0e8027b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 9547b021fb23ff1d5f01b9ee6c772cd0ed0e418272b50b7c19cb027d834959ee.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 2891bf2177849dd187453284984390c98e566c460ae29e14caa613eed0ffc927; certificate  
sha256 : d411bce1773196b0a371d0c53784c36e5cfa940a0f5e585457e8870eac44e862; external-validation  
sha256 : b33b374e06772c01f754cf4ed40bae3240ad691f7b5af77ff2f7df918a90cabc.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 6ceea2a4da91df377be1eed9db511dd03014673bae311cb2edf645c211d4c80d. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256 : ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
 sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-compartment-provenance-001-authority-correspondence: predicted biology:source-bound-compartment-carrier\_\_predecessor-to-successor-boundary-map\_\_division-fusion-and-transfer-recorded\_\_lineage-window-declared; source-derived biology:source-bound-compartment-carrier\_\_predecessor-to-successor-boundary-map\_\_division-fusion-and-transfer-recorded\_\_lineage-window-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:446d5cf91c6ff056fbf601cf4d756adc1251cd4c9e912749c599a6ef55be172e; derivation seal sha256:3f96a7a7c2ae95d312aa2928d06937196d6218d52d108337ed85f2ba5ba380dd; engine receipt sha256:8db42cb9a92c6ddef570c53aed2672fd8f4714357daec1cd587e945d4955f26d at receipts/engine/model\_admitted/SFT-BIO-COMPARTMENT-PROVENANCE-001-8db42cb9a92c6dde.json; empirical validation sha256:c9c0ea427d1d1768e97cef5ae2d1f40e76834de26411f0c418d04063988300fb; measurement receipt sha256:c7807b41057bdec2ab13e966161f14118af1d2e414cacd0da886e630258b99f; isolation sha256:8853a3e1a92b8052eab9c053e45045b0b76e8ec10ea1a99bdbc97347a24f8248; custody sha256:1fcb0d560da168d005ea098ff0ec6885413a27860309b53208dbf4d7cf6fffb93.

## 12. Metabolism Resource

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

### 13. Metabolic network

Claim identity: SFT-BIO-METABOLISM-001

**Question and exact theorem.** Metabolism is a compartment-bound transformation network that records every resource, product, catalyst, transfer and retained constituent relation required for maintenance.



`complete-transformation-network__resource-to-product-transition__catalyst-content-and-boundary-ledger__condition-and-time-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-COMPARTMENT-PROVENANCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                          | Exact elimination / retention basis                                                                                                                                                      |
|--------------|------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | complete-transformation-network      | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                         |
| relation     | relation-imported-fitted-or-erased | resource-to-product-transition       | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.          |
| organization | organization-collapsed             | catalyst-content-and-boundary-ledger | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|-------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | condition-and-time-bounded                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record      | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance  | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality  | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-transformation-network\_\_resource-to-product-transition\_\_catalyst-content-and-boundary-ledger\_\_condition-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces metabolism is a compartment-bound transformation network that records every resource, product, catalyst, transfer and retained constituent relation required for maintenance. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:1fa5554a3a0b70805e5b65ca6a31363672dd32e10dc5a3020c6b91dd407d7dc6.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5e3d35510923c05c641d049f25eacf432c78f4c071a312c88041f0ea5172d6b0.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:c7965cc9ee65d2b7fe10a5db2dab122baaa403d3fd65ee1df8bab69abf0ab485.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:7a797e86dc6f127b86071fc30d32b3c52828cc4df72b99fcad54c1f1b983dab0.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:6f64c68defd0a879027ba7e0806ac6d2d07c62914d45f2c4385cbfd7e5a85fd2; certificate  
sha256:0bf0a6f38e5244345da11811cb621f97bce868de16acc2e6eaadc739222977b4; external-validation  
sha256:ff8b4af474e0c852434de353ee0196a0d3295dd2eaba36c091fab50ac24e9c52.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells;  
membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular  
function; cellular component; biological process and relations.

Comparison record:

- sft-bio-metabolism-001-authority-correspondence: predicted biology:complete-transformation-network\_\_resource-to-product-transition\_\_catalyst-content-and-boundary-ledger\_\_condition-and-time-bounded; source-derived biology:complete-transformation-network\_\_resource-to-product-transition\_\_catalyst-content-and-boundary-ledger\_\_condition-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal  
experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256:b651bf6d8f12552da0ce1d5690afdf02e6759a3b44f5a7652f17180f59398952; derivation seal

sha256:a713e478ef89e1383e98cde2436c99890d81d4d9d21996b5f412a9f59a8899f2; engine receipt  
 sha256:d6478476c899e8a4995d0332764480abbed0f3d6953144652993ba226b445415 at  
 receipts/engine/model\_admitted/SFT-BIO-METABOLISM-001-d6478476c899e8a4.json; empirical  
 validation sha256:413016a61a0ff16380c36f81d24c77b4aa67ed5a903a593e1b96ce89e5d18771;  
 measurement receipt sha256:dae5f7261610008b3599a111691371d49ba206c5443bf3c10864d9ff03468e26;  
 isolation sha256:0237c5a9368994d6e81e669f321e60cfa829b91722e8770cc70eb395e1c1db1a; custody  
 sha256:fd23689283571aca13f5337b562277b7cca8976ad182a6838eef68ca4be31684.

## 14. Biological resource uptake

Claim identity: SFT-BIO-RESOURCE-UPTAKE-001

**Question and exact theorem.** Resource uptake is a source-bound permitted transition from an external support into a biological compartment with identity, amount, route and condition retained.

external-resource-and-boundary-carrier\_\_external-to-internal-transfer\_\_source-id  
 entity-and-path-retained\_\_availability-and-time-bounded\_\_complete-state-transiti  
 on-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adv  
 erse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-METABOLISM-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | external-resource-and-boundary-carrier        | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | external-to-internal-transfer                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | source-identity-and-path-retained             | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | availability-and-time-bounded                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: external-resource-and-boundary-carrier\_\_external-to-internal-transfer\_\_source-identity-and-path-retained\_\_availability-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces resource uptake is a source-bound permitted transition from an external support into a biological compartment with identity, amount, route and condition retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 18da43e41c1ff9952194e41655bb25a47c3f8304b60dd4ea5f3aa44a55794af4.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 6c3950d75253e84710bf0021c0d5c945e0bf3bfd8cc4d0bbdfa1f9573b6506ec.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 49ad80f08ac3a0bc7e8509cf793cb8e33226571bba1db55ad45c2b2fbd08d5ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: ddc7587b811f3a892af8b842bf50daa89bf7ff9994ff2f2bf2408f0f440b8249.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: f0dff81b769ec60fd52306637242c49326435ce043b399a8fa6f7739fa84f4d4; certificate  
sha256: 7e648b297c995d803228ae492c0d09515fae4ead2a1f90a9e62a53df3c136832; external-validation  
sha256: 6f9b712b5c36dc1dd6cb383f9907a78f9abb4ab4f9369c250e7d9e3254bbb122.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-resource-uptake-001-authority-correspondence: predicted biology:external-resource-and-boundary-carrier\_\_external-to-internal-transfer\_\_source-identity-and-path-retained\_\_availability-and-time-bounded; source-derived biology:external-resource-and-boundary-carrier\_\_external-to-internal-transfer\_\_source-identity-and-path-retained\_\_availability-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value

- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:c08eb4e6c052e917116e6ac8ddadde20e0d9fc197b3699065c9b43f46cda44b0; derivation seal sha256:9dc00579bdcfdcb002d1dc608dad4e05a3e46260031730df20bd2d76a6c526c1; engine receipt sha256:b967bbb1af45e237bfbdb7a44afd92156b328f7821f5369aeb0a5bbe3e8366354 at receipts/engine/model\_admitted/SFT-BIO-RESOURCE-UPTAKE-001-b967bbb1af45e237.json; empirical validation sha256:a8f0247e31d0c587bea5149cc05734488401df1061aa190bcf6a2beafcce8fbf; measurement receipt sha256:ffdda0c244afe6945c4c1c719275faa0e742924c92bf79bb6a0300151a0aba61; isolation sha256:7a04b1858c463bff85a2200f484c32aa112b9978a7396a0b040e6ce8928e0a7f; custody sha256:9451de38d5953d067fae5cbf9de68701d96a39afdbf5e6250c638f7dd97e3d38.

## 15. Metabolic transformation conservation

Claim identity: SFT-BIO-METABOLIC-TRANSFORMATION-001

**Question and exact theorem.** A metabolic transformation is admissible only when the complete constituent ledger balances and catalysts, resources and products remain separately identified.

complete-reactant-product-carrier\_\_balanced-transformation-relation\_\_atom-and-held-label-ledger-retained\_\_reaction-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001



- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-RESOURCE-UPTAKE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-reactant-product-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | balanced-transformation-relation              | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | atom-and-held-label-ledger-retained           | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | reaction-boundary-declared                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-reactant-product-carrier\_\_balanced-transformation-relation\_\_atom-and-held-label-ledger-retained\_\_reaction-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- carrier: the biological carrier is explicit and nonempty; passed true.

- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a metabolic transformation is admissible only when the complete constituent ledger balances and catalysts, resources and products remain separately identified. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:089fba2f4d763d34e5660dd3fc7b552e0853fbbd7acb694ab291284c12bfcc81.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a0dff93659a3e7eb51b56c9a5bb2fddf40b45bd6d6b582438b592f3ad778dbdf.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:be5da43ad2d165f419fc478baa830d5d77bd46f18aa85b76fac1172de05b398d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:80c9e7f361d1f50f6a6ab531721ce28dce48da1842d395836039940fcdc6ae11.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:17b210d5e8eb3bf28e009089eb64fd13e6cc6ba0100747e5e43021e93c4526bf; certificate  
sha256:993b16bb1078815488a4b5d693535e275b4e651a29269306fede925dd6bd26c9; external-validation  
sha256:669a12b03698fc67ea9d7d44829f3c9fabf6ed2df765026c6a45541a1eff9876.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-metabolic-transformation-001-authority-correspondence: predicted biology:complete-reactant-product-carrier\_\_balanced-transformation-relation\_\_atom-and-held-label-ledger-retained\_\_reaction-boundary-declared; source-derived biology:complete-reactant-product-carrier\_\_balanced-transformation-relation\_\_atom-and-held-label-ledger-retained\_\_reaction-boundary-declared; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained

- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:2115a126988b46ed61cc950e04c26e8d1721320ddcba4ac630cc58f2639a7d09; derivation seal sha256:ed2cdc8ccdad27d79233cc18dfc614f8aa274d83fef180b966d6854c7576b564; engine receipt sha256:0f1ee13bab8ff013eb1e629c8ea7f95d710817fb54744617342881f9e019ffa3 at receipts/engine/model\_admitted/SFT-BIO-METABOLIC-TRANSFORMATION-001-0f1ee13bab8ff013.json; empirical validation sha256:ba8df7739eff3d5d0f503a8dab2a92275b3c5cf335af97976b21be145ab76cb0; measurement receipt sha256:5cd31967085365c03863d1f26d709e3d6d2b84f27245e6e0a67d823f0eff779c; isolation sha256:094a1ccc13f5066988ae949a9b45b2f7f7a4ec05769ee543f25bcc7e7970f8e3; custody sha256:dee4fd829374186fb804f4f9c4d62f3c1e59e579772b81f777eb04f44ffa08d0.

## 16. Metabolic process coupling

Claim identity: SFT-BIO-ENERGY-COUPLING-001

**Question and exact theorem.** Biological process coupling joins otherwise nonpersistent and resource-releasing transitions through one retained transfer record; an unrecorded gain is rejected.

joint-uphill-downhill-process-carrier\_\_shared-transition-coupling\_\_complete-transfer-record-retained\_\_condition-and-direction-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-METABOLIC-TRANSFORMATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | joint-uphill-downhill-process-carrier         | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | shared-transition-coupling                    | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | complete-transfer-record-retained             | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | condition-and-direction-bounded               | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: joint-uphill-downhill-process-carrier\_\_shared-transition-coupling\_\_complete-transfer-record-retained\_\_condition-and-direction-bounded\_\_

`_complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule.`

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier**: the biological carrier is explicit and nonempty; passed `true`.
- **relation**: relation and organization remain separately reconstructible; passed `true`.
- **boundary**: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces biological process coupling joins otherwise nonpersistent and resource-releasing transitions through one retained transfer record; an unrecorded gain is rejected. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 3179655630407d0ed039c716dcae05c01e75d6ee060e0a5bfdba67d25810b4ea.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9f9c396fe43192c1a39720e70b5bc4e007b1db56a0643c574729f0a79cb539bb.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: c1b563e5dba6d88c1ab191e5201431c297410ed7fc480529b7d64a81864a5d05.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 1571117fb227ae7c6ae50ae6f76ee4d98886c7512d7c55bcd3dcfe279b92b938.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256: 0719b124d5f93b20a037b3f9386802e3d1967a06d74ae7f75360df045f7182cb; certificate  
sha256: 29b9ae95e179ff48ddd65d2157c52a9e9a25b25674f58bb11a4c261a563ac043; external-validation  
sha256: 4f9af53a3d41b9bfcf758f18b751d0eb39e17df9086a321aecf6d8461f5500ce.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: 827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;

sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-energy-coupling-001-authority-correspondence: predicted biology:joint-uphill-downhill-process-carrier\_\_shared-transition-coupling\_\_complete-transfer-record-retained\_\_condition-and-direction-bounded; source-derived biology:joint-uphill-downhill-process-carrier\_\_shared-transition-coupling\_\_complete-transfer-record-retained\_\_condition-and-direction-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:124a87a25d44dd9999a65892cf0d5a0360d74b89c692fb1f4bb6875aadaaf112e; derivation seal sha256:5c6fcdd87763aa8e03f0002ce15962c8802705182359841a35e3f3e3a80fd0d7; engine receipt sha256:01e85868d0ff5ebca68aba0cacb4521615fc96982027aab2a347154370309465 at receipts/engine/model\_admitted/SFT-BIO-ENERGY-COUPPING-001-01e85868d0ff5ebc.json; empirical validation sha256:8e3e8cd222c7975452201f80347729e8ca59c8765cc9c4865c0a9a9dbad2b75d; measurement receipt sha256:ac2e57c6ab237197cb813f92f9739cc08be0db53c100fa2e6949b2917874e41c; isolation sha256:5f7c2a4fc20cbb48149f5c43412eddd405ed8c9620d3013582394854fad96d07; custody sha256:cccdbe53a563dc9b8f24aaa0cbaab6c14405e67e9925489d3dd10d6efa3d46.

## 17. Metabolic by-product and waste export

Claim identity: SFT-BIO-WASTE-EXPORT-001

**Question and exact theorem.** Waste export is a recorded transfer of nonretained metabolic products across the compartment boundary; no product may disappear from the constituent ledger.

byproduct-and-boundary-carrier\_\_internal-to-external-transfer\_\_mass-identity-and-impact-retained\_\_system-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ENERGY-COUPLING-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | byproduct-and-boundary-carrier           | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | internal-to-external-transfer            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | mass-identity-and-impact-retained        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | system-boundary-declared                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record   | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |



| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: byproduct-and-boundary-carrier\_\_internal-to-external-transfer\_\_mass-identity-and-impact-retained\_\_system-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces waste export is a recorded transfer of nonretained metabolic products across the compartment boundary; no product may disappear from the constituent ledger. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 758256d11897a08e00203a534dcb08ab08ab7994720220501335c81bc5357783.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b9f728e3977fa9f0ac51bafb47b6be1eee74d287d767e6512cd04d04f8fc3931.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 432bad812a1107359220cbaf5a0a7af37d7dbfcae8bd8de6782d05fb26c60dfb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 22b5873742e5f3509f658386868a3ca46e3715fcba71728f6ef36eeefd5cb249.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 0e705fc5066581d9397f493dac297d6f743e58f408783a5294229910a20c42ad; certificate  
sha256: 645fc11d463b92c6d6a7c437f671b5a0ddfbde385a747cdd7b54d190c19c7f1a; external-validation  
sha256: 96102f2287a928cdce412590bf0dc3fe64a5d06948c6932d055235e740fd2a63.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-waste-export-001-authority-correspondence: predicted biology:byproduct-and-boundary-carrier\_\_internal-to-external-transfer\_\_mass-identity-and-impact-retained\_\_system-boundary-declared; source-derived biology:byproduct-and-boundary-carrier\_\_internal-to-external-transfer\_\_mass-identity-and-impact-retained\_\_system-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

### Immutable evidence identities.

Pre-source seal experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:efdfc3171af8cfb1915704e415c774998010955fc5f3236b101be8b2b6744d08; derivation seal sha256:8b3b2204b4dcd5a4b331e6a665e7673004bfc58f686f2dbd3be8c9edbe495cc9; engine receipt sha256:7fce9323e170ea5f1948f56c25331618289f899c9abd0a400aa51837ee30667a at receipts/engine/model\_admitted/SFT-BIO-WASTE-EXPORT-001-7fce9323e170ea5f.json; empirical validation sha256:82d7aae3fec50ea77cc18fdc85d53d7e137855f6169db47955ab4e12272ef75; measurement receipt sha256:f54187e5dfd21da13c6019a49810e09632ed17de04df5d03a27240df08a870c3; isolation sha256:37cc09fd827cee442d99aca593a95c9d70d59f81051a3a5ffaebb6eda02ad39c; custody sha256:86a397a2a84877182a6d3e8f5bcd56ad9983800231cde7d41f8ae5ee2682bae.

## 18. Finite biological resource closure

Claim identity: SFT-BIO-RESOURCE-CLOSURE-001

**Question and exact theorem.** Resource closure is always relative to a declared organism-environment boundary and finite interval and requires every necessary input, transformation, storage, export and loss path to be recorded.

organism-environment-joint-carrier\_\_uptake-transformation-export-cycle\_\_all-required-inputs-and-losses-retained\_\_finite-window-closure-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-WASTE-EXPORT-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis    | Eliminated form               | Forced form                        | Exact elimination / retention basis                                                                                                              |
|---------|-------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | organism-environment-joint-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | uptake-transformation-export-cycle            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | all-required-inputs-and-losses-retained       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-window-closure-test                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: organism-environment-joint-carrier\_\_uptake-transformation-export-cycle\_\_all-required-inputs-and-losses-retained\_\_finite-window-closure-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces resource closure is always relative to a declared organism-environment boundary and finite interval and requires every necessary input, transformation, storage, export and loss path to be recorded. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Metabolism is treated as a complete, compartment-bound transformation and transfer ledger. Energy language cannot hide missing matter, unrecorded waste, unspecified resources or an impossible closed organism detached from its environment.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 276f8f6f6ab802b33abfe33f95b7a108f1a9283eeeb916e7e795a0f2c7956694.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e3e586aa2333d1d2672b70993585af3be0ee2760c3d508f38584376c6a0ac727.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:76bbcf75c9d1e300910c92bfabf06d7ccb63c26f59d122498436cc99609aef41.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:d7214062acdf500d1f48cfe8a3aaff80145f8d1b20cb8778b332fd3bfffde627a.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:bbca7b524779a91dab545f41c71defb546514dd60b1970be02bd55ff754e6aed; certificate sha256:3dce4cecf9cab53f1b972c16d37702e5a1b022720cb54e4997e31b2927e2fc8; external-validation sha256:5d4c3302f0af56af6e8a98af1984d12b2915e64058851207aa70734fe367c64c.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:827005b6496bb210df75d8b96ad1f7955b7393d0a35c1c097b83ed2b510bcd5f. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell; <https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html; sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.
- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium; <https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html; sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-resource-closure-001-authority-correspondence: predicted biology:organism-environment-joint-carrier\_\_uptake-transformation-export-cycle\_\_all-required-inputs-and-losses-retained\_\_finite-window-closure-test; source-derived biology:organism-environment-joint-carrier\_\_uptake-transformation-export-cycle\_\_all-required-inputs-and-losses-retained\_\_finite-window-closure-test; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor

- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:4b40bac3a04a89fd27aedfe652e20ce19b2ad3ab1a87c6af248f3963134b984b; derivation seal sha256:ce3b9d703953e18c88e352e8dd0e3217442435d3b854abcc483d8bdddec95f; engine receipt sha256:72758f467ea389cdb7b6a346645361f81f7cf04c4a8b07cf52f71331cfc21114 at receipts/engine/model\_admitted/SFT-BIO-RESOURCE-CLOSURE-001-72758f467ea389cd.json; empirical validation sha256:8d040fbc3175ea33fde873f145a02f0079b76a34ae25e0883798495076a0061a; measurement receipt sha256:50982f53479047f4c771c7fb081d0373b7820e4f97fb9c0fa7833042a21cca7e6; isolation sha256:e3e7f68cb8c221716a29fde30a08d57e0395cd27ad634afb7fa5e010985eaddb; custody sha256:71dc022d0edf8ada93b713b222d3d948cc58add3c01f5a17c209d0d0d9146890.

## 13. Inheritance Replication

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

### 19. Information-bearing inheritance

Claim identity: SFT-BIO-INHERITANCE-001

**Question and exact theorem.** Inheritance is the reconstructible retention of a biological distinction from predecessor to successor through an identified carrier and generation relation.

lineage-and-carrier-state\_\_predecessor-successor-retention\_\_heritable-distinction-and-context-held\_\_generation-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-RESOURCE-CLOSURE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | lineage-and-carrier-state                     | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | predecessor-successor-retention               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | heritable-distinction-and-context-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | generation-boundary-declared                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: lineage-and-carrier-state\_\_predecessor-successor-retention\_\_heritable-distinction-and-context-held\_\_generation-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**



- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces inheritance is the reconstructible retention of a biological distinction from predecessor to successor through an identified carrier and generation relation. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 7701c5d538b47413717a90a22e3c51db52c72f2048279d602b285f07c59c67d9.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 4bb2249d2feacd14adf4ab559b41b67ba6a01e1f5b01ddfc3b264c59cd9d0590.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: bcac5428169967ed10f9b948c4d9c36a44e345e609a7ba23a217a5ae8bee3726.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: bc74a726e22e0cc6d38a63192f592b264f1be166874ec3beb2e4d58c404a0e7c.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 5d5c02605e4e43dc3d39ababdbd1f7553c96d56ffbf077bf8b0406bfc1cb3d0f; certificate  
sha256: e5f645ecae5af8c90b2767cef081dfb808c345fa2e811b372f14a6b84697260b; external-validation  
sha256: f492d009eb079d05f5578979e1cb5a906efb9cc682c499411e31db41dcc3fde7.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aafc13b3c1a43d83e. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-inheritance-001-authority-correspondence: predicted biology:lineage-and-carrier-state\_\_predecessor-successor-retention\_\_heritable-distinction-and-context-held\_\_generation-boundary-declared; source-derived biology:lineage-and-carrier-state\_\_predecessor-successor-retention\_\_heritable-distinction-and-context-held\_\_generation-boundary-declared; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained

- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:b6f98109cddb7012a51ecf4a99acdcd8aa014b7d743d9a931ba22ae5e548e9c3; derivation seal sha256:785b6f42876d211a5305466f07c103fcde8dde75f2a6287a29af87df456b5bef; engine receipt sha256:bda67ea3373d60bc39193d3e349cb4134dbc67a74c75ff195cdcec01ff9fa2ed at receipts/engine/model\_admitted/SFT-BIO-INHERITANCE-001-bda67ea3373d60bc.json; empirical validation sha256:93527d46b1bd8adcbc61018a3360975e1a9ee60bbc50fc73938c05f22b5f56e6; measurement receipt sha256:becdbf0a2d9afac559c5b57d27633f04ea305a3006acbd643d9fab5a345f849; isolation sha256:800143c48f9a354b404d3a4ab4a305a75a823a125afdc144ac88565669e8addb; custody sha256:65ff8010c195e9d54fc4d6bebb6519db4df64323e7c69e87aaf237a0b9b9dc56.

## 20. Template-directed copying

Claim identity: SFT-BIO-TEMPLATE-001

**Question and exact theorem.** Template copying generates a product word through a declared positionwise relation while retaining template, product, direction and every mismatch.

template-and-product-carrier\_\_positionwise-complement-or-copy-relation\_\_template-product-distinction-retained\_\_finite-word-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-INHERITANCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | template-and-product-carrier                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | positionwise-complement-or-copy-relation      | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | template-product-distinction-retained         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-word-boundary                          | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: template-and-product-carrier\_\_positionwise-complement-or-copy-relation\_\_template-product-distinction-retained\_\_finite-word-boundary\_\_c

complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces template copying generates a product word through a declared positionwise relation while retaining template, product, direction and every mismatch. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : c070877f905a086425d18d33ed67adad9e66e63bc052515d5f54683c6fcf407d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 6d77d626aad946c433875782af84ee3582744785af8583813116dd3a83c56a04.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 7a8a38bd7cda8bb407f696fca2aef9802876979cef961c1a226e2847264d6770.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 3f1328e2150bb91fe193b179ede26513c6b2019347537ec05e00c07a7964be25.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256 : 6a2d30f5b849be053fb09f25d7a8881f2ec377de4ed7d218b2932ee197ddb339; certificate  
sha256 : 982a51b62fcb82a218a8acf83e9e6500117ac7ab29b0bce3eaa6234981e2f5ea; external-validation  
sha256 : 59c930cb336ab219cf17ea1e51e6eed248de662da8477fb4344f93564261d68.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256 : 9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aafc13b3c1a43d83e. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010** - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256 : 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- **SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;

sha256 : ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-template-001-authority-correspondence: predicted biology:template-and-product-carrier\_\_positionwise-complement-or-copy-relation\_\_template-product-distinction-retained\_\_finite-word-boundary; source-derived biology:template-and-product-carrier\_\_positionwise-complement-or-copy-relation\_\_template-product-distinction-retained\_\_finite-word-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:499fcfb346f621acfff5e9544056f51d43e3b05402dca1d4480a9ce193667935; derivation seal sha256:bee96b76e5f1bbfb91b0ab2b71b21f5c1cb16db50b16fd692455e42ea6c60a30; engine receipt sha256:61253b914236e5d5311c56897776efd2be36082b311f1dbfda780aafc7d2dd04 at receipts/engine/model\_admitted/SFT-BIO-TEMPLATE-001-61253b914236e5d5.json; empirical validation sha256:db9825bbcef24c300693130d642bccaa56580197fb43b62a01b2d469e39c345c7; measurement receipt sha256:28adc69372a19bbb05bbbec0ca23262ecdfe8113826753365d9309310ee7cb07; isolation sha256:53691fdc91bb09e2c75112a470fe82bb250d6cd6b6ecf158fc52d7bf6ddfd9b7; custody sha256:9d0e6a7fadedf08d9ceddeaebfdbb2036bfd1d5638732f3c5a94a87352fdf104.

## 21. Biological replication

Claim identity: SFT-BIO-REPLICATION-001

**Question and exact theorem.** Replication is a resource-bounded process that generates at least one distinct successor carrying the declared hereditary relation while retaining predecessor and copy provenance.

complete-replicator-and-resource-carrier\_\_one-predecessor-to-distinct-successors\_\_copy-lineage-and-resource-ledger\_\_generation-and-capacity-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-TEMPLATE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | complete-replicator-and-resource-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | one-predecessor-to-distinct-successors   | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | copy-lineage-and-resource-ledger         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | generation-and-capacity-bounded          | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record   | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |



| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-replicator-and-resource-carrier\_\_one-predecessor-to-distinct-successors\_\_copy-lineage-and-resource-ledger\_\_generation-and-capacity-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The Fold supplies distinguishability and successor organization, but Biology additionally requires a resource-bounded predecessor-to-copy process and retained lineage. No universal per-generation doubling is asserted for every biological replicator.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:eca2786a9edf18710fc6072cf74d503803a0c933d0cc0554eb5eb212d58d4576.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:db91212c72ff19289d83b769513e6a1605936b6817d0816aafec819d0a1fbab4.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:bf47e40ce6a8cfc0516cdec8f5c7303b0c34b3426a1c95b6d3d1d2e33f820581.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:e2ee3a160801dfd54fd95e275d4cb4821f7f01bce4252387b6c072b72c3501dc.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:98efb6d451f67988732f774b07a51ca32b0590fd91f8c65af56bfd1f64aaa5af; certificate  
sha256:eda71c37c1afc408f50acb422daefd09063434fe5573ee4d544fcfc0d803170c; external-validation  
sha256:65730a8e9d215a1b5aef1abc012d5cc27fa68be71df2d3761d16123e9e056237.



**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aaafc13b3c1a43d83e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bfff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-replication-001-authority-correspondence: predicted biology:complete-replicator-and-resource-carrier\_\_one-predecessor-to-distinct-successors\_\_copy-lineage-and-resource-ledger\_\_generation-and-capacity-bounded; source-derived biology:complete-replicator-and-resource-carrier\_\_one-predecessor-to-distinct-successors\_\_copy-lineage-and-resource-ledger\_\_generation-and-capacity-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:01f983e5c8b521a68ca9bed5c5c1468eecef32936f97838c86e0e44e0d2ef51f; derivation seal sha256:07865883ffddb3c40cb710d244229b4ff902e92fee0efbcd03e40e611331e7ee; engine receipt sha256:786da8a317ddf2f962210ed732dc9cdd68011cf4045ea288a44fd5825bb159fd at receipts/engine/model\_admitted/SFT-BIO-REPLICATION-001-786da8a317ddf2f9.json; empirical validation sha256:90692386f18fc65a5fdd6be551de9839661b60cdb38f8c180d24a262bb2ca80a; measurement receipt sha256:532b741238bf16e32835661598ae4ad07116e747f23c98e7dfcb43ec55dd9db8; isolation sha256:f84dc55e3e8d9c1b76ce16b34c1b98da22e4a975217b89600ad285bc8fe1fe04; custody sha256:0efb39aa775a2c9bf1d1794e6bb918bcd8b651d8fefb3971ee3383d1e984b7a.

## 22. Replication fidelity and error

Claim identity: SFT-BIO-COPY-FIDELITY-001

**Question and exact theorem.** Copy fidelity is the exact part-to-whole count of agreed positions over a declared aligned word; every mismatch and unobserved position remains in the record.

aligned-template-product-words\_\_positionwise-agreement-census\_\_matches-mismatche  
s-and-unread-positions-held\_\_assay-and-length-declared\_\_complete-state-transitio  
n-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adve  
rse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-REPLICATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis    | Eliminated form               | Forced form                    | Exact elimination / retention basis                                                                                                              |
|---------|-------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | aligned-template-product-words | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | positionwise-agreement-census                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | matches-mismatches-and-unread-positions-held  | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | assay-and-length-declared                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: aligned-template-product-words\_\_positionwise-agreement-census\_\_matches-mismatches-and-unread-positions-held\_\_assay-and-length-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces copy fidelity is the exact part-to-whole count of agreed positions over a declared aligned word; every mismatch and unobserved position remains in the record. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:5b21ae0626adce63321f7337107f5c4eec81d436785d34b41e5ded8b377cd38e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0d2afd17e5868a94cdb82cbdaa73e4b208ea7c435a77724f7f15c7b621772e97.

- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:10a4a58cd89765ec0c3f08fa24f59057e88a98c8b81f754d7764960500148b03.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:603c27024bd608c8b36314e52213c2c602eb8d7b40a2e82bb3f328f8df10ff8a.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:cb941b5065d87a34b707d87bb8d5e9cfc66480ca3494d0c24dd3d4387ae8edfb; certificate  
sha256:004f5f8f0e7629af33367ae3618d8352c0da7abbc3027ddf4e3978e489a80857; external-validation  
sha256:d040c65a929d84cd573d334ddde5fe1a8506a824feb3824f087656bdc755c5c2.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aafc13b3c1a43d83e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-copy-fidelity-001-authority-correspondence: predicted biology:aligned-template-product-words\_\_positionwise-agreement-census\_\_matches-mismatches-and-unread-positions-held\_\_assay-and-length-declared; source-derived biology:aligned-template-product-words\_\_positionwise-agreement-census\_\_matches-mismatches-and-unread-positions-held\_\_assay-and-length-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition

- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:c7eb57f98dfe3c7900a0e1e9a586f10df33d2f1f4fa29c65e82e58bb67aa9b62; derivation seal sha256:360f57d68c42c1e35b7d47e76cc41e5b4dd334a6ab7075fb166ba3b67a07d969; engine receipt sha256:02f163116212c08a6f3c238f7b007daa26b02948159a246ce7cbab7258c537aa at receipts/engine/model\_admitted/SFT-BIO-COPY-FIDELITY-001-02f163116212c08a.json; empirical validation sha256:efb7b7f2935720d19c66a021f44bdeb45e7bd7baf4aa002bdf1f77a3c6047e99; measurement receipt sha256:3c6aacd4105c13fdb592409c09ab02918c8f81e60b7f32f510ed47ad8379816c; isolation sha256:ece517e4a865532e23eb7c837964ddc8f16b33a64fad0617b3f1e3269ad5bae4; custody sha256:4cc4b74ce063fba12cdf62a69f20d7edcc8159cdd0675fe367c92ab80e8cc649.

## 23. Heritable copy variation

Claim identity: SFT-BIO-HERITABLE-VARIATION-001

**Question and exact theorem.** A copy difference is heritable variation only when its exact state and location persist through a registered successor relation rather than appearing only as transient damage or observation error.

variant-product-and-lineage-carrier\_\_copy-difference-retained-across-generation\_  
\_location-state-and-origin-held\_\_observation-window-declared\_\_complete-state-tra  
nsition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and  
d-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-COPY-FIDELITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | variant-product-and-lineage-carrier           | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | copy-difference-retained-across-generation    | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | location-state-and-origin-held                | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | observation-window-declared                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: variant-product-and-lineage-carrier\_\_copy-difference-retained-across-generation\_\_location-state-and-origin-held\_\_observation-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the biological law; passed true.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a copy difference is heritable variation only when its exact state and location persist through a registered successor relation rather than appearing only as transient damage or observation error. Specimen-, organism-, population-, method- and condition-dependent magnitudes



remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : a9a1530d73e269cc814fb906897198c159aaf0b7505cae9cafeadac5263847e9.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : a6d129e0967f3ee4a93d70f83d729f7a142bf036686cc88a2d5b541f6285d3df.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : fe2c90466d6f5f0c15170a95274f82c1bf026eff78b51cbf08326cc51d0f588a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : f992c0377c721175cc5a3cc71d9db7394e41ba308a8919eac76489f8a7ea7c87.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : eca4beb180e7037e8bb68360e751caa2b8957b1aa9f995926dfa2a1032a1735b; certificate  
sha256 : 93e76fe91c90461f11b8311c6bfa360e99f9be4499ef7d5d387c0745d29082314; external-validation  
sha256 : 45ab46d925b107bd7da566148bec6a14b7599a0914f16cd408df65af3a33b216.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aafc13b3c1a43d83e. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256 : 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256 : ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-heritable-variation-001-authority-correspondence: predicted biology:variant-product-and-lineage-carrier\_\_copy-difference-retained-across-generation\_\_location-state-and-origin-held\_\_observation-window-declared; source-derived biology:variant-product-and-lineage-carrier\_\_copy-difference-retained-across-generation\_\_location-state-and-origin-held\_\_observation-window-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.



- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:69e89508871b2e13d48c0a99a273061ad82c49cb676ca7b4a4b44b19c9b672e0; derivation seal sha256:9f9735ba8052ce9c97c0d0dcdb2a20130dc503baf4ca8ffd6ead6ee8c586a6b; engine receipt sha256:bf30cba5ca7777237d9989a3a820981b8fb1bfddc05d0418e9d39187a215f50f at receipts/engine/model\_admitted/SFT-BIO-HERITABLE-VARIATION-001-bf30cba5ca777723.json; empirical validation sha256:cf5ad7eebe11c12e0f9be07c60f75c269bae75460de6adf9befa33b6e2021a77; measurement receipt sha256:2bbad29f598e342c544793b9b0ab7180b2b5bef5caefbae881e5e93c9700af1b; isolation sha256:da76e8fbf07ea4edb3083ae9c67801e4c9c5eada217a13727c9116e43af853e1; custody sha256:5bd4e12f6d3d7000cd0cc8fbdedde95905f55d35733886618f8eab67dca9bb65.

## 24. Replication lineage

Claim identity: SFT-BIO-REPLICATION-LINEAGE-001

**Question and exact theorem.** A replication lineage is a finite directed predecessor-successor graph with every sampled branch, merger, missing observation and generation boundary explicit.

finite-predecessor-successor-graph\_\_parent-child-generation-edges\_\_branch-merger-and-sample-history-retained\_\_depth-and-sampling-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-a-dverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-HERITABLE-VARIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | finite-predecessor-successor-graph            | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | parent-child-generation-edges                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | branch-merger-and-sample-history-retained     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | depth-and-sampling-declared                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: finite-predecessor-successor-graph\_\_parent-child-generation-edges\_\_branch-merger-and-sample-history-retained\_\_depth-and-sampling-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a replication lineage is a finite directed predecessor-successor graph with every sampled branch, merger, missing observation and generation boundary explicit. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Inheritance is a predecessor-successor relation carried by an identified word or structure. Template, product, fidelity, mismatch, variation and lineage remain distinct; a Fold preimage count cannot by itself prove biological replication.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 61264e6b615167c9058a715ce3147ff81fe760b0adbb9c040782345905630974.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e3bdfccd6d1111c28ee889f920ca106e854aceb352b6e7596f87a1d977958710.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 89d37fe5bd95fcce6595584856484c87452ec04f2928231d926cd69b16ac9d02.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: fe41d977c77e93cec038729c1dfc3c432e24fdeeddde1354a10fb26167d5d146d.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 72c0d47a01b573674c2fcef5a717db150efa01c3c5da2ac0cfbaa0a9b452a071; certificate  
sha256: 056975a0ee91e8d72ea34c6898fb3675068688ff8b2ffe367d04db5e8dcd07c0; external-validation  
sha256: 4973ed7290744a250fccca286cea4212767ac411e6378e07bab4b57d227f82927.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 9bd322119aa867be3d69d0d06a1b8d9f980723c82c27965aafc13b3c1a43d83e. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-cell-origin-metabolism.html;  
sha256: ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; scope: cells; membranes; compartments; metabolism; replication.

Comparison record:

- sft-bio-replication-lineage-001-authority-correspondence: predicted biology:finite-predecessor-successor-graph\_\_parent-child-generation-edges\_\_branch-merger-and-sample-history-retained\_\_depth-and-sampling-declared; source-derived biology:finite-predecessor-successor-graph\_\_parent-child-generation-edges\_\_branch-merger-and-sample-history-retained\_\_depth-and-sampling-de

clared; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:79c1e7bf2557f6bdd3d905fb55a7ea36c49af7afffd3140815369ffc5fa6182e; derivation seal sha256:8e9747ef592eb7aa1f78f1ba6a072edc519ebba4ada69516a951778ecb9b48c7; engine receipt sha256:ab8ac6e3ff4c35a7389870a76995ccd049e29980d43773686ee4efbcedc52391 at receipts/engine/model\_admitted/SFT-BIO-REPLICATION-LINEAGE-001-ab8ac6e3ff4c35a7.json; empirical validation sha256:a25cdb577c0756dcb98af3dc888fc1b7bcc73858666179cb9a7416ceb8b05199; measurement receipt sha256:6c642fc33bb46b56ed6ae17bb89779e4ab549d37580a8e4b8ef55bd00beb3f1e; isolation sha256:c9ff5d8121a3118285fa0fa53707fbbe14ce94573f853e696550fe950fcd1f5e; custody sha256:cf5fc79e1e602a0345ce7fcf0fb6cda9cc413eacba3d3600e34bc7f491f6895a.

## 14. Variation Evolution

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary. Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

## 25. Biological variation

Claim identity: SFT-BIO-VARIATION-001

**Question and exact theorem.** Variation is the complete retained set of distinguishable biological states within a declared population and sampling boundary.

population-state-carrier\_\_within-class-distinction-relation\_\_individual-lineage-and-state-held\_\_sampling-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-REPLICATION-LINEAGE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|----------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | population-state-carrier                 | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased     | within-class-distinction-relation        | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed                 | individual-lineage-and-state-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded        | sampling-boundary-declared               | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record       | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing               | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.            |

| Axis       | Eliminated form                    | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-favorable-instance | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule   | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: population-state-carrier\_within-class-distinction-relation\_individual-lineage-and-state-held\_sampling-boundary-declared\_complete-state-transition-control-record\_root-bound-forward-forcing\_positive-finite-successor-and-adverse-closure\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces variation is the complete retained set of distinguishable biological states within a declared population and sampling boundary. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary.

Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 6c724bf5766feb055ceff8a02443a6c4937cd7c26a6e4614b91dd0f90a741523.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 83f01959608fc6fcc4fb22324000711dfa00b304bb45f0d9d0f30acfcfe4561.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 7c3318b4c674a8b5620e0c40d2b99570e757bd60ca09d9a33790f123db8a3641.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: e1dcaa2282177e38e699c37f2cb6980f4eb02eafe843295ae65e95dd12a41ae8.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: b902aee23e7417d53dea60020315f3e188983124fdfcae4b7f9b5a7520d4bc89; certificate  
sha256: f23538b2c9ef0adef7382910fab31eb3ac09f23a77ccf83886a22ab3f4bdec1c; external-validation  
sha256: c6ca5f528f683ac6f85fb030f43b0ae3ff9618b3132456114853d66b1f076b5b.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after

derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
 sha256: 49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-variation-001-authority-correspondence: predicted biology:population-state-carrier\_\_within-class-distinction-relation\_\_individual-lineage-and-state-held\_\_sampling-boundary-declared; source-derived biology:population-state-carrier\_\_within-class-distinction-relation\_\_individual-lineage-and-state-held\_\_sampling-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256: c2ab429f898c76f76a4309d25800f0d068e811fb66d1af21d986e4f0c3e0f827; derivation seal sha256: 37891da03add4351969e66202429e199d547ca7312ba4cf88c8d187cb66a5916; engine receipt sha256: b954170326296f4004da08e53cc32ceafabd09e84aca0bd274cf5151d294a888 at receipts/engine/model\_admitted/SFT-BIO-VARIATION-001-b954170326296f40.json; empirical validation sha256: 881222935b5b2f2be5c6746e32f8603cb553fe28c154a35b729933b01fe58733; measurement receipt sha256: 3376b9c79b7fc0f8423ea26eadf9609c4ca2d0d3166275df436c45399e9eeb48; isolation sha256: 9038b427501321105d1bd8c8939bdda181ed06c355c8b5bed75edb733019ce14; custody sha256: a267970aef045b76a83555f9ddb54b55d07544c4a263b66774038c044f3816e.

## 26. Heritability relation

Claim identity: SFT-BIO-HERITABILITY-001

**Question and exact theorem.** Heritability is a population- and condition-bounded relation between predecessor and successor distinctions and does not by itself identify a molecular cause.

related-generation-carrier\_\_predecessor-successor-covariation\_\_environment-and-carrier-context-retained\_\_population-and-generation-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-



adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT - ROOT - THERE - IS - NO - NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-VARIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | related-generation-carrier               | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | predecessor-successor-covariation        | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | environment-and-carrier-context-retained | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | population-and-generation-bounded        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `related-generation-carrier__predecessor-successor-covariation__environment-and-carrier-context-retained__population-and-generation-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces heritability is a population- and condition-bounded relation between predecessor and successor distinctions and does not by itself identify a molecular cause. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary.

Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: ca0e9abccdf0880f9d9551a1d6741617fe686063238cbfcfc1de51e6567bc34.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 570eb8a3d93678185fbb65a419cab3db42b7c706705e17f75181ea7bb1002fc.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: b6b745d8d20c6f51ec86dae7bf1d432179dada97f46c55698083709bf40b794a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: cb72591dcde90f6a7084748e807ff3c020bc64b61cf7118264711baa13b185bc.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:ace0d732d46ced5cde74ac825ed95a8acc4d5dde25e8a0ae3602b51c6baabe44; certificate  
 sha256:1936e101e664824a9c84094f7734ea58c0bc6796377a6fc5d06ab3aba707fd5b; external-validation  
 sha256:0d1da3972d90b66b52a1a2c80f74ef1344750ec3affbed2a09400c99cad1d28d.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
 sha256:49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-heritability-001-authority-correspondence: predicted biology:related-generation-carrier\_\_predecessor-successor-covariation\_\_environment-and-carrier-context-retained\_\_population-and-generation-bounded; source-derived biology:related-generation-carrier\_\_predecessor-successor-covariation\_\_environment-and-carrier-context-retained\_\_population-and-generation-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
 sha256:8d341168789ac59220a02d990c921e6511a514631ec4b0d80195cbc1f0294685; derivation seal  
 sha256:e5f5ce1bbce7bb5ef54626e859afcfa939c0c6b7a1d8d7d4f4bd09de85e4ee6a; engine receipt  
 sha256:4bf4352533206993a667ffb2502a0a73963659b877192d2f0bdf28c33fc7a99a at  
 receipts/engine/model\_admitted/SFT-BIO-HERITABILITY-001-4bf4352533206993.json; empirical  
 validation sha256:4024f7cb41a705555c98b35591787ddc2b0aa3b77134eda8935961a29892aa13;  
 measurement receipt sha256:349e21aa58ff00085057acc24c7ab706379276bc3408b2828bd0cd4b419669b7;  
 isolation sha256:007ab252e996367bd11360924968789b86c54f2c939ee4e23ee5267fc73d7daa; custody  
 sha256:27a013203cca5f6da4cb822310350d5007f20f4443c57a245f0475fd66cea1.

## 27. Differential reproduction

Claim identity: SFT-BIO-DIFFERENTIAL-REPRODUCTION-001

**Question and exact theorem.** Differential reproduction is an exact comparison of retained successor counts among biological types under one declared environment and generation window.

typed-lineage-carrier\_\_type-specific-successor-count\_\_exposure-survival-and-fecundity-ledger\_\_generation-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-HERITABILITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                   | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | typed-lineage-carrier         | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | type-specific-successor-count | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | exposure-survival-and-fecundity-ledger        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | generation-window-declared                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: typed-lineage-carrier\_\_type-specific-successor-count\_\_exposure-survival-and-fecundity-ledger\_\_generation-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces differential reproduction is an exact comparison of retained successor counts among biological types under one declared environment and generation window. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary. Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 7509f9fde3643349ba9d7201a09fb3085a7b7421e42752bb6f76046c6aa22641.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e66234e6981ffff36eeee1812ef93a22318cca0355df779461cf6ae9d1e1a65fe.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:249559a742d29bd91451abc7f870a7ae8036d25367ba691b54d1dbc009c8bd5c.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:ddf5a615b19447b8f6281b1723455966ff9296dde6ed38f9ee828d9862f940b5.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:85335a45dbc1035b19963288b571caaba487fc0568aac3eeec6f34081b5c51cb; certificate  
sha256:0c7e3a816f59ea51679f3655b0b543719082b7dd0fdeafc39aee582285342a77; external-validation  
sha256:aaa9ddab0c98d979ff4b73f8a43b9668064b40cda7cc7226f8345616a55e60e6.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
sha256:49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-differential-reproduction-001-authority-correspondence: predicted biology:typed-lineage-carrier\_\_type-specific-successor-count\_\_exposure-survival-and-fecundity-ledger\_\_generation-window-declared; source-derived biology:typed-lineage-carrier\_\_type-specific-successor-count\_\_exposure-survival-and-fecundity-ledger\_\_generation-window-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256:e2ffc0ca838bbb78d9d070741bf549122c1426552caabd5a6513d90675112db3; derivation seal  
sha256:9cef70018bac067530c1e5d8f238580fcdc9f9a234e419c2de02d5b5c847f4e3; engine receipt  
sha256:4dbd6e4ba6ea52aa8b4ab09ea6a1804fbd34ad7ed57014981982ce701960501 at receipts/engine/model\_admitted/SFT-BIO-DIFFERENTIAL-REPRODUCTION-001-4dbd6e4ba6ea52aa.json; empirical

validation sha256:20213dcf616097e958d562bcc39fa1a2f747abb5891f5b66db3c61942cd23d2a;  
 measurement receipt sha256:bc333fc352dac4a20f7c973bc5300005ca53d2aff2b2edc5ea044e14acc9318e;  
 isolation sha256:69ddce8dce2787db9d0d6e6a68c79772e4a3ea826d46ec9db9d5aff47ef197f6; custody  
 sha256:b8d85855d9b1475a62570ddf1b87f876dfc4bfa4e5b1ae19b7515b32c21f29c5.

## 28. Selection

Claim identity: SFT-BIO-SELECTION-001

**Question and exact theorem.** Selection is the recurrence of a heritable variant together with reproducible differential continuation relative to registered alternatives under a declared environment.

heritable-variant-population\_\_variant-linked-differential-continuation\_\_competin  
 g-lineages-and-environment-held\_\_finite-generation-test\_\_complete-state-transiti  
 on-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adv  
 erse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-DIFFERENTIAL-REPRODUCTION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.



| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | heritable-variant-population                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | variant-linked-differential-continuation      | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | competing-lineages-and-environment-held       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-generation-test                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: heritable-variant-population\_\_variant-linked-differential-continuation\_\_competing-lineages-and-environment-held\_\_finite-generation-test\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces selection is the recurrence of a heritable variant together with reproducible differential continuation relative to registered alternatives under a declared environment. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary. Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:cc3627cec1593084447bad4f997e5e65d65ab0f19e54fd7a3dc3a9521d92695b.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d8a1aa1fe74df040081cc120cb5d1ae05c047bae48e26c46d837ede5b3d6f868.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:8301513aa3eac2c7637ef1e0830b346934280505928f4ad66bc0550656b8318d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:6f7bec6519556af17288a9118073a572aad0a1ffe844e5ff310348555218f421.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:9d75f32663d35aecbf1826ea2c2ddd9658a016461ea55ede747fe37d036d38c4; certificate

sha256:93fff1779a5e2658a918cb9383d3f567eca6e9b8d2a13beaf5078b74cd389a276; external-validation

sha256:bf543f13dd3736897cd23828553319966c6bba70f7aadb6b01a1475f5e8dd2d9.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1** - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
 sha256:49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-selection-001-authority-correspondence: predicted biology:heritable-variant-population\_\_variant-linked-differential-continuation\_\_competing-lineages-and-environment-held\_\_finite-generation-test; source-derived biology:heritable-variant-population\_\_variant-linked-differential-continuation\_\_competing-lineages-and-environment-held\_\_finite-generation-test; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result

- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:11fa8b55dc4c0b3981e926655145b2c93107db699307e3600c234930a22a4a7f; derivation seal sha256:f652b8865151f4fe434186702d80148fdedee3bcc0fa9c202e0803595f7b7b9e; engine receipt sha256:ad24aa695baf0f2105af2a4422093a3e61b41805ff42c8fbe133744d0967d4e3 at receipts/engine/model\_admitted/SFT-BIO-SELECTION-001-ad24aa695baf0f21.json; empirical validation sha256:38012c5b09eeb9add13ad7f29b9c90f92e83af79377f551678845ebbfbd7205; measurement receipt sha256:215ab9ae1e9b38b29cd547ef15ae81b9d6a77d291eb84777cc9e937176a7878b; isolation sha256:204cd3bd31ee9ad25bdaa2b638ba1647b0fd9fbc3969963ada6b1273e91f8e46; custody sha256:5f8360211c8d855220f6d6a97832199e2094437617ec93286a4e28a0b26df959.

## 29. Fixation and loss boundary

Claim identity: SFT-BIO-FIXATION-001

**Question and exact theorem.** Fixation is whole support of one retained variant in the declared population; loss is structural absence of that variant from the same support, never a universal guarantee from benefit alone.

complete-population-support\_\_variant-count-to-population-count\_\_all-sampled-line  
ages-retained\_\_finite-population-generation-bounded\_\_complete-state-transition-c  
ontrol-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse  
-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-SELECTION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-population-support                   | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | variant-count-to-population-count             | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | all-sampled-lineages-retained                 | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-population-generation-bounded          | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-population-support\_\_variant-count-to-population-count\_\_all-sampled-lineages-retained\_\_finite-population-generation-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** Fixation is exactly whole support of the registered variant in the declared finite population. It is an observed state classification, not a theorem that every beneficial allele must reach that state.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary. Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 76ade3498a6fc026a4f5bb262d44e6a4476b8fee2388a41bf67fe001fbce058c.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 180aaf20ff2965c015ea0c37cee1cda76b32e6fa709f2371b5cc28609156445b.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 92cc96f130f0621334d18df4d0d3712381113e714ceabfe5bc5f861e303c0d64.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 5b48e5c7bee5525198c9100fc163e324c6e2cf057ac45b8a0f451c2f67f0afd4.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: cef09818d86b163670727926d240c2806043cd572eb86332970f0dd3a4d83917; certificate  
sha256: 166bf84c7b6eed3ce36adcb2c7d20cfa05003a824a1553822b6f611f80efffb5c; external-validation  
sha256: 3f751bbdae308b4b14f2d35af8feb81e46530d87741a04ab658a7e152d72a924.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
sha256: 49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-fixation-001-authority-correspondence: predicted biology:complete-population-support\_\_variant-count-to-population-count\_\_all-sampled-lineages-retained\_\_finite-population-generation-bounded; source-derived biology:complete-population-support\_\_variant-count-to-population-count\_\_all-sampled-lineages-retained\_\_finite-population-generation-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage

- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:dfe610e8f36b3cd60ea973305775dbca7ad11fb9770a6ca702039355920d3279; derivation seal sha256:b7a67748061861b9cfdca93b87109cea1e4a3441ef5e6e560e3a0e378ecafbf1; engine receipt sha256:5433bc60179895f6ea7c0aafb31fe07de2b418fadc79ca23b33bcda503172cef at receipts/engine/model\_admitted/SFT-BIO-FIXATION-001-5433bc60179895f6.json; empirical validation sha256:c64741b4d3018107ac0368acafe52b904d76d2af85a0d845332b531f711bd5af; measurement receipt sha256:4d0033a4c8dae6cfffac34d3339aa9a4133ce01ea91d5772d701fdb8156babed5; isolation sha256:ff9781171311ad1ba9d9f6d1ddc63376864d1611a12f79bbebd76932dd094077; custody sha256:b68a1ba3f257a756237ee79353411ddd04cd40db0a87395f68ce0b1836144c0f.

### 30. Selection and drift distinction

Claim identity: SFT-BIO-DRIFT-BOUNDARY-001

**Question and exact theorem.** A change is attributed to selection only when the registered variant-continuation relation survives the declared controls; otherwise finite sampling recurrence remains an explicit drift-compatible alternative.

```
replicate-population-carrier__state-change-to-fitness-association__sampling-and-
randomizer-provenance-held__finite-replicate-boundary__complete-state-transition
-control-record__root-bound-forward-forcing__positive-finite-successor-and-adver
se-closure__no-extra-rule
```

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001

• SFT-BIO-FIXATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | replicate-population-carrier                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | state-change-to-fitness-association           | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | sampling-and-randomizer-provenance-held       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-replicate-boundary                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: replicate-population-carrier\_\_state-change-to-fitness-association\_\_sampling-and-randomizer-provenance-held\_\_finite-replicate-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the biological law; passed true.



**Scientific meaning.** Genetic drift is preserved as a genuine finite-sampling alternative. A selection claim closes only when variant-linked differential continuation survives the registered controls; otherwise the engine does not award a selective cause.

Evolutionary laws retain the population, environment, heritable distinctions, successor counts and finite sampling boundary. Selection and drift are competing generated explanations until controls distinguish them; benefit alone does not force fixation.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : d80a0d5a0819a682a070dc6d9154164ef5a100995d9a858b3e9a3faa204f67bc.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 639923c4f2390d0ba83ae73c8ae71b2dd1e9fbe336b2674c2caf00584957f2ce.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 3e5024e6abf4660bb25c9a61601049bbabda74ba8874ea7c294ca3e8da891ec0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 9d3328403acce7b4285fc4e0ce5a522c0f958d84b9852353f90a1ac7989bceb.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : a6f913f5bba62d034cfc391232ea3564d1598bae7b3ac23eccd97750e6ec385d; certificate  
sha256 : c259ab21785b5bfc63d2a166fd87e6e9a7d425de15994ac4216908c281ed6df4; external-validation  
sha256 : 1f2926b402240ff5f4ceb7127c88ae751a23535d6628ab3dbb9f9e912e0e8794.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : ecbd114a38ac55b9762b533c42f569d89464e1038f29e9c71a27d704c0a0adb9. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-evolution-genetics-v1.html;  
sha256 : 49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; scope: heritable variation, mutation, selection, population size and drift.

Comparison record:

- sft-bio-drift-boundary-001-authority-correspondence: predicted biology:replicate-population-carrier\_\_state-change-to-fitness-association\_\_sampling-and-randomizer-provenance-held\_\_finite-replicate-boundary; source-derived biology:replicate-population-carrier\_\_state-change-to-fitness-association\_\_sampling-and-randomizer-provenance-held\_\_finite-replicate-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value

- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:e6bdfa0368d11e86988eaec9630036e5fff50609d60f283018f969ede693bfcf; derivation seal sha256:66f4bf19f979a8c1d23a50aa08bb00325a8f2a5e4aabb62f877e4b5204dbcb1b; engine receipt sha256:86df52d952985ced70d264231c72484aec2ee99bac822c5d9e3b70a7c8c3552d at receipts/engine/model\_admitted/SFT-BIO-DRIFT-BOUNDARY-001-86df52d952985ced.json; empirical validation sha256:1a412e83696b7d8e019a78e9393de3433125ef8fad4fcc292734bb00dba76fb3; measurement receipt sha256:8a7c2c3e4bbd3b823341d4e7bab7a936d3c29bd0fb0d02cdcc62e9ea80108e26; isolation sha256:d2a6c805c5457db98035566b8b9a3e66319b60161c787922ecf4021fc325a0e3; custody sha256:178317c9675c8b2d54db6fce15493c4b9f14e6235aa9e309fb41faa3c2f9424a.

## 15. Gene Genome Regulation

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

### 31. Nucleic-acid symbol alphabet

Claim identity: SFT-BIO-NUCLEOTIDE-ALPHABET-001

**Question and exact theorem.** Two independent held distinctions generate exactly four nucleic-acid symbol labels at the DNA coding boundary.

two-distinction-symbol-carrier\_\_ordered-pair-label-generation\_\_four-code-symbols  
-retained\_\_DNA-alphabet-boundary\_\_complete-state-transition-control-record\_\_root  
-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-DRIFT-BOUNDARY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | two-distinction-symbol-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | ordered-pair-label-generation                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | four-code-symbols-retained                    | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | DNA-alphabet-boundary                         | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: two-distinction-symbol-carrier\_\_ordered-pair-label-generation\_\_four-code-symbols-retained\_\_DNA-alphabet-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.
- `four-symbol-alphabet`: two held distinctions generate exactly four distinct ordered labels; passed `true`.

**Scientific meaning.** The exact construction uses two independently held distinctions. Their ordered product has four and only four labels. NCBI's genetic-code table is opened after sealing and confirms the external DNA coding boundary; it never chooses four.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: b2b7fd24f6f50334f54095cf9e853bde107f7c6beec67413f56db83c05ede2bb.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: aae52631e8ef53301b706bc3607847e3cfd5598c0d49a9365c96ee51959ebc3.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: faab68ba4ee4c7e66e0425f9d21fbfb4890ce20801f5bc1aa36f77fe82e1e352.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 092977d104c2657e693c306ff832185f5ad381a91986579b5458619b0dc17a0b.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256: 8a5043f511744aecc5ae8a302ee5e092e55fc9e64c99789cc26d71df15f7ad8e; certificate

sha256: 1a6a0208a759e52b855bde66a5744d5197bdc978af74b020965048c3d3cfba5; external-validation

sha256: 5cfbcf9a0f712e0df05daf6d081d7f7cdca5e526e03339b9bd09bcaa479fc3b5.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: 6f250bdb68f5cafdbda4d7ff5a890fd1fd247154849a51ade1665ba5b62dfe7d. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-GENETIC-CODE - NCBI Taxonomy; <https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genetic-code.html;  
sha256: c329091f012ae379949dc3a39f1961d2338a7b38ddad60245f9c12d8a31db514; scope: complete genetic-code tables and ordered base positions.

Comparison record:

- sft-bio-nucleotide-alphabet-001-authority-correspondence: predicted  
biology:two-distinction-symbol-carrier\_\_ordered-pair-label-generation\_\_four-code-symbols-retained\_\_DNA-alphabet-boundary;  
source-derived biology:two-distinction-symbol-carrier\_\_ordered-pair-label-generation\_\_four-code-symbols-retained\_\_DNA-alphabet-boundary; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:e3afdeac9f79951f16e8eb3c6e61076cb3239298a1e5fc57cf651e9ebba8213d; derivation seal sha256:13a7017054fc4a7f8132de2bb90a1d83723e1b25879dbeb2f4d90098b0b03e07; engine receipt sha256:39e582a3d655540a7f6ff19c813e7ee9e246c97d49e062515b9cc50bfd5e0b80 at receipts/engine/model\_admitted/SFT-BIO-NUCLEOTIDE-ALPHABET-001-39e582a3d655540a.json; empirical validation sha256:5b88fd9297da135ebbef1785ba4caf8cf978c7f1f505fc46a7bbc7b059312e42; measurement receipt sha256:6b36e083fc42d9b9a261441de19bdbba935e903f6b5dd37b8fb9784052a9089d; isolation sha256:6f053876ba2824f93453476a4313f57bd4dc65da87d6287392f440c14e7cafe4; custody sha256:c307c276ec4236baf22036d3c04238dc8894bbfe8159e6ca0c8254b8a798c1c2.

## 32. Triplet codon word space

Claim identity: SFT-BIO-CODON-001

**Question and exact theorem.** Three ordered positions over four generated symbols force exactly sixty-four distinguishable codon words.

four-symbol-word-carrier\_\_three-position-ordered-composition\_\_position-and-symbol-identities-retained\_\_complete-word-census\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-NUCLEOTIDE-ALPHABET-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | four-symbol-word-carrier                      | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | three-position-ordered-composition            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | position-and-symbol-identities-retained       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | complete-word-census                          | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: four-symbol-word-carrier\_\_three-position-ordered-composition\_\_position-and-symbol-identities-retained\_\_complete-word-census\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.
- **sixty-four-triplets:** four symbols over three ordered positions generate exactly sixty-four words; passed **true**.

**Scientific meaning.** The complete ordered product of four generated labels over three positions contains  $4 \times 4 \times 4 = 64$  distinct words. The census stores all 64 exactly, with no floating value, continuum alphabet or imported codon table.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 256965349dcde2cdf50749405ec56876f2f83f33c411a95185370db1b286903a.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 466ecc5a98c5f6d5dc6f8923d9a9a608b41688c381c8d16d4c998b03acbd9535.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: e6aa56169cbc1ece2047e1c1a583c2db64f0ec81e636b301efeb5b98167be5a1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 17a85c84c8601b2e9a68099792370c3e1ed9627429780401922ef3091c039b7e.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 545cc708833fa2878d47adcee64884f7ad68117bc57853972c34a5a6c3d603ab; certificate  
sha256: efd8219f2a4b72a4cd1ade0e36539d01454d601d05b0f4042073cc2ebb16132; external-validation  
sha256: 5ee84aaa90b02ad001633cc875e3d4bd49b45ea145e8c3d0958eb72b224fb2db.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity

sha256: bcb4bb29a428d2635213bd6cfdc2d790afa6be319498252ebafa39e29b8037b7. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-GENETIC-CODE** - NCBI Taxonomy; <https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genetic-code.html;  
sha256: c329091f012ae379949dc3a39f1961d2338a7b38ddad60245f9c12d8a31db514; scope: complete  
genetic-code tables and ordered base positions.

Comparison record:



- sft-bio-codon-001-authority-correspondence: predicted biology:four-symbol-word-carrier\_\_three-position-ordered-composition\_\_position-and-symbol-identities-retained\_\_complete-word-census; source-derived biology:four-symbol-word-carrier\_\_three-position-ordered-composition\_\_position-and-symbol-identities-retained\_\_complete-word-census; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:99e8c003a1287eed95a2dfffb51425ed87ddeab4caa2252c962fc06c822aa166e; derivation seal sha256:b9799da737e9851ad382569ba4868e8926e83d7bf7ac8d64aed08388a4fc31a8; engine receipt sha256:fd2c8f05f16667ba8eb4cee73b2a8fef5ed9fecaa0194b8991464dbe0dfb5940 at receipts/engine/model\_admitted/SFT-BIO-CODON-001-fd2c8f05f16667ba.json; empirical validation sha256:4b31da0e2e6638b9dfdab3d66c74fd31e3fe170aecff5d3257e021de1b0afa76; measurement receipt sha256:ccdd6fb4eb08beb5d6bf5f1319f9b5351b82406be929d928921f983e97807ebe; isolation sha256:02a4ecea2010ebb04feeff2d35f7e0e1d464c896f6a07c403a105e2d8c41c29c; custody sha256:f5b5a7c4b22ad8f78adac5a344fc0cd8f7f445fdd5a1607b778c25b2f2020d77.

### 33. First-two-position codon boxes

Claim identity: SFT-BIO-CODON-BOX-001

**Question and exact theorem.** The sixty-four codons partition by their first two positions into exactly sixteen boxes of exactly four third-position words each.

complete-codon-support\_\_prefix-equivalence-partition\_\_third-position-variation-retained\_\_complete-partition-census\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CODON-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-codon-support                        | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | prefix-equivalence-partition                  | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | third-position-variation-retained             | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | complete-partition-census                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |

| Axis      | Eliminated form                      | Forced form   | Exact elimination / retention basis                                                                                                                                                               |
|-----------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extension | free-fit-exception<br>-or-extra-rule | no-extra-rule | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor and depth independence.** Sole survivor: complete-codon-support\_\_prefix-equivalence-partition\_\_third-position-variation-retained\_\_complete-partition-census\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.
- **sixteen-four-word-boxes:** prefix equivalence produces sixteen boxes of four words and partitions all sixty-four exactly once; passed `true`.

**Scientific meaning.** Prefix equivalence over the first two positions yields  $4 \times 4 = 16$  boxes. Holding the third position leaves exactly four codons in each box and partitions all 64 words once. This proves the combinatorial box structure, not a universal amino-acid assignment for every genetic-code variant.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 88d664879dbfd75c23d0ec6fba81ed5ecc420337bf70ca7f54e2318c9bd5dd77.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f5451f5672a5a4802e51fd4ad4c45656e44f13747526733b61b0150fd98c7397.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 8eb0f4f3a5970c7f7247a94b3712a9961a3ecf9bdeeba933dd13c0016b0a3f0a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 25a87c8315e284912f01db00def14c4a74b08092d8ef34251e10ced232600fca.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 39f806d93b68d88255f7783dd51b40faf3135715fad3bcecc620a3f948147f19; certificate  
sha256: 37b4e64a47c4476ab2aa6def3f3d8368080540ccbaec36004c0e262a38ca70fa; external-validation  
sha256: 75f7884fcb0f05b98ba2e2713b83103814edcd1ea245c77717fe26c3add2297a.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: bc60baf9eb8fe4bcac8014ba329ba5a4444e7ae954fd8899a78d34e343ff4281. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-GENETIC-CODE - NCBI Taxonomy; <https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genetic-code.html;  
sha256:c329091f012ae379949dc3a39f1961d2338a7b38ddad60245f9c12d8a31db514; scope: complete  
genetic-code tables and ordered base positions.

Comparison record:

- sft-bio-codon-box-001-authority-correspondence: predicted  
biology:complete-codon-support\_\_prefix-equivalence-partition\_\_third-position-variation-retained\_\_complete-partition-census;  
source-derived  
biology:complete-codon-support\_\_prefix-equivalence-partition\_\_third-position-variation-retained\_\_complete-partition-census;  
exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256:af43128fb83a33629ad488a96d9fea23a5d5e1c5aefe89f47a97949fd9123d11; derivation seal  
sha256:a54eddafe0b8dbe8d83bdb906805f592a81927885a7ad760f1f3d62276691904; engine receipt  
sha256:15585592b4edb63a9d8dacad3d81b9dc4f06782a239cdceff805007b48114794 at  
receipts/engine/model\_admitted/SFT-BIO-CODON-BOX-001-15585592b4edb63a.json; empirical  
validation sha256:0544895969669d073b142bbbe820f637f77f48c76c5841729f944adec1b4c549;  
measurement receipt sha256:7c5d9253d208e3a3825de507caffbfabc0b6ebccb59ff9c9f466a2ff17f29bb0;  
isolation sha256:dcd5a4ea8c265b777f05db95097c538046a872c23c1b2c1a3b9b0f7d02389bd3; custody  
sha256:d29c406a2602059a16e66fe1c263a01ed918b3226fa8635292e9e6e990729a6a.

## 34. Gene as expression-bearing hereditary interval

Claim identity: SFT-BIO-GENE-001

**Question and exact theorem.** A gene is a source-bound hereditary interval whose expression or regulatory relation is demonstrated with orientation, context and evidence boundary retained.

source-bound-sequence-interval\_\_interval-to-product-or-regulatory-relation\_\_orientation-context-and-provenance-retained\_\_annotation-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CODON-BOX-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                                 | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | source-bound-sequence-interval              | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | interval-to-product-or-regulatory-relation  | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | orientation-context-and-provenance-retained | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | annotation-boundary-declared                | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: source-bound-sequence-interval\_\_interval-to-product-or-regulatory-relation\_\_orientation-context-and-provenance-retained\_\_annotation-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a gene is a source-bound hereditary interval whose expression or regulatory relation is demonstrated with orientation, context and evidence boundary retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 8b24cd7e9d81484815fc28e10e2de7410c34a310e681b31f468ae78d4d24ed32.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 85c58826c4fa0056083e77d37a3752bfdd7a974002fa48da4dbd8998fa3a139b.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 1c33c164ace7102fb71c723d46730256fb3eaca16b39f7df15aafbcca459aaf3c.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 2612004fddcf2e9ada7193de0b4d823933d0ec143037bcb2e75bb682f59a2cb6.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:2759b1b29a13100a661f6b0eb3a864125d178022bbba096b0e877fdcd9b90ec8; certificate  
 sha256:2c6c3dd34c92044a4b3fccb742a04cc98df2aac2cc30eafad371066c632e59a9; external-validation  
 sha256:0269050c18913a184ae6e1b69fa9a44f449d7988453fd192a224af5b013e8f13.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:bb795ba687f8884ecc5a2c9f7ac175ef81ec49ab748bdb40069f9640bd0db02a. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1 - Sequence Ontology Consortium; [https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256:dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.
- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE - Gene Ontology Consortium; <https://geneontology.org/docs/go-annotations/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html; sha256:b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.

Comparison record:

- sft-bio-gene-001-authority-correspondence: predicted biology:source-bound-sequence-interval\_\_interval-to-product-or-regulatory-relation\_\_orientation-context-and-provenance-retained\_\_annotation-boundary-declared; source-derived biology:source-bound-sequence-interval\_\_interval-to-product-or-regulatory-relation\_\_orientation-context-and-provenance-retained\_\_annotation-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
 sha256:da94aa3194091727390c82417462c3be29795af4d1f4a5423165ed5dc6bd0b4e; derivation seal  
 sha256:765be48fc978ca8200740c4ce61f1f76419220d7e09303b4a60c78f41ce50b38; engine receipt  
 sha256:1f647ee1b8fdbf32fcb5fca9934a4d7b3dcd300fb73b4a069cd1fd3ddfdc2321 at  
 receipts/engine/model\_admitted/SFT-BIO-GENE-001-1f647ee1b8fdbf32.json; empirical validation  
 sha256:0b3c146fe32601934a053d4607634bd8ae07a8acd3e493ab9a2ba63296f7d42c; measurement receipt



sha256:56387976cb718efe5adae29af0b75d99fe1f1a65e5a4e40187bba5ad2ac3d014; isolation  
 sha256:36811cd2a853b445e7a7b8651d75991f17a34109e34d01259171aa99a583ca10; custody  
 sha256:7b4629df45fb6fb6d62e687f92c7036aee8f9b953920d21311c4446b746cbc87.

### 35. Genome organization

Claim identity: SFT-BIO-GENOME-001

**Question and exact theorem.** A genome is the complete declared hereditary carrier support of an organism or cell with order, linkage, compartment, copy state, gaps and assembly provenance retained.

complete-hereditary-carrier-support\_\_ordered-and-linked-locus-relation\_\_chromosomal-compartment-and-copy-state-held\_\_assembly-and-sample-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-success-or-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-GENE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-hereditary-carrier-support           | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | ordered-and-linked-locus-relation             | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | chromosomal-compartment-and-copy-state-held   | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | assembly-and-sample-declared                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-hereditary-carrier-support\_\_ordered-and-linked-locus-relation\_\_chromosomal-compartment-and-copy-state-held\_\_assembly-and-sample-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a genome is the complete declared hereditary carrier support of an organism or cell with order, linkage, compartment, copy state, gaps and assembly provenance retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:d2647f260665157e067554102d0a45f885092b92a04980402f286dc86561dbb8.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e262c8318cf31fe42caf5dc185be14865a3965d3b4ab9cb5bbeab3923e0e2bf1.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:2fa4bcd2c258d41964e1898ddab4d325d7014b6444450e6608159236796768fa.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7f402f86c3fe27233160b804fd3324ec4d4431e0df627b298e7220f2e4037cc2.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:b91df81951dc2bf5cfc270f10fdcf82f14f46e2a8b17f9f58d99aa253fd7801b; certificate

sha256:43e342da9d2d7f4db34ce0343947bbc79d17883909eb32fce7d22938ee056a8; external-validation

sha256:e4f2866ce021fb4289f90a5d73468d3d27bfe8364e77ccb1d4d6449d96ddb0eb.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:32629665264ddd4a5dd352fbec1c75ab6780c26958f8c962199ac5a659603c59. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1** - Sequence Ontology Consortium;  
[https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256:dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.
- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html; sha256:e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.

Comparison record:

- sft-bio-genome-001-authority-correspondence: predicted biology:complete-hereditary-carrier-support\_\_ordered-and-linked-locus-relation\_\_chromosomal-compartment-and-copy-state-held\_\_assembly-and-sample-declared; source-derived biology:complete-hereditary-carrier-support\_\_ordered-and-linked-locus-relation\_\_chromosomal-compartment-and-copy-state-held\_\_assembly-and-sample-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage

- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:891206a908bcabfb523e57d4f58f6fc7449e0860f6db441182d29643f58965c3; derivation seal sha256:f4a102e885ed28376c2c5946da2ff21d4d061634115f3e364ec61e7ffd017dd3; engine receipt sha256:6c0797aa33cb2bd27abc50954ed0ebdfc9daf9cae65ebba4cf1905f43bb77de0 at receipts/engine/model\_admitted/SFT-BIO-GENOME-001-6c0797aa33cb2bd2.json; empirical validation sha256:902eb9ac80ba0f6ed43ee8242211f17623f1d0e2ff015f52629f8c0a8ba09446; measurement receipt sha256:cbe5d5583c067b3a31c7f7dffb500d403b51e130455836248033a1c738d4ba56; isolation sha256:f67f67307eaf4c6ca0a7a8216c10a78cf9e157f8a37188afa1298751c5fceed; custody sha256:1bb70e047d5fe71f9b875ebdfc5adb174763c7ed8b8f904fa4dfe4d5aa4ac3fd.

## 36. Expression and regulation

Claim identity: SFT-BIO-EXPRESSION-REGULATION-001

**Question and exact theorem.** Gene expression is a recorded transition from a hereditary interval to a product; regulation is a reproducible condition-dependent difference in that transition with regulator, context and time retained.

gene-product-condition-carrier\_\_condition-dependent-production-transition\_\_regulator-context-time-and-output-held\_\_assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-cl  
osure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001

• SFT-BIO-GENOME-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | gene-product-condition-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | condition-dependent-production-transition     | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | regulator-context-time-and-output-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | assay-bounded                                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: gene-product-condition-carrier\_\_condition-dependent-production-transition\_\_regulator-context-time-and-output-held\_\_assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the biological law; passed true.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces gene expression is a recorded transition from a hereditary interval to a product; regulation is a reproducible condition-dependent difference in that transition with regulator, context and time retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

The code is generated as an exact finite word structure before external tables are opened. Gene, genome, expression and regulation remain evidence-bound sequence and process relations rather than names borrowed from an annotation database.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 2f44b8605d5ccf27b4612d64d970f69d24ab2eab66762a352c36b957c948e809.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : be4e58a663e7d787568f5b1754ff9c6e3a4737ef8d7b978594e98e1581172b0d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : aedd7ddc9dd6c8108dcad5a9e87f94e7584889ddabba6d08c6de41051f2ebc93.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 722a67e78744e01859a111482214eb6f305d0bd3da7eda9c0f6264aa1ea83087.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 065c255481f53681063a9372221b7d7b4f77efe306a7e6f55416d331d975c7e4; certificate  
sha256 : 3ad51e91204dbbf0aa814713d6036b3de8e11440302b017012a2e4a08ee5e200; external-validation  
sha256 : 4a878647f0cda6209ee7b1dd6405b7ef6149cc56bceb97b301e3254bbaf69310.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 92cad67126fa0e2e3f109da353a4166b9c3243876a5e2e34e8c94c845c8899fe. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes; <https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html; sha256 : e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.
- **SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE** - Gene Ontology Consortium; <https://geneontology.org/docs/go-annotations/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html; sha256 : b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.

Comparison record:

- sft-bio-expression-regulation-001-authority-correspondence: predicted biology:gene-product-condition-carrier\_\_condition-dependent-production-transition\_\_regulator-context-time-and-output-held\_\_assay-bounded; source-derived biology:gene-product-condition-carrier\_\_condition-dependent-production-transition\_\_regulator-context-time-and-output-held\_\_assay-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:9e43850252a549989ac8539b184885097b492a5ab892f4d634c691d0699c36a6; derivation seal sha256:e6d01b4cad82777a7636b4382030730e1e28e85f673e9708f86c848720c6cc; engine receipt sha256:c3d503b8e21516a9ef94990d71b37e651441b2b040e32be6e261a863eab6cf2d at receipts/engine/model\_admitted/SFT-BIO-EXPRESSION-REGULATION-001-c3d503b8e21516a9.json; empirical validation sha256:78d53bea10eee6c4aff4b18a33e1d7b30a8279ea8321ffe2bc27ecf0d45b687a; measurement receipt sha256:8f0af19f14319b2e92fc909533cbc88726a5bd700ab44755831a7dc404590294; isolation sha256:bd097b3bc12b295a157d662f6e16893ab97f11d37f04f1a0a206e30dff9a500c; custody sha256:3528d918b22365db0224c66f87259b59a2d6876adee3b456ac3388ddfa441777.

## 16. Protein Structure Function

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

### 37. Protein sequence identity

Claim identity: SFT-BIO-PROTEIN-SEQUENCE-001

**Question and exact theorem.** A protein sequence is an ordered residue word with every position, linkage, modification, ambiguity and source record retained.

ordered-residue-word-carrier\_\_peptide-linked-position-relation\_\_residue-modification-and-provenance-held\_\_sequence-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001



- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-EXPRESSION-REGULATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | ordered-residue-word-carrier                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | peptide-linked-position-relation              | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | residue-modification-and-provenance-held      | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | sequence-boundary-declared                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: ordered-residue-word-carrier\_\_peptide-linked-position-relation\_\_residue-modification-and-provenance-held\_\_sequence-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a protein sequence is an ordered residue word with every position, linkage, modification, ambiguity and source record retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 02022e7ee2f7383c57fa269de280c463e59ba27227c129a67e63fe776d9fd899.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 99b11a138d9e243fe3e9f3fe5919c41d0d4c6f34ee4fee2d5293ffa8d0516310.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 61d04d27ac41125a67eebd975fa941832f35c160dfd857148007aa4728c8fe1d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 9bd7f3d20ceddca413b9a8a55b68bb02f2b8292ef9885097ec712df6ff48d637.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 10c88201bbb9aad9cd438cbeae0ca7016045e7a88f8f79714cfc2bbebfca720d; certificate  
sha256: 8c6e372e2baa115405ab5d0175949ca2f7b40a8df727740ebb8226184a6614a9; external-validation  
sha256: fc904cf29548624102c67133621dc590be743912f735b66824148af7224dae83.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 235bdc4ad83b399a2f7cf741e06c97c1b083d54d9a62702e66fe4af38cdef180. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

#### Sources:

- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1 - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html;  
sha256: ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.

- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1 - Sequence Ontology Consortium;  
[https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256:dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0fffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.

Comparison record:

- sft-bio-protein-sequence-001-authority-correspondence: predicted biology:ordered-residue-word-carrier\_\_peptide-linked-position-relation\_\_residue-modification-and-provenance-held\_\_sequence-boundary-declared; source-derived biology:ordered-residue-word-carrier\_\_peptide-linked-position-relation\_\_residue-modification-and-provenance-held\_\_sequence-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:612284a3289d59240c55fd8272b0131bb5c59b5daa66b994cabf06fffae54f0c; derivation seal sha256:4dd96a4bc24649c3d37d98be74957abce178ef885f214c2d56ce5a91baae8435; engine receipt sha256:3789bf44006ea0f149389546b5ca94ed4f38fb4272ce37212c66ed9451965c71 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-SEQUENCE-001-3789bf44006ea0f1.json; empirical validation sha256:e10571348943ab4a80b46b7ba52b1963c50fe84410cc4359f2c1dd92eaf9aa90; measurement receipt sha256:c9f4ab4e7197241a387f487d689c687a2e64907620ac8584a26fb621fb88a05e; isolation sha256:aa5a2d8c8d437a767a79f2bf1f17fc23e02c476edb7e4e6ef1d2c9afa3aed360; custody sha256:48b2c1f67dabb905ac9515bf7ab5db50227a8c2ca76c507a59cded8a27ac01cb.

## 38. Protein folding relation

Claim identity: SFT-BIO-PROTEIN-FOLD-001

**Question and exact theorem.** Protein folding is a condition-bounded transition from sequence support to a recurrent structural class; it is a generated descent relation, not exhaustive random search or an imported unique-shape premise.

sequence-condition-conformation-carrier\_\_transition-to-recurrent-structure-class\_\_path-ensemble-and-condition-held\_\_resolution-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-SEQUENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                             | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | sequence-condition-conformation-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | transition-to-recurrent-structure-class | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | path-ensemble-and-condition-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | resolution-and-time-bound-ed            | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: sequence-condition-conformation-carrier\_\_transition-to-recurrent-structure-class\_\_path-ensemble-and-condition-held\_\_resolution-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The law closes directed, condition-bounded reduction into a recurrent structure class and rejects exhaustive random search. It does not claim that every protein has one rigid conformation or that the old two-step 3/4-to-1 path is a measured folding trajectory.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 41f38e12759034a681cc4ddfed52fe239c377dcef0a24481b2f1b188811eac92.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c04ba14ee26642441aa7bc85fe8eafc32b566cf8b9818c91402caa371330296e.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: b592d2ac698c66097545ffdc48e8bae9b4aee8615c847368ca48b1389f263b30.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: d2ecb5634038d1eb974c6affb81d1298f0954cf88c3933361ce1455f510c15ab.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: fd0bc8f8b1da01db9066a2c58907d285ec784480d8e940f33ad2e4d91fedda4d; certificate

sha256: ea1d7f0c822bbbc3064c4fb43f562e403f47ce78a2c8cad3fff42c2864a7781b; external-validation sha256: 8f54fbb9b31d97c30f98fdf8f511195f9fa4ffd62893abc6e0f9c363ccd7be5c.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: 235bdc4ad83b399a2f7cf741e06c97c1b083d54d9a62702e66fe4af38cdef180. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1 - NCBI Bookshelf, The Cell; <https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html; sha256: ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.
- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1 - Sequence Ontology Consortium; [https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256: dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.

Comparison record:

- sft-bio-protein-fold-001-authority-correspondence: predicted biology:sequence-condition-conformation-carrier\_\_transition-to-recurrent-structure-class\_\_path-ensemble-and-condition-held\_\_resolution-and-time-bounded; source-derived biology:sequence-condition-conformation-carrier\_\_transition-to-recurrent-structure-class\_\_path-ensemble-and-condition-held\_\_resolution-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256: 85ff6cef6aed6684617b0454f0c152d3d625b99ef83912d50dcb1bfac7f85cde; derivation seal sha256: ff985b69821252a0ed7d4ef7d46353fab4915129d8d7e7eb2444708dfcabf102; engine receipt sha256: e9c16744ae02fcdae406a2a454c9eecd909027ec6aaa88693b2426b8ca29e09 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-FOLD-001-e9c16744ae02fcda.json; empirical validation sha256: 008dd3d2b2df4d2e9ecbfd02b2a36d136ddad12838d58b2f12ce192b57e76eaf;

measurement receipt sha256:4aaf0f8a855fc8a59eccc9c6595d145ee91927b8f2099bacc1d888358099805d;  
isolation sha256:1138479558cb21a2eb4a43b05052fcc870f97b0ef7c9d5443e4b9bc2d53fde8f; custody  
sha256:d3f755fd317e95bc51988f584e7382a1917b4ba74a89728bd7f4b90cd90891c1.

### 39. Protein conformational ensemble

Claim identity: SFT-BIO-PROTEIN-ENSEMBLE-001

**Question and exact theorem.** A protein state is an observed recurrence class over a conformational ensemble; one structure may not erase retained alternate conformations or intrinsically disordered support.

complete-observed-conformation-support\_\_interconversion-recurrence\_\_populations-times-and-conditions-held\_\_method-resolution-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-advers  
e-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-FOLD-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.



| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-observed-conformation-support        | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | interconversion-recurrence                    | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | populations-times-and-conditions-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | method-resolution-bounded                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-observed-conformation-support\_\_interconversion-recurrence\_\_populations-times-and-conditions-held\_\_method-resolution-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** Alternate conformers and intrinsically disordered support are retained. The RCSB 1COP deposit itself contains a solution-NMR ensemble, directly demonstrating why one coordinate file may not erase conformational multiplicity.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:50e74fd0a06a98139212ef64b15743a5ce0c4b2925e81cde9a7331f879434e74.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 5a9589657fdaff9f53896154097b26ac7b0bab533c4c3941c9ce3ccaf6eab932.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 60f5ce45eedff359ee21f153f6b9d22fc66fdfc714289e873cc4908cf883015d.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 01797e8589b6610839fc134223e5fd4d918fb1ab1218bc72fd7c72b946b2ac14.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256 : 7e22954620f0c933d54d04e45b042abef37aaf8663e47d5221a74be97514d535; certificate sha256 : c9dcf52f3e662c90bfe5fed6eb2a5d7a769e66db714db811e0cbfb0097ca7eb6; external-validation sha256 : 79f13942f80c284e854d2c61a1c2bd758a5958f7a3cebf8910a823c371a46f45.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256 : 235bdc4ad83b399a2f7cf741e06c97c1b083d54d9a62702e66fe4af38cdef180. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1** - NCBI Bookshelf, The Cell; <https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html; sha256 : ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.
- **SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1** - Sequence Ontology Consortium; [https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256 : dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.

Comparison record:

- sft-bio-protein-ensemble-001-authority-correspondence: predicted biology:complete-observed-conformation-support\_\_interconversion-recurrence\_\_populations-times-and-conditions-held\_\_method-resolution-bounded; source-derived biology:complete-observed-conformation-support\_\_interconversion-recurrence\_\_populations-times-and-conditions-held\_\_method-resolution-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor

- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:1d189a0031df6b0a60e67bf646116d10bf91d1ebca5fe655775038f131852512; derivation seal sha256:5dfffe0a8e77dd89613a22e45d050cf47709b8de9434bd1820b68c90885ac66bf; engine receipt sha256:47dd4ee2964bd7acfebe732314a5cb600499fa0523fb09ba6cf0a0c70dec2401 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-ENSEMBLE-001-47dd4ee2964bd7ac.json; empirical validation sha256:22e8371702e7d3c8cd57b528585270df9da2274577f186c8c2264d2e17f2575e; measurement receipt sha256:00586a09884af21084016c4aee3c0a8e3e09f6820eff5479eb1a5624bc39645e; isolation sha256:4a9963c49fb77c22d221bbbde2698e1a8cbefd8b8fbe8820d50005ef4fb06676; custody sha256:7f5d7c75b13436d2cfcf424673783ae60d52a6256ddedbf7f6cb5982a94c5d53.

## 40. Protein function

Claim identity: SFT-BIO-PROTEIN-FUNCTION-001

**Question and exact theorem.** Protein function is a reproducible protein-dependent difference in a declared biological process under explicit context and controls, not a label inferred from shape alone.

protein-context-process-carrier\_\_protein-dependent-process-difference\_\_sequence-structure-context-and-controls-held\_\_assay-and-organism-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-ENSEMBLE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | protein-context-process-carrier               | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | protein-dependent-process-difference          | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | sequence-structure-context-and-controls-held  | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | assay-and-organism-bounded                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: protein-context-process-carrier\_\_protein-dependent-process-difference\_\_sequence-structure-context-and-controls-held\_\_assay-and-organism-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the biological law; passed true.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces protein function is a reproducible protein-dependent difference in a declared biological process under explicit context and controls, not a label

inferred from shape alone. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: bd5cc40642345838a9cdd560b559fdea491712bc365e4c102651caa3e79e99cb.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c1181e4abcb3548ee7b001f3d9b875d897a13a55e8b8a182a846bd9ee8c6d3b.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 8c8a43b286e25fcc197f21684a65cc53e64dc1dde72644d1cb4dc64ca949e435.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: cb8ec411df63ae60bc6d5730b0ca6f854c2cf551eb4c4a6808aca09051623c61.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 90cb479c4262b718b48f324f188c04172d265b286057853e318d49ccf4f78dae; certificate  
sha256: cf4ec1513d1d85438bc152904c630437e596c7879d6ded045d4926ecd2c45d80; external-validation  
sha256: 6bdd6976d6c0b31fa3339f15c2be431bd2a53e26f0f3724b7f5033a64a7f7294.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 235bdc4ad83b399a2f7cf741e06c97c1b083d54d9a62702e66fe4af38cdef180. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1** - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html;  
sha256: ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.
- **SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1** - Sequence Ontology Consortium;  
[https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256: dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.

Comparison record:

- sft-bio-protein-function-001-authority-correspondence: predicted biology:protein-context-process-carrier\_\_protein-dependent-process-difference\_\_sequence-structure-context-and-controls-held\_\_assay-and-organism-bounded; source-derived biology:protein-context-process-carrier\_\_protein-dependent-process-difference\_\_sequence-structure-context-and-controls-held\_\_assay-and-organism-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:36255b8898b54d7bde403fa0856a61f80d5e1bc652c0dbdf0b0cf825491221be; derivation seal sha256:d9c4f81336f4767d193738a05ea6ae2d9da8e897453c8105d5131ade35dc6a04; engine receipt sha256:946afec8074bb304b554d02c2b91056966d13d930868d873a389d21ac5435252 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-FUNCTION-001-946afec8074bb304.json; empirical validation sha256:5f029448b1be879e982949865ad1386d5610fe898feb73600c1d7f582a3c15cc; measurement receipt sha256:9ecf349acae6a91cde1930c53e4905d1c2ba0a93cf41fadc17d90dbd98d0ef99; isolation sha256:0395666260e1cccc1bf8aa7181ed08c464f06f9634289da31ae5dd50970a4462; custody sha256:353ce7e37dfdf710808f4aef81829a159855330e302ef1d2e423ef403357af9b.

**41. Protein interaction**

Claim identity: SFT-BIO-PROTEIN-INTERACTION-001

**Question and exact theorem.** A protein interaction is a reproducible specific joint-state relation whose alternatives, stoichiometry, condition and observation method are retained.

two-protein-context-carrier\_\_specific-joint-state-recurrence\_\_alternatives-stoichiometry-and-condition-held\_\_interaction-assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-FUNCTION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | two-protein-context-carrier                   | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | specific-joint-state-recurrence               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | alternatives-stoichiometry-and-condition-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | interaction-assay-bounded                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: two-protein-context-carrier\_\_specific-joint-state-recurrence\_\_alternatives-stoichiometry-and-condition-held\_\_interaction-assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.



Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a protein interaction is a reproducible specific joint-state relation whose alternatives, stoichiometry, condition and observation method are retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 38bda3a4a6cb5a1e06c82bb5ec1bae3e29039b5e3f38e2d158849bb8670ccc3e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 58aa44ba5a941c30a886172612c5dfffb6c9fa0a57ccecd8605355bb8a376e4a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: a9aaf58865fc19b7d32d30163ff85d21cdfd6e343e052f7681be33d4f3f7f9ec.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 2bd06a9a78127a0e89d86d438d3f3d8174171dc696668865f0690cd009b2928c.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 02dbb4916db477d196a18e85613407a08942a3b0a11490385ac7ac8b8cc3d465; certificate  
sha256: 5aec82da6015c25c3c925f7aa42cfde89a53c33bb0dab56e3fbf7d8a024ce642; external-validation  
sha256: bbe82b5df1c32fdc24057da2e3f5cb0e657f8a8f827d51a4768256fc1430a6b2.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: 235bdc4ad83b399a2f7cf741e06c97c1b083d54d9a62702e66fe4af38cdef180. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1 - NCBI Bookshelf, The Cell;  
<https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html;  
sha256: ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.
- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1 - Sequence Ontology Consortium;  
[https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo); snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-sequence-ontology-obo-release-v1.obo; sha256: dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; scope: sequence features, gene, genome, transcript, polypeptide and mutation classes.

Comparison record:

- sft-bio-protein-interaction-001-authority-correspondence: predicted biology:two-protein-context-carrier\_\_specific-joint-state-recurrence\_\_alternatives-stoichiometry-and-condition-held\_\_interaction-assay-bounded; source-derived biology:two-protein-context-carrier\_\_specific-joint-state-recurrence\_\_alternatives-stoichiometry-and-condition-held\_\_interaction-assay-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:7c4f20db7825bf6f39c80c70a9396f9c050eec9b6d8f38cd2e5f5325f3d24d21; derivation seal sha256:9e3c8e6263182265958f9880b45549ecac2489d593fa715d2de2380f56a25186; engine receipt sha256:fd81c6551ea90e53f7976f2426dac503ddee3269368fac9798aa43f4776c5a94 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-INTERACTION-001-fd81c6551ea90e53.json; empirical validation sha256:963c8265608a3f1881f267f79aae3ef04ae1b0fbb4404c1d19b5313d279b85b0; measurement receipt sha256:dd9c066afd656df8ad423a8f8e2fe0bcb0a595ccdf0cb7218116dee3bc921c2; isolation sha256:8f4b1642673795c00d0b2bc7de6fb6942e9dc23c49c193fc519286f969daa968; custody sha256:2e68d6e598ec3ad2d7ee66be7c95a4235a34849e1619ee2b23a7009215e4d7f4.

## 42. Quaternary protein assembly

Claim identity: SFT-BIO-PROTEIN-QUATERNARY-001

**Question and exact theorem.** Quaternary assembly is a generated multi-chain joint state that retains chain identities, stoichiometry, symmetry, interface contacts and adverse steric alternatives; no fitted docking score may select it.

multi-chain-labelled-carrier\_\_interface-compatible-composition\_\_chain-identity-symmetry-and-contact-held\_\_structure-method-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-INTERACTION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|----------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | multi-chain-labelled-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased     | interface-compatible-composition         | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed                 | chain-identity-symmetry-and-contact-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded        | structure-method-bounded                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record       | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing               | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.            |

| Axis       | Eliminated form                    | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-favorable-instance | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule   | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: multi-chain-labelled-carrier\_\_interface-compatible-composition\_\_chain-identity-symmetry-and-contact-held\_\_structure-method-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The primary 1COP deposit confirms a Lambda Cro homodimer and its experimental NMR provenance. The prior 10.266 and 12.539 angstrom docking errors, imported 3.2 angstrom clash floor and Euclidean six-dimensional score are not admitted; the V3 theorem is the exact assembly evidence boundary, not a successful coordinate predictor claim.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: f89d3e22d8b6d40f6eb110bdaf372e93d147255e2cdd51f46c8e753a4168e5bb.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: da2d72f1a4a7ffa12fc7ce6d1ab104ba41daccad2c8380703d2170d85192279d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: af5cd43f86c461dff8bca5528e95bfcd908f8c6d4d39eb76e4fbbc3f3e09ca97.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: f28b35b9aa94efb3aa852ee5db9f41c03cd87eabf80e48ab33f3ae1aab6388b6.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 9b4b3bbc4e59a4b79fa0799a5004119ea40d2c11437ba62522312c2f4f201388; certificate  
sha256: cd88ac228a7ebe07d30bfccc52fa4ebb40e159f83794aa44052ea1d7530a73fe; external-validation  
sha256: 631fa256788de405a5062f3308f1a38fecfe50008c95438c80cfc1abffa21800.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity  
sha256: 590c01f8e58d6baa79fc1a4674da65f48fca9de2f4be65aabc95998f12041595. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

## Sources:

- SFT-BIO-SRC-RCSB-1COP-CIF - Worldwide Protein Data Bank / RCSB PDB; <https://files.rcsb.org/download/1COP.cif>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-rcsb-1cop-cif.cif; sha256:6eddefaa66f6191a3080ca3b0dd333c9e7859448d6aaf17b79b60a3e32ad9a3c; scope: primary Lambda Cro NMR ensemble and homodimer assembly.
- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1 - NCBI Bookshelf, The Cell; <https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-protein-folding-v1.html; sha256:ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; scope: sequence, folding, intermediates, chaperones, function and assembly.

## Comparison record:

- sft-bio-protein-quaternary-001-authority-correspondence: predicted biology:multi-chain-labelled-carrier\_\_interface-compatible-composition\_\_chain-identity-symmetry-and-contact-held\_\_structure-method-bounded; source-derived biology:multi-chain-labelled-carrier\_\_interface-compatible-composition\_\_chain-identity-symmetry-and-contact-held\_\_structure-method-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:674715994d82003ab575b47cd244906798da58f8bcd2ac71396467f8492fd3d; derivation seal sha256:24c5dfc9e444211d0e9376b07a5a1b104491ec2dff945e3af6569080b118c783; engine receipt sha256:3b62cc207a3f30a23e590c94e4b20247a59ee7006c187e74fc94e78d3cf180f0 at receipts/engine/model\_admitted/SFT-BIO-PROTEIN-QUATERNARY-001-3b62cc207a3f30a2.json; empirical validation sha256:87caf1a6241d0bdd9c395e0a11d5c045a5b6ec0c3d1f6ad7f43e4e13d5bc6c7b; measurement receipt sha256:e88a9920fcfe1421bdaf89863c766fe94b65cfc1afa3282724f39e2faa4d4c31; isolation sha256:515df57fc0645b081062ce137de0bc1c2d8a97fa8b33bcb2bd5bd2d3b8025294; custody sha256:d6e9ff3fbdd202447baebc450b57cab7d6185f811baa4c63bb419a8abfb3b44e.

**43. Biological homochirality**

Claim identity: SFT-BIO-BIOLOGICAL-HOMOCHIRALITY-001

**Question and exact theorem.** Biological homochirality is a source-bound recurrence of one molecular hand across a declared biopolymer class; Chemistry supplies chirality while Biology owns its lineage-maintained selection.

chiral-biopolymer-carrier\_\_same-hand-recurrence-along-chain\_\_opposed-hand-control-retained\_\_organism-and-polymer-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT - ROOT - THERE - IS - NO - NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PROTEIN-QUATERNARY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                      | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | chiral-biopolymer-carrier        | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | same-hand-recurrence-along-chain | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | opposed-hand-controls-retained                | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | organism-and-polymer-bounded                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `chiral-biopolymer-carrier__same-hand-recurrence-along-chain__opposed-hand-controls-retained__organism-and-polymer-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The external record supports predominantly L-amino-acid proteins and D-sugar nucleic acids while explicitly preserving that the origin mechanism remains unresolved. Weak-parity language therefore cannot be presented as if it uniquely selected the observed biological hand.

Protein sequence, folding, ensemble, function, interaction and quaternary assembly are different carriers. No opaque predictor, fixed clash cutoff or one favored coordinate model is admitted as a biological law.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b4a5079e3b1ad57fd2160f316f6cd82494be92c3c521d8fd8ca201c777670142`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bc7b2be8bb0789c55d4aca38d6de84a757107f17c2f8662188fd1dba32659787`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:69edcaf0d189d53f190ddc0885aacfcfe5d12aab7c6277ef0c42b6910bd097f1`.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : b039abf8bb54b78a548392358318b396cdc813f7f7a35244a3ac0e49cf294922.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 1c40eda5ea1567bdeb99d4d3ff2fe44a2ae46ec5d6a8b2951c2f597d71b00479; certificate  
sha256 : 884459a4a589b54ce6faed613a7441db2fe27b3974b3dd089547f0a0e8860419; external-validation  
sha256 : 002f42a4d963c18ae8b53e0c86bfc8550904677c7c2631b76e26044dc1003dfb.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 6e784638cd7c31e42c8810377694fb412b772fd5419b592bf3db454c1fd2f410. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-PUBMED-HOMOCHIRALITY-35969306-V1 - US National Library of Medicine / PubMed;  
<https://pubmed.ncbi.nlm.nih.gov/35969306/?format=pubmed>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-pubmed-homochirality-35969306-v1.html;  
sha256 : 8980303d43e13f638eccaf22e24edc141fc51b26037c5e9336facdea4f5c9bca; scope: observed  
L-amino-acid and D-sugar predominance with mechanism unresolved.

Comparison record:

- sft-bio-biological-homochirality-001-authority-correspondence: predicted biology:chiral-biopolymer-carrier\_\_same-hand-recurrence-along-chain\_\_opposed-hand-controls-retained\_\_organism-and-polymer-bounded; source-derived biology:chiral-biopolymer-carrier\_\_same-hand-recurrence-along-chain\_\_opposed-hand-controls-retained\_\_organism-and-polymer-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal  
experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256 : 24c5bc3360da23f70bde918f22d07f1f6fc8d9c7a011c50c841b32950fde2276; derivation seal  
sha256 : 2b94a30c8773db0531d1d22e7d5c9f727bd8498e2cab741f7356c0d16d15e3a1; engine receipt  
sha256 : 1ab0ed8c242bddfe3a54ac4652b01a65cfc8197e5af16c9660b722c7982ead31 at receipts/engine/model\_admitted/SFT-BIO-BIOLOGICAL-HOMOCHIRALITY-001-1ab0ed8c242bddfe.json; empirical

validation sha256:683243b56e49975e5a83732304a503e9c0a28e6c31615365dee9bc9a3e5bb393;  
 measurement receipt sha256:8e8d3e0d7f3ba1657abb70dbc0f16d18481949c96bd62fda51d3052a8fba9c28;  
 isolation sha256:0574dc0b80da84be1824444800ed721c9d274b3c041c2722c2b12fc00179245f; custody  
 sha256:dbf82280091f749156e396fc44e8e30bcb19b91aba3bc6513d1801109a1870c6.

## 17. Cell Process

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

### 44. Cell identity

Claim identity: SFT-BIO-CELL-001

**Question and exact theorem.** A cell is the least compartmented living organization jointly retaining boundary, maintenance, hereditary continuation and regulated process integration.

compartmented-living-process-carrier\_\_integrated-maintenance-inheritance-relatio  
 n\_\_membrane-metabolism-and-lineage-held\_\_cell-cycle-window-declared\_\_complete-st  
 ate-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-succe  
 ssor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIOLOGICAL-HOMOCHIRALITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | compartmented-living-process-carrier          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | integrated-maintenance-inheritance-relation   | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | membrane-metabolism-and-lineage-held          | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | cell-cycle-window-declared                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: compartmented-living-process-carrier\_\_integrated-maintenance-inheritance-relation\_\_membrane-metabolism-and-lineage-held\_\_cell-cycle-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a cell is the least compartmented living organization jointly retaining boundary, maintenance, hereditary continuation and regulated process integration. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:ceb27a28af88a91eee0e9ec19ad0e53ae876c58d21f2a98366dd8536be707112.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ca2094c3c5fce41a6079afca5c8a278d07ecf1c0cb6736244f29a22d6fbb6a64.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:fe3e1a9a0c6d00c38ad299fdbc7f7fadc69521f6a44532fc14693cf2074d91124.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:11f9c60cbc4d531d382d8f70c18351d18dcfdcc492887c3e077f797c5bbabcf2.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:242b20d778b97a0dc5cdf8a9e9ff0f87151bd5e86f8e52cee4957e2eedf6c0f5; certificate  
sha256:c0358730c025900078edbfab6e33a9609c0471688cf5abfcbc92aaa02a8268c; external-validation  
sha256:73ac79a43838807f866f8accfd4ae07ec36238d150df20e71b00c44e3b63456a.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:3db2901281d67477b0d30cf6d6933fff4414efbdfc4ff66a5395d59b4239bce3. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-cell-001-authority-correspondence: predicted biology:compartmented-living-process-carrier\_\_integrated-maintenance-inheritance-relation\_\_membrane-metabolism-and-lineage-held\_\_cell-cycle-window-declared; source-derived biology:compartmented-living-process-carrier\_\_integrated-maintenance-inheritance-relation\_\_membrane-metabolism-and-lineage-held\_\_cell-cycle-window-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold

- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:d35a9c11f46e5ef5a659a1d4a2c2575b10c9b43ae86437199b0a7c2e0151eaaf; derivation seal sha256:1d26c017e0797f3868ce080da520650401599c67904aa4e92e37445634d56b27; engine receipt sha256:3a8ab83a75bac852b533de00acf6851ff4ecb7f6ca6177d5358a3b7b4692194c at receipts/engine/model\_admitted/SFT-BIO-CELL-001-3a8ab83a75bac852.json; empirical validation sha256:586dca1d0421bb7951bac0b4513688d9c130ce5cd4edf1eb1f5586ab099db69b; measurement receipt sha256:261e38df7ee0d9b671854d1cf8c31969a25996a914318cafa06169a5911f8b20; isolation sha256:9ad08661719f9fefaf7c0032743c9d3d432d18a3ad44940ee55efa831d9bf21f; custody sha256:f624e2fc02642edb1af0af33d744cb0c437f03ff7c420c5a27f297aef42f77bd.

## 45. Cellular transport

Claim identity: SFT-BIO-CELL-TRANSPORT-001

**Question and exact theorem.** Cellular transport is a counted species-state transition through a declared membrane route with direction, driving relation and intracellular fate retained.

cell-boundary-species-carrier\_\_directed-cross-boundary-transition\_\_route-gradient-resource-and-fate-held\_\_time-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001

- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CELL-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | cell-boundary-species-carrier                 | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | directed-cross-boundary-transition            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | route-gradient-resource-and-fate-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | time-condition-bounded                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: cell-boundary-species-carrier\_\_directed-cross-boundary-transition\_\_route-gradient-resource-and-fate-held\_\_time-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.

- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces cellular transport is a counted species-state transition through a declared membrane route with direction, driving relation and intracellular fate retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 29af76bcf95ae438f7abbbac7d7feb5bcdd2f4087762b6dc5ce82c3a16b338a.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e7c7dd7494caf5a1d1bb7c814ce71340cf20c83f1ac7d8e2de107cc36d383e53.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 758abd6d3fb80a3638440de1940253a60f2b55082bb260b637b1bbd87097bee1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 1e106efaf5e38ae2b0881ac05ced2967c989de4f14bd2284e6c30718196df7ce.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: b7c75de711bca5ea1dd85abc6f27d0f69bbd360d1d6865d5ae5b28f84c2ef817; certificate  
sha256: 885e5e1f2bba94ae9663690560b387c08e5638eb2a6f25ed1e7177252bf950e0; external-validation  
sha256: 840bddf858419f5a0d9b1026f12fa5df04520bec54f5b0948288ea17a8928e19.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 3db2901281d67477b0d30cf6d6933fff4414efbdfc4ff66a5395d59b4239bce3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- **sft-bio-cell-transport-001-authority-correspondence:** predicted biology:cell-boundary-species-carrier\_\_directed-cross-boundary-transition\_\_route-gradient-resource-and-fate-held\_\_time-condition-bounded; source-derived biology:cell-boundary-species-carrier\_\_directed-cross-boundary-transition\_\_route-gradient-resource-and-fate-held\_\_time-condition-bounded; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology



- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:5a2802b62f10e4b9233a69d67c1020a9c2ca030202d04bfff51131afe8116344; derivation seal sha256:89810c4d649fe8f0379e36c79bc166816acf48a1078ab6b364c810f5ed2a628b; engine receipt sha256:5f041a52c3628067d9741750665bb18a2bb4475055df0992be750af8748b824e at receipts/engine/model\_admitted/SFT-BIO-CELL-TRANSPORT-001-5f041a52c3628067.json; empirical validation sha256:d08217717cc5464ec906fc51d3a61a4aa8385d38fb7147da02f10144719b58b7; measurement receipt sha256:f48458338620d08502d99df07c05a20f79ca5b75a38c0878a96a315f8bccf49e; isolation sha256:884acb9bb99e71ef7b937022e7ed024be68f3a3d4dd09a1f49b3e8edf81f1f46; custody sha256:a78ff09a2487c2818e9a0cd74094c2865a68ddb2e2b52c95af42df64d33ef954.

## 46. Cell signalling

Claim identity: SFT-BIO-CELL-SIGNALLING-001

**Question and exact theorem.** Cell signalling is a reconstructible source-to-receptor-to-response transition chain whose intermediates, controls, timing and alternative routes remain explicit.

source-signal-receptor-response-carrier\_\_source-to-response-causal-chain\_\_intermediates-controls-and-time-held\_\_assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CELL-TRANSPORT-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-signal-receptor-response-carrier       | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | source-to-response-causal-chain               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | intermediates-controls-and-time-held          | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | assay-bounded                                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: source-signal-receptor-response-carrier\_\_source-to-response-causal-chain\_\_intermediates-controls-and-time-held\_\_assay-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces cell signalling is a reconstructible source-to-receptor-to-response transition chain whose intermediates, controls, timing and alternative routes remain explicit. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 666edc55c5c0f64457ef75a2397aa20261a04925f92466b18389e7ad9f7ea548.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b471c77652075a65271c680c3e378a1d9f3c50a2676217dc7777426017f18305.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: b4cc15004061ea98a49eabba2dd081632273b94093e2201810da6407b9190a78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 93af62ffcf10750c2bf25e3cf113041dd2b82263dc49c1cf5e0aae0d63871cc3.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: b6acb2a2b0def33e923f1081094079557341932b52cb905d8e05790cccb0b71f; certificate  
sha256: be0abecccc6b597475e4570b87d2bd07eaabc708198c1b5cc9316123b076e05f9; external-validation  
sha256: 0c94919f85cf4fb1d4ceab0759935858bb29ae555eac385d34da31789df0d657.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 3db2901281d67477b0d30cf6d6933fff4414efbdfc4ff66a5395d59b4239bce3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- **sft-bio-cell-signalling-001-authority-correspondence:** predicted biology:source-signal-receptor-response-carrier\_\_source-to-response-causal-chain\_\_intermediates-controls-and-time-held\_\_assay-bounded; source-derived biology:source-signal-receptor-response-carrier\_\_source-to-response-causal-chain\_\_intermediates-controls-and-time-held\_\_assay-bounded; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction

is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:654bf3a289cf2e72412172dc7ed9b4fc41301499363697acbfc4ca8ff095545f; derivation seal sha256:b04186962baf136159a7da78338ae956498d0621a1a9adec023388b1e01c8ff6; engine receipt sha256:13c962f012563386ecb1b0368ac5c1a155f625035cec94afd26071e414dc5b57 at receipts/engine/model\_admitted/SFT-BIO-CELL-SIGNALLING-001-13c962f012563386.json; empirical validation sha256:ac9bb5b17163103ae1496f36ff727b5c6248386848bafa9ef8396b0fe32a6b13; measurement receipt sha256:3aabc4f49290fb73a0078bbde692321ade237375c25313c172e3179fef8d8c4a; isolation sha256:e25f8c92f8c43bc9aa4aa96e9b42ddaf677ee4ed9a5f1ad52db64c7dc27c893e; custody sha256:8db82744d69a7bac1e49e2270f4c9fc94aa6354ab32468a291b5073cd2b0c6fc.

## 47. Excitable-cell threshold transition

Claim identity: SFT-BIO-EXCITABLE-THRESHOLD-001

**Question and exact theorem.** An excitable-cell event is a regenerative state transition after a declared threshold is crossed, distinguished from subthreshold and refractory responses; no universal voltage is imported.

resting-excitable-state-carrier\_\_threshold-crossing-to-regenerative-transition\_\_subthreshold-and-refractory-controls-held\_\_cell-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-success-or-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CELL-SIGNALLING-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | resting-excitabile-state-carrier              | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | threshold-crossing-to-regenerative-transition | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | subthreshold-and-refractory-controls-held     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | cell-condition-bounded                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: resting-excitabile-state-carrier\_\_threshold-crossing-to-regenerative-transition\_\_subthreshold-and-refractory-controls-held\_\_cell-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The exact law is regenerative transition after a declared threshold, with subthreshold and refractory controls. Half-One is structural balance in the proof language; it is not silently reinterpreted as one universal membrane voltage.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 3ed34b017de8e0290e7481007040e1f222aadf6e6c21ee65d6c1836e1eff4fae.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 33d56d726655e4b5a064958e4ee39d949199159d446a99518a361b6a86363568.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : b95402f0c5147de17df890d0ac18cc3d356066be3c0ac2c72e47509430a30c64.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : bab1c31365d1e7d24d8cadfe874b1aa31ef2c44062e8d7227372aea20a620435.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : f131d0ffa281f89e653b1526419a9e55fed0bfb838e95c86bcce7989169fb7ab; certificate  
sha256 : 30e45fa614cba25cf7c28ca9c5bc50fa5ad6416f0699e915f6137f93eb6f9815; external-validation  
sha256 : 44b79d27d638400cb12b3fa0611c2f2832eb0118fa3ce57b33cc3eaa76c247cb.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : 3db2901281d67477b0d30cf6d6933fff4414efbdfc4ff66a5395d59b4239bce3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256 : a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-excitability-threshold-001-authority-correspondence: predicted biology:resting-excitability-state-carrier\_\_threshold-crossing-to-regenerative-transition\_\_subthreshold-and-refractory-controls-held\_\_cell-condition-bounded; source-derived biology:resting-excitability-state-carrier\_\_threshold-crossing-to-regenerative-transition\_\_subthreshold-and-refractory-controls-held\_\_cell-condition-bounded; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained

- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:f6269da93522f346c7514b9867a6ffc0223c47c4e0d4ae6a3d899f6797c6f540; derivation seal sha256:a1680893041aef5fe15f29dfe86a1eaf4cc0d40031753a127ff7f63d4f8229e1; engine receipt sha256:4282cbeed88f9d120d7ce1489a7c96c4325c1d966e1e56d2d86701c1fcb6065b at receipts/engine/model\_admitted/SFT-BIO-EXCITABLE-THRESHOLD-001-4282cbeed88f9d12.json; empirical validation sha256:91cb5e32ee19a90b469d404c2e1f3b1aac0766d35ff110f353cd76573115bbb2; measurement receipt sha256:a11762bf9edf1fbbdd1a2be377792ca6d3cf820c63b51b2205d15150beea8bd1; isolation sha256:59af72a1e9349e906ba59ad63bd14da27c7f2937e8a8a634aa83e8f81fc416f4; custody sha256:99d5ebd7f39fee5b4b8b251a53115d7f8eadbf11b42ce0fc9ac4d3093d4b46e9.

## 48. Cell division

Claim identity: SFT-BIO-CELL-DIVISION-001

**Question and exact theorem.** Cell division partitions one lineage-bound cell into successor compartments while retaining the complete hereditary, cytoplasmic, boundary and error-allocation record.

one-cell-lineage-carrier\_\_one-predecessor-to-partitioned-successors\_\_genome-cytoplasm-boundary-and-errors-held\_\_division-cycle-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001



- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-EXCITABLE-THRESHOLD-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | one-cell-lineage-carrier                      | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | one-predecessor-to-partitioned-successors     | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | genome-cytoplasm-boundary-and-errors-held     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | division-cycle-bounded                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: one-cell-lineage-carrier\_\_one-predecessor-to-partitioned-successors\_\_genome-cytoplasm-boundary-and-errors-held\_\_division-cycle-bounded

`__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule.`

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier**: the biological carrier is explicit and nonempty; passed `true`.
- **relation**: relation and organization remain separately reconstructible; passed `true`.
- **boundary**: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces cell division partitions one lineage-bound cell into successor compartments while retaining the complete hereditary, cytoplasmic, boundary and error-allocation record. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 97e69a342860b468b05f69f7f37d07769459a11bac1fcdb046f8198773ea4217.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f5f93fa944c1d31af1319167d35c6c65afea5bf2e0c3dd11413c5457a87cc8e1.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 764db4b7bfe5be7a46c80f0ba9bb641d80182ecc2653931d75b9b65f877f818c.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 2fc3f93b6003870ed70838b18f8cd64c529a3de973c8b267cdd477ddb6580192.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256: 4e2cf93fc24f0e2b5529bcf1e4dd9f935cf6d03b4a9c63856edbc49076054547; certificate  
sha256: f54f6954c494cfae800fb4716423175c6b04ab3cee396301e880c045664c1218; external-validation  
sha256: 36d007c661de56a04dfcb42c9d9041ca08a8e2e795137ca92dde569d6c626f6a.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: 3db2901281d67477b0d30cf6d6933fff4414efbdfc4ff66a5395d59b4239bce3. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION** - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
sha256: a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- **sft-bio-cell-division-001-authority-correspondence**: predicted biology:one-cell-lineage-carrier\_\_one-predecessor-to-partitioned-successors\_\_genome-cytoplasm-boundary-and-errors-held\_\_division-cycle-bounded; source-derived biology:one-cell-lineage-carrier\_\_one-predecessor-to-partitioned-successors\_\_genome-cytoplasm-boundary-and-errors-held\_\_division-cycle-bounded

rier\_\_one-predecessor-to-partitioned-successors\_\_genome-cytoplasm-boundary-and-errors-held\_\_division-cycle-bounded; exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:563e9b27c2c4d5d6f41c88098762e561da7e5dd21356c77db59d0f7593886d54; derivation seal sha256:3f792ec3b86e6eb76aa3752939f8a8fc531f1f0a391e288e763958c5ce9cb6b8; engine receipt sha256:812e3a48b5dde43facb3f3b5491fe5b68ed0951ec8961c8ce8893945eaeb5965 at receipts/engine/model\_admitted/SFT-BIO-CELL-DIVISION-001-812e3a48b5dde43f.json; empirical validation sha256:125fd6e88b4b8ef38368e93a692e091c838a2ee206b6eb1825f30b86a2b70fab; measurement receipt sha256:bb1bce14edbbddbd88bc2094869409592e9cd399e057344152126b46dedcc171; isolation sha256:872375fdbbcbf2882ab3ce606ede1e06cd236773fc5819fde7de0fddc6c3d3a82; custody sha256:e17832305d1588cd3a81961848b171384a02c660e054eccccb40b4a5a5fe8a6a.

## 49. Dysregulated cell continuation and cancer boundary

Claim identity: SFT-BIO-DYSREGULATED-DIVISION-001

**Question and exact theorem.** Dysregulated cellular continuation is a lineage-supported failure of normal division, death or differentiation control under a declared tissue context; cycling alone is not sufficient to identify cancer.

lineage-control-state-carrier\_\_division-death-differentiation-control-comparison\_\_mutations-context-and-clonal-history-held\_\_tissue-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CELL-DIVISION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                       | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | lineage-control-state-carrier                     | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | division-death-differentiation-control-comparison | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | mutations-context-and-clonal-history-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | tissue-and-time-bounded                           | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record          | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                        | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure     | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |

| Axis      | Eliminated form                      | Forced form   | Exact elimination / retention basis                                                                                                                                                               |
|-----------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extension | free-fit-exception<br>-or-extra-rule | no-extra-rule | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor and depth independence.** Sole survivor: lineage-control-state-carrier\_\_division-death-differentiation-control-comparison\_\_mutations-context-and-clonal-history-held\_\_tissue-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** Persistent cycling is insufficient to identify cancer. The admitted boundary requires lineage-supported failure of normal division, death or differentiation control in tissue context, with mutations and clonal history retained.

A cell closes boundary, maintenance, inheritance and regulated process integration. Transport, signalling, excitability, division and dysregulation are separate state-transition questions with condition and adverse controls.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: cd4feaa1109c4398d7c35d4f7df25942642e8f81b7ffcea535278948ae16e7ec.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 00d745d59d0af687bea41c7fec865732772a791b4834fa8f1d120619370b672e.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 83eb301331781f3cdb481fd6075b912d33c17d8c9a481467d244b4a1be82c519.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: cd84356377a81c14135859ecc022d797a91f33ad0f7ee617645d18a14b928725.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: e3dd7fe90eb1d1bd76e5f40d2677df8f1ea9c3768ece2b12253b5d739d3915cb; certificate  
sha256: 6bab449b8642ea49956b32bbf589c7df13f0146e5011287aa41d6c6f409aaf4b; external-validation  
sha256: 4b1931bebc10113d40197218fe71c9ac4727ef2354dcefea71c9709734281599.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 36bd8b728f61aad6980a1848938524ab315b3fefe328b7fc2a3607e2b9a851b8. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-AGING-SENESCENCE-V1** - NCBI Bookshelf, Developmental Biology;  
<https://www.ncbi.nlm.nih.gov/books/NBK10041/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-aging-senescence-v1.html;

sha256:920f58581ed5a30448598733f38a1b2eb85682fb9901cf5c1a5606513f6adb4c; scope: aging, lifespan, regeneration and cancer distinction.

- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION - Gene Ontology Consortium;  
<https://geneontology.org/docs/ontology-documentation/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-go-ontology-documentation.html;  
 sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; scope: molecular function; cellular component; biological process and relations.

Comparison record:

- sft-bio-dysregulated-division-001-authority-correspondence: predicted biology:lineage-control-state-carrier\_\_division-death-differentiation-control-comparison\_\_mutations-context-and-clonal-history-held\_\_tissue-and-time-bounded; source-derived biology:lineage-control-state-carrier\_\_division-death-differentiation-control-comparison\_\_mutations-context-and-clonal-history-held\_\_tissue-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:204fa25d824391b0cec9c46397d2bb353e40ebf9317fd5c78d8f592459cccbab; derivation seal sha256:41566c0eb10a28636992d0d600cda861e43c2008bc4a173c31bbd8580d04ec83; engine receipt sha256:cfb86089a23b4c3240ff65c7275c09855138bb729d08c8367f93d5ef8929c2d7 at receipts/engine/model\_admitted/SFT-BIO-DYSREGULATED-DIVISION-001-cfb86089a23b4c32.json; empirical validation sha256:3f34a54ec4d1370f1ff2bb88acbee8871987b1f94300fd1c2aa7edf3260c1d5e; measurement receipt sha256:351ba35fe52bfcd067af8f3453b6cb35243e27c23ce7510fd04656d84485c8b9; isolation sha256:1b25ee9baca56fc9bd74ee0ead084e5b652dff8c8b7014424fd890240b65adc0; custody sha256:1b0a4fa457a56acb5ae8698bbbbc2eb30f3e6df57f99ac7b5e85f6e6860a89b8.

## 18. Multicellular Development

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

### 50. Multicellular organismal composition

Claim identity: SFT-BIO-MULTICELLULAR-COMPOSITION-001

**Question and exact theorem.** Multicellular organization is a connected lineage-related cell composition with retained cellular identities and regulated shared processes.

connected-cell-lineage-carrier\_\_coordinated-cell-composition\_\_cell-identities-boundaries-and-shared-processes-held\_\_organism-stage-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-DYSREGULATED-DIVISION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                    | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | connected-cell-lineage-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | coordinated-cell-composition   | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |



| Axis         | Eliminated form                        | Forced form                                          | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | cell-identities-boundaries-and-shared-processes-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | organism-stage-bounded                               | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record             | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                           | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure        | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                        | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `connected-cell-lineage-carrier__coordinated-cell-composition__cell-identities-boundaries-and-shared-processes-held__organism-stage-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces multicellular organization is a connected lineage-related cell composition with retained cellular identities and regulated shared processes. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4f117588113b792f9251d40269272f1ed04678404f7b9f2af3892acb5b760587`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:61edbd81d38711c12a134607aed43c3f74c25feca1cec67e6439515715f87226`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:f4df4767c52f0ca23a45483f97ea9adff87b0bec143da293a03da05295829d2f.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:b33ae30bfff68a61edbc3a33b8be61cbbb499d4bb88f5d9ce1dbd0d180e79d3c.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:74f349a3106edb68608d87e7cfb3f1ada3cf3026c53003caa9f8b9656bc0df32; certificate  
sha256:9fd67921dca1316de4e64a45909cfbb0112857ef42a43f35b828c94205844e40; external-validation  
sha256:91f5a7d029c75edd7aaaa5fba707f401d9134262c7d7d9a299d8ac2febe52cde.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:d0a3ca396e000724f91dbc8542903cdb1071b459ce6655616f01d9a798f47328. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1 - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html;  
sha256:e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-multicellular-composition-001-authority-correspondence: predicted biology:connected-cell-lineage-carrier\_\_coordinated-cell-composition\_\_cell-identities-boundaries-and-shared-processes-held\_\_organism-stage-bounded; source-derived biology:connected-cell-lineage-carrier\_\_coordinated-cell-composition\_\_cell-identities-boundaries-and-shared-processes-held\_\_organism-stage-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:e5cb3474b2e7c7fb7e4c89d1b1d18ac6bc68b837112e7f2223c5e53f7a4fbb5f; derivation seal sha256:b033cdb37ca6092182cf087990767e089e4130d1eee02799ed181b1d355ed22b; engine receipt sha256:afb5f7805ecda6f297068f7532eb5683361d0dfd5b46f1d2175e8587a5bab3f4 at receipts/engine/model\_admitted/SFT-BIO-MULTICELLULAR-COMPOSITION-001-afb5f7805ecda6f2.json; empirical validation sha256:a1592714fa2cd81f5a2954ce8e1c9cca220e8ebffe17f86efea8ac753c556195; measurement receipt sha256:98ef590d724775539b1aee45bec79fde93a4a7dfba023cb5c985f363d63f04ab; isolation sha256:be3d12e952c910eb8945db289b56558f40320c06fd0eb31ddacf476542ab9d22; custody sha256:90386e67600312b7b319eb04f6d71d0244f742650508f5458976494d032d7e08.

**51. Cell adhesion and tissue boundary**

Claim identity: SFT-BIO-CELL-ADHESION-001

**Question and exact theorem.** Cell adhesion is a specific persistent intercellular or cell-matrix relation that retains component identity, contact geometry, mechanical context and adverse controls.

cell-interface-carrier\_\_specific-persistent-contact-relation\_\_adhesion-component  
s-mechanics-and-controls-held\_\_tissue-condition-bounded\_\_complete-state-transiti  
on-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adv  
erse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-MULTICELLULAR-COMPOSITION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                     | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | cell-interface-carrier                          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | specific-persistent-contact-relation            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | adhesion-components-mechanics-and-controls-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | tissue-condition-bounded                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record        | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                      | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure   | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                   | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: cell-interface-carrier\_\_specific-persistent-contact-relation\_\_adhesion-components-mechanics-and-controls-held\_\_tissue-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- carrier: the biological carrier is explicit and nonempty; passed true.
- relation: relation and organization remain separately reconstructible; passed true.
- boundary: observation boundary cannot silently become the biological law; passed true.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces cell adhesion is a specific persistent intercellular or cell-matrix relation that retains component identity, contact geometry, mechanical context and adverse controls. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 2855976b60b15be75e9a62f11acd2075ad0ea8be319ca98f9c0e2a2cde4c62cd.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : bb9e9014c7b2f9641249a20bc391ba1d7114bc1173f8f3bd4339d4afb8e4b6ee.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : ca8bac676b72c8588a0782c05fa76825b92c54a033e2db5fa6fcadc49005a741.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 5732d7b2a932974a64646c9b11e6f5092e3f01192730d2e4bbbf740b29052777.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 36832c3410218a1e24d3ac65a3770e7c6893309c12147c5be2732acfd09e881; certificate  
sha256 : 9cd96a5b24d5e5d2cd554f68952d70190bc7440e8d54abdcee27c508a59f6715; external-validation  
sha256 : f1692d06c560966f1b97e06c17e3b07cb12eabdb9a5bc39cf0c581d21d03326e.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : d0a3ca396e000724f91dbcb8542903cdb1071b459ce6655616f01d9a798f47328. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html;  
sha256 : e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.
- **SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010** - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256 : 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- **sft-bio-cell-adhesion-001-authority-correspondence**: predicted biology:cell-interface-carrier\_\_specific-persistent-contact-relation\_\_adhesion-components-mechanics-and-controls-held\_\_tissue-condition-bounded; source-derived biology:cell-interface-carrier\_\_specific-persistent-contact-relation\_\_adhesion-components-mechanics-and-controls-held\_\_tissue-condition-bounded; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:202a39231af1c2cc2aa354d247b40a06246a7ce1b208e15ba692425c850b7d1d; derivation seal sha256:b191105b775953e1a34ac440b1be6737ba71fffb0e9d1f8d05a166a2baf01dfe1; engine receipt sha256:0c854c871fd66f0b15e416e1da98cf127e8e562c49d095c2da11396e178f2e8c at receipts/engine/model\_admitted/SFT-BIO-CELL-ADHESION-001-0c854c871fd66f0b.json; empirical validation sha256:99c48d50189885d075523f0bfb12d87e915f684f101380bee84a81a136264b0d; measurement receipt sha256:31fee17535a2a5d2dfa6d3264bd0c78779fdedb058c01eb40a8ee7f8fa1f7190; isolation sha256:9e8d737a4f528fd2d81069f924a550b9998a76e123e3b29e56ce11ac8300dbe9; custody sha256:1c5efc3babcb6fa936284193cd2331eba82a7544bde667f60bf4016d2837c1b.

## 52. Cell differentiation

Claim identity: SFT-BIO-DIFFERENTIATION-001

**Question and exact theorem.** Differentiation is a stable lineage-related divergence of cellular state and function while hereditary carrier identity and transition provenance remain reconstructible.

common-lineage-cell-state-carrier\_\_stable-state-divergence\_\_genome-identity-expression-and-function-held\_\_developmental-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-CELL-ADHESION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | common-lineage-cell-state-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | stable-state-divergence                       | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | genome-identity-expression-and-function-held  | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | developmental-window-declared                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: common-lineage-cell-state-carrier\_\_stable-state-divergence\_\_genome-identity-expression-and-function-held\_\_developmental-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.



**Operational witnesses.**

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces differentiation is a stable lineage-related divergence of cellular state and function while hereditary carrier identity and transition provenance remain reconstructible. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : d2775142c176462f862a8d6580ef8cc284e1251ae05847354395cd6e2d19f42b.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 73c76927b5143c3e5b22b987784cb3061ea5fb136d6c738e6aa9ff6350ed6828.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 38bd419b8de69cb57357b4a84841c7aa620c4668959927705a1876b89d2c2743.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : a1b80f19e27e0e90e470963416888a2261eac2b30a553f2f902a6d8c51fa5492.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : a7a6d3f978f76ea47a9edf59113a87721a6dea5e0b80d56b6a57b09bd0408231; certificate  
sha256 : 671919364a72d25d35d2c1e3cae9891922bacf9828a4393da19ff4b2dbdb88b4; external-validation  
sha256 : 5db39f7c9c3b0e924eb48a94f5ae9319f4bd45886a1cdaccb691a291265abcc6.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : d0a3ca396e000724f91dbc8542903cdb1071b459ce6655616f01d9a798f47328. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html;  
sha256 : e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.
- **SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010** - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256 : 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- **sft-bio-differentiation-001-authority-correspondence:** predicted biology:common-lineage-cell-state-carrier\_\_stable-state-divergence\_\_genome-identity-expression-and-function-held\_\_developmental-window-declared; source-derived biology:common-lineage-cell-state-carrier\_\_stable-state-divergence\_\_genome-identity-expression-and-function-held\_\_developmental-window-declared; exact match **True**

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:659670bc22195474e0a2b78ce8ba386bc9f12a96516cc796f601acd2f89ab02d; derivation seal sha256:3590cdd6d46e7dc9f3a059adf66098ab0677131026416d8d576fb1d6a6c6cc3a; engine receipt sha256:10fc2b713e757614a0b216a741cb8b11c7760e871f099e903e45d2eee24840ad at receipts/engine/model\_admitted/SFT-BIO-DIFFERENTIATION-001-10fc2b713e757614.json; empirical validation sha256:58be8452f59dc878ad023e24827d421d27eb894bc42030a298b1b1f5205f5ac6; measurement receipt sha256:42dfaddff58327c908def6be8d1136ea14f987f19da7a5fc5236e15f882dd8d5; isolation sha256:e0ba597471b91ede3350c4abc10c125c9511ff76a352e115901b3b0a27437d68; custody sha256:e7ceb6ac91f46aff539df5303befef941a268252bada761814beed05c9e50e1a.

## 53. Developmental lineage

Claim identity: SFT-BIO-DEVELOPMENT-001

**Question and exact theorem.** Development is an ordered lineage of division, movement and differentiation transitions from an initiating cell to a multicellular organization with every branch and stage boundary retained.

zygote-to-organism-lineage-carrier\_\_ordered-division-differentiation-transition\_\_branching-time-position-and-state-held\_\_stage-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-DIFFERENTIATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | zygote-to-organism-lineage-carrier            | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | ordered-division-differentiation-transition   | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | branching-time-position-and-state-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | stage-boundary-declared                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: zygote-to-organism-lineage-carrier\_\_ordered-division-differentiation-transition\_\_branching-time-position-and-state-held\_\_stage-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces development is an ordered lineage of division, movement and differentiation transitions from an initiating cell to a multicellular organization with every branch and stage boundary retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 9b1475e25e47a4b752f7d0d0627f86b4f64680770c49aa5ee6e44cddfe38c703.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 17204abd74484449f9f577d4f8b8a84d59cc22c651945859cc257d4b4e56ab16.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 068e6f57a31e6d161bed6a0747cd412dce61cbe6879cf13c2a198b68f8a165b4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: e1565ae47a9b9c4b6910841bb222b6aa19e783cf7320d8683204b3f13d80feec.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 4df308ed05b41972bf1b9392329572636a942c932497181b936dc20167b8a07d; certificate  
sha256: cd8a3c3451b741406e41b036a7a6251f00b2981da6513c4f1d566c1129186969; external-validation  
sha256: 30197f4dad20b3bed5c93a073b7a9bdf4daef3baad29868c804ca272d016d1cb.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: d0a3ca396e000724f91dbc8542903cdb1071b459ce6655616f01d9a798f47328. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

#### Sources:

- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html;  
sha256: e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-development-001-authority-correspondence: predicted biology:zygote-to-organism-lineage-carrier\_\_ordered-division-differentiation-transition\_\_branching-time-position-and-state-held\_\_stage-boundary-declared; source-derived biology:zygote-to-organism-lineage-carrier\_\_ordered-division-differentiation-transition\_\_branching-time-position-and-state-held\_\_stage-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:5edef799cc9461bf7a26ff4aded46710046e516bc546d0a9500665eec09690ab; derivation seal sha256:ce895e177ec0b4a052ae54b2cb1a571424cf15cbf3c50c4e3f0000b575f8ac60; engine receipt sha256:66f84782d6c8a70c2ec7a9e98676aba00e65565588b01c40bd2374bcb01a8c1f at receipts/engine/model\_admitted/SFT-BIO-DEVELOPMENT-001-66f84782d6c8a70c.json; empirical validation sha256:8f12d02f61928d99d1e2faebfea203b3c0a6f02de12cd062f7f241d6cbdf1032; measurement receipt sha256:c9b88df507c7ccd4febc651631e97e1c9198e1fa35d31f1d8ab8c73db01e57a6; isolation sha256:0e342aff4306d79c385f3bf3305f80f80788a0cfb98560f6b1ea6b6b35a29804; custody sha256:fa7ec18a4dfa96a7982f0bc0803aa65779dce577b79ee2799eca2c994e93805.

## 54. Morphogenesis

Claim identity: SFT-BIO-MORPHOGENESIS-001

**Question and exact theorem.** Morphogenesis is the reconstruction of global biological form from local cell-state, adjacency, movement and boundary transitions over a declared developmental interval.

cell-state-spatial-carrier\_\_local-transition-to-global-form\_\_neighbour-boundary-force-and-time-held\_\_scale-and-stage-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-DEVELOPMENT-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                            | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | cell-state-spatial-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | local-transition-to-global-form        | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | neighbour-boundary-force-and-time-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | scale-and-stage-bounded                | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: cell-state-spatial-carrier\_\_local-transition-to-global-form\_\_neighbour-boundary-force-and-time-held\_\_scale-and-stage-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces morphogenesis is the reconstruction of global biological form from local cell-state, adjacency, movement and boundary transitions over a declared developmental interval. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : c03566c4e01268caac77627bee806b60a9177dbb623668c113848f91d07a7ab1.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 1aa2df5d4852b6f95b4ca16d508df69cac2ddec852689754ecc2ce0f181b7bda.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 305135d95c50e4c2fca049782209474e93db5564e661bc8b8c9780993b6490dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 87a8816f9d4d980995e6b4e9f3f1affcce80d614becdcd804163d70baf57ea6e.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation



sha256: ca4867335237309beb2eb47abf391f641edb77d5525832e43652d95d6bd7c362; certificate  
 sha256: 39e3b46ca7292450526f989b5ba69343325b74803a4bc9a793b32e0ec4c079c0; external-validation  
 sha256: ad39ca73bf9dca65c72338d83177aebbb0c4052b0f8252521a57be679bb824a9.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: d0a3ca396e000724f91dbc8542903cdb1071b459ce6655616f01d9a798f47328. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1 - NCBI Bookshelf, Genomes;  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html;  
 sha256: e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
 sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-morphogenesis-001-authority-correspondence: predicted biology:cell-state-spatial-carrier\_\_local-transition-to-global-form\_\_neighbour-boundary-force-and-time-held\_\_scale-and-stage-bounded; source-derived biology:cell-state-spatial-carrier\_\_local-transition-to-global-form\_\_neighbour-boundary-force-and-time-held\_\_scale-and-stage-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
 sha256: d2d01d5e148ae9e2b93b3c59d50fba97a1a7a5edb3515f51a8096152248d9a67; derivation seal  
 sha256: 6699560c3af3dc86c3912c42acc8d6e23ad7aa3bf58fe098c8ecd6cfd61777ee; engine receipt  
 sha256: 39da10edb31c18b4b00f4a69a4806aa5c46aa6543c1bb6d7e50d5c423e12f529 at  
 receipts/engine/model\_admitted/SFT-BIO-MORPHOGENESIS-001-39da10edb31c18b4.json; empirical  
 validation sha256: 38873723eccaf0c7535d264ca95edf0b630d5bffa33fa6bd6c4ae438afad4131;

measurement receipt sha256:586b622216ba40e951467dcc73146720603373fe84a500113dbc1137b78eb05a;  
isolation sha256:44572aa786435b6cb131e6a019207ce4efe9c27dcfebd60833a5dcc68bf20fcc; custody  
sha256:981ffb69cbaf90fdf25837df38feb3b02e9863e6638e00f2fb1b2ebce081bd0a.

## 55. Biological regeneration

Claim identity: SFT-BIO-REGENERATION-001

**Question and exact theorem.** Regeneration is a lineage-supported transition from a recorded loss state to restored organization and function, with residual source, path, extent and failures retained.

damaged-organization-lineage-carrier\_\_loss-to-restored-organization-transition\_\_  
residual-source-path-and-function-held\_\_extent-and-time-bounded\_\_complete-state-  
transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor  
-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-MORPHOGENESIS-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | damaged-organization-lineage-carrier          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | loss-to-restored-organization-transition      | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | residual-source-path-and-function-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | extent-and-time-bounded                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `damaged-organization-lineage-carrier__loss-to-restored-organization-transition__residual-source-path-and-function-held__extent-and-time-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces regeneration is a lineage-supported transition from a recorded loss state to restored organization and function, with residual source, path, extent and failures retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Multicellularity is reconstructed from lineage-related cells, interfaces, differentiation, ordered development, spatial morphogenesis and regeneration without erasing cellular identity or developmental failures.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:50777e373f3ec1a540f95be3a9dc0a7a712e1ca5eb9cf56db20d95e38ed5b205.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ec8def0c9b69b63c074b931dbd109c07862f1288ec6a438def43c1ce295124b1.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a0725ad7564f2ee682c7771946de7615ed825afad372297f1ec11bbe96cac5a0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:a2aa87ff4e94999e263612965d4c71d0f62d4896723535a5725af1a80f7ff58c.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:5fd54d67a1f9caad5f94ad247a62cf1c087d2cb3b87a428c125f0f88404b6308; certificate

sha256:e75fc10a5a56b02081d62dbc34c16a8087ea15938c1b787798273703e260416e; external-validation

sha256:aa925b8db5b56b672fdb8ccea5eb5d2a1787abc7c9ffaa8a2984e0e1e2fbd978.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:7fe714af7329c963d446b8e65a0e375fee78b544792107c7d9b56c51a31f95d1. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-AGING-SENESCENCE-V1** - NCBI Bookshelf, Developmental Biology; <https://www.ncbi.nlm.nih.gov/books/NBK10041/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-aging-senescence-v1.html; sha256:920f58581ed5a30448598733f38a1b2eb85682fb9901cf5c1a5606513f6adb4c; scope: aging, lifespan, regeneration and cancer distinction.
- **SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1** - NCBI Bookshelf, Genomes; <https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-genome-regulation-development-v1.html; sha256:e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; scope: genome expression, signalling, differentiation and development.

Comparison record:

- sft-bio-regeneration-001-authority-correspondence: predicted biology:damaged-organization-lineage-carrier\_\_loss-to-restored-organization-transition\_\_residual-source-path-and-function-held\_\_extent-and-time-bounded; source-derived biology:damaged-organization-lineage-carrier\_\_loss-to-restored-organization-transition\_\_residual-source-path-and-function-held\_\_extent-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage

- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:a315a8f8f2728646e2a7b9bad4a761cff3ab2cf7c4f311cc9cec577d4e2bc7ff; derivation seal sha256:5ab0d3912aa4e5329877748f030d5181db38dd5d1fa517994910b98ecbd7b798; engine receipt sha256:1181aa8c81e16a17dca6a612b879a6866d759cea7dc0eaaee0a189475b20cf1bb at receipts/engine/model\_admitted/SFT-BIO-REGENERATION-001-1181aa8c81e16a17.json; empirical validation sha256:2c5ef27ff341c5bf6d0b5544cc1becc4b8d09faea91205442ae04792435fd6; measurement receipt sha256:8769cbfb84a558c19e867236ec5ffa9ba27538f919e3f5b212b36ab0061fa1aa; isolation sha256:2cfcdf03c783a5db522a8d22c9c85878bb08f34e6dfed18bf519059dbef13bc5; custody sha256:447ab47932e8ebe44db4d2d0bdbb7169f1b38ecfcc9098d5b2fcbb76c287847.

## 19. Organism Physiology

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

### 56. Organism identity

Claim identity: SFT-BIO-ORGANISM-001

**Question and exact theorem.** An organism is an integrated lineage-bound living whole whose parts participate in coordinated maintenance, development, reproduction or continuation within a declared life cycle.

integrated-living-lineage-carrier\_\_part-to-whole-functional-integration\_\_boundaries-life-cycle-and-environment-held\_\_taxon-stage-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-REGENERATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | integrated-living-lineage-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | part-to-whole-functional-integration          | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | boundaries-life-cycle-and-environment-held    | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | taxon-stage-condition-bounded                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: integrated-living-lineage-carrier\_\_part-to-whole-functional-integration\_\_boundaries-life-cycle-and-environment-held\_\_taxon-stage-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.



**Operational witnesses.**

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces an organism is an integrated lineage-bound living whole whose parts participate in coordinated maintenance, development, reproduction or continuation within a declared life cycle. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: ecbeb3741b6391217c24f2a597bbf9b1885b713292df61e1b8da15a136f84239.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 93b2eec23ee89b44df26bcdc26ba84ba384dcc589fa41847cf629a50debe34bb.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 2ad02db54c88d60ab2e9ebc53c9ef04ea066cb888d98d743f88318b11b1548f5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: b285d0ff173d9ac9a2daae201b12522d1322a5dce1acb2649b62b7ad277d776f.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 16e9534daac4d1e9cf3d134dfc859bcb44f1b4cff307edc731ab788c2944eb63; certificate  
sha256: 78001288444a1a4d936fb0a559322a5c679e2b8bb7ca6fb7963d71e91192100d; external-validation  
sha256: b2ddd44c07b411be570f3007513e6c5bc731d348238c3172e6b9baf7d67933e4.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 85414cad516009b74adbc771c07c4274eedf202cb6d54e1d0713c062d827394a. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1** - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-homeostasis-v1.html;  
sha256: 687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caee4c7c24; scope: physiology  
and homeostatic control loops.
- **SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010** - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA  
inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- **sft-bio-organism-001-authority-correspondence:** predicted biology:integrated-living-lineage-carrier\_\_part-to-whole-functional-integration\_\_boundaries-life-cycle-and-environment-held\_\_taxon-stage-condition-bounded; source-derived biology:integrated-living-lineage-carrier\_\_part-to-whole-functional-integration\_\_boundaries-life-cycle-and-environment-held\_\_taxon-stage-condition-bounded; exact match **True**



- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:13c164965c3ba39ac17bac459315f846ce3261ca9c19396f0b965cad31abe1e2; derivation seal sha256:efdc573b18b22ff5caea77dbd74e3ee4d9864b8b210a2d8c6b797cb48bd6d116; engine receipt sha256:5be8b1f41d5fabdd92a7319dbec75c1f71eefd1bb6a75781b354926d6132a234 at receipts/engine/model\_admitted/SFT-BIO-ORGANISM-001-5be8b1f41d5fabdd.json; empirical validation sha256:f82ef8e4495a161395d6e4f9d9f5ddbdf7cdebed85608bec043d4da452e0edb6; measurement receipt sha256:2d6b444ab55e749844f6c1b69e942598086b21ec04447032a80d878156882916; isolation sha256:c0b0c4ae7342f225e3a2a47ece8b1692458fea205e63ab146a96566392cb4900; custody sha256:5ead68a3001158c062076b573ae9e7e087ac065452cdf0194fa170f984512445.

## 57. Physiological process

Claim identity: SFT-BIO-PHYSIOLOGY-001

**Question and exact theorem.** Physiology is the complete coordinated transition record by which organismal parts maintain or change a biological function under declared conditions.

```
organ-system-state-carrier__coordinated-function-transition__parts-flows-control
s-and-time-held__organism-condition-bounded__complete-state-transition-control-r
ecord__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure
__no-extra-rule
```

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ORGANISM-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | organ-system-state-carrier                    | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | coordinated-function-transition               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | parts-flows-controls-and-time-held            | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | organism-condition-bound                      | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: organ-system-state-carrier\_\_coordinated-function-transition\_\_parts-flows-controls-and-time-held\_\_organism-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces physiology is the complete coordinated transition record by which organismal parts maintain or change a biological function under declared conditions. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 5fbbf29308ce2e219c66264e36d76249ca393b9a5500ec6ad7238347e6497c853.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 414caa8b7f6fd6d759d892305a2f435e3014f55e1e1aa031a6f73a25a800061a.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 4791dd5ee2dccfa23e743060336874992f0461551bb58b9db2ba8624cb1d721e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 7d8a2a885d7915c368fc0d97835fc1cabf29dfb3a66d2bfff0357887f474b9e03.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 07c28796fd96b52793366293db8fd3af03178e3d95067e3341ceaba12e4a3f04; certificate  
sha256: 743c30943bc6114633c57da5041844b5725d71d0890d83ee2ca0fc48efee0edc; external-validation  
sha256: 3cc8131f003be478f627b9d88444ed9f270e2fba6278204692b8a61b86c43cd7.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 85414cad516009b74adbc771c07c4274eedf202cb6d54e1d0713c062d827394a. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-homeostasis-v1.html;  
sha256: 687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caae4c7c24; scope: physiology  
and homeostatic control loops.

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-physiology-001-authority-correspondence: predicted biology:organ-system-state-carrier\_\_coordinated-function-transition\_\_parts-flows-controls-and-time-held\_\_organism-condition-bounded; source-derived biology:organ-system-state-carrier\_\_coordinated-function-transition\_\_parts-flows-controls-and-time-held\_\_organism-condition-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:349d30bb6886aa11de8767a6c2ff59b9759c77cebbe6535a2586f3a4bc0ddd67; derivation seal sha256:273a710b8a98b604c2ec046725a41d2888ad1ed2840dd5355cb5055086f445f6; engine receipt sha256:efb61f71851c4c5a4af6e38baca1bb5a0524983ae7990d06a44899b1269dd0c6 at receipts/engine/model\_admitted/SFT-BIO-PHYSIOLOGY-001-efb61f71851c4c5a.json; empirical validation sha256:6a6d6f1702560f46bea6606b2188aaa591966021a07edb230a6d29fe6d113534; measurement receipt sha256:8322f4fc21688f95954f37217ca3fe8f4bc3e832a6d827ffe3f8202a2dc66948; isolation sha256:02bce3ade8a855bde448b26d215773f6e828e4f9ee5dbf8dbc863cd30a8c1ff2; custody sha256:cd466cd8720169a47097e1d40b49cd438f6af34d7187807ea884a6eba0942597.

## 58. Homeostatic regulation

Claim identity: SFT-BIO-HOMEOSTASIS-001

**Question and exact theorem.** Homeostasis is a closed sensing-and-response recurrence that returns a biological variable to a declared viable range while preserving delays, overshoot and failure cases.

regulated-variable-control-carrier\_\_departure-sensing-response-return-relation\_\_set-range-delay-and-failure-held\_\_condition-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-PHYSIOLOGY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                                | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | regulated-variable-control-carrier         | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | departure-sensing-response-return-relation | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | set-range-delay-and-failure-held           | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | condition-and-time-bound                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: regulated-variable-control-carrier\_\_departure-sensing-response-return-relation\_\_set-range-delay-and-failure-held\_\_condition-and-time-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces homeostasis is a closed sensing-and-response recurrence that returns a biological variable to a declared viable range while preserving delays, overshoot and failure cases. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 752fa457b92e1e688d0430731906cce0d7fc0bb44d042bd0bfa9285a092ef1aa.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 25d84d06ef76a488dda68ba6432ca2894284d1e5c7fb9cddb958122cb56df63f.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 155f7b236f908d9c258a78c45c9cb01da271bd8833ac541427a44e00a3d5a0c4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 4a1de3a2c7ce7638154792679483c5601d07bbc3cbf89445af0c610b17f5aa86.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:ccb7c848fbe2c8d81974b31f4bdbd08a5efbf568a1a650108c3a099e33b41fba; certificate  
 sha256:9ed537568c060257869bcf50da3cc7d52725e058e9441ec9b4f7a81f198e4a01; external-validation  
 sha256:53dfa753e6b186aad5cef32b6024bc2117220782d48f99c38b4ae42f44565219.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:1a7a2cfc17e76e95c77511047ba8d6f0e5f182cc28b157d37599608ff00d1db2. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>; snapshot  
 experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-homeostasis-v1.html;  
 sha256:687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caae4c7c24; scope: physiology  
 and homeostatic control loops.

Comparison record:

- sft-bio-homeostasis-001-authority-correspondence: predicted biology:regulated-variable-control-carrier\_\_departure-sensing-response-return-relation\_\_set-range-delay-and-failure-held\_\_condition-and-time-bounded; source-derived biology:regulated-variable-control-carrier\_\_departure-sensing-response-return-relation\_\_set-range-delay-and-failure-held\_\_condition-and-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
 sha256:4f4881256e53c3a1ea94b7ccae301dfe557c7d9c31495aa960528d2d15f001e1; derivation seal  
 sha256:9aff1051ef2d145a32bb717af46f64fc8f92d462a17066a7789f048b0bcd8c16; engine receipt  
 sha256:6d125e5d47a334da623d6b6859f89cfa85852efa2364a81b5875766c67bf151 at  
 receipts/engine/model\_admitted/SFT-BIO-HOMEOSTASIS-001-6d125e5d47a334da.json; empirical  
 validation sha256:8244aa2958300b9425b873064eed66c728c0c1e0c027481146f09fe185c182c2;  
 measurement receipt sha256:bf81a82ba2968c33f013b7d2b6bf900ec60e901ce4f7fe4e8e46717fcdcb0a1d;  
 isolation sha256:7237e9ad674045d7d79a9e081ee684b19ca6380a8bbd182e7c4da0861d1dde9a; custody  
 sha256:71f18c4ccace9a8a0a5f479a623a9e9a096110f6436c184309f0f7ccef2ca79.

## 59. Organismal reproduction



Claim identity: SFT-BIO-REPRODUCTION-001

**Question and exact theorem.** Organismal reproduction generates a new lineage-bound organism through declared parental carrier contributions and development, with variation and failures retained.

parent-offspring-lineage-carrier\_\_generation-of-new-organism\_\_carrier-contributi  
ons-development-and-errors-held\_\_life-cycle-bounded\_\_complete-state-transition-c  
ontrol-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse  
-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-HOMEOSTASIS-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                      | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | parent-offspring-lineage-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | generation-of-new-organism       | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                       | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | carrier-contributions-development-and-errors-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | life-cycle-bounded                                | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record          | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                        | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure     | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                     | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: parent-offspring-lineage-carrier\_\_generation-of-new-organism\_\_carrier-contributions-development-and-errors-held\_\_life-cycle-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces organismal reproduction generates a new lineage-bound organism through declared parental carrier contributions and development, with variation and failures retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: e424d42f8acd78017466eae18bbe3da39222f0dab1bbb3903f003bc385e61337.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b0aca8ebcf084c8d4c0c33469ae72abc3a46311ad548516e2cdd519f1ca852b9.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:4fda5ff33bd7e759fa846253e699577b461601706f0240fbe5803315e0b14bce.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:34521b812445d208f80b5e45e8d7754a1ceade3e21311e4943822bb3ca3950fa.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:f5bd9418aca698aee45d6a4885a3985869eb13d03c7cc1164ac35d08c185c638; certificate

sha256:fbaf755b431f42f1ad51f5ab53aa3ef7b7a47f8740f63c819c457bfde4bec7862; external-validation

sha256:2b17945d61108bea9400106918c96f1c5d47a16dc225f18a55f900badf0d1f7f.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:85414cad516009b74adbc771c07c4274eedf202cb6d54e1d0713c062d827394a. Target opened after

derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-homeostasis-v1.html;  
sha256:687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caaee4c7c24; scope: physiology  
and homeostatic control loops.
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA  
inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-reproduction-001-authority-correspondence: predicted biology:parent-offspring-lineage-carrier\_\_generation-of-new-organism\_\_carrier-contributions-development-and-errors-held\_\_life-cycle-bounded; source-derived biology:parent-offspring-lineage-carrier\_\_generation-of-new-organism\_\_carrier-contributions-development-and-errors-held\_\_life-cycle-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:1054d32b84c565c536d0ee146ed0f162153bb184caf8b3d27763d3370159f972; derivation seal sha256:be2a8aee4317148a702308fa561cd4ea091c7bbb3e7bf0f141a1d3ac514b3261; engine receipt sha256:b391636734ce9f838d52feafefbfefeeed4e809e4f70c14abb172373e77ace862 at receipts/engine/model\_admitted/SFT-BIO-REPRODUCTION-001-b391636734ce9f83.json; empirical validation sha256:ecac68d3ed98ab1f83778288cd01ff7646e1b453b230a738c4ff991db21c5fbe; measurement receipt sha256:e0eec80ed010a2710439c5e0f836aa122a8110c50bd8d33db349103673cc4099; isolation sha256:c00b29da052c3a56145d3b3a380dbd25c7bd2ffd5ffdc442852824f2a6f03367; custody sha256:e77dc61c1ecb1aa7712c34c890440684a3704ac3423c0f4621c4bc015ac178f0.

**60. Biological life cycle**

Claim identity: SFT-BIO-LIFE-CYCLE-001

**Question and exact theorem.** A biological life cycle is the complete ordered recurrence of organismal stages including development, reproductive continuation and terminal loss at a declared taxonomic boundary.

stage-transition-carrier\_\_ordered-recurrent-stage-relation\_\_development-reproduction-and-death-held\_\_taxon-environment-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-REPRODUCTION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | stage-transition-carrier                      | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | ordered-recurrent-stage-relation              | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | development-reproduction-and-death-held       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | taxon-environment-bounded                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: stage-transition-carrier\_\_ordered-recurrent-stage-relation\_\_development-reproduction-and-death-held\_\_taxon-environment-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological life cycle is the complete ordered recurrence of organismal stages including development, reproductive continuation and terminal loss at a declared taxonomic boundary. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 7c72ca2aad4761ccea52df5729639b961225484db7eae67a49abf72b4a7581d4.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9ae94907b87d6e84451c0fcacf305bd9e873ede5757918836e09291ab1db5514.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 15414716b45ca11c1ca556739c2a2cc685a957d08b2439a84e75e862d5695a96.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: f134e9a1f36f9cf6142b97800bcb432a9acd359af5384732d76759996cfef12.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 094ae023ead8f3ea9c2726a282e00af1665a724ba4fe17f6f2aba9f3dbbf82bf; certificate  
sha256: 1ec7a4f6ee7acd562f48538f0cafbb99f46499e1ef8eeff44387b9252b0353489; external-validation  
sha256: 99fc62d41c26e6ffa549bc7b8ca8ce0fb331e5e312f9d1ae834f796a7a244433.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 85414cad516009b74adbc771c07c4274eedf202cb6d54e1d0713c062d827394a. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1 - NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-homeostasis-v1.html;  
sha256: 687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caae4c7c24; scope: physiology  
and homeostatic control loops.
- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 - US National Academies / NCBI Bookshelf;  
<https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-nasem-unity-biology-2010.html;  
sha256: 62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; scope: DNA  
inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.

Comparison record:

- sft-bio-life-cycle-001-authority-correspondence: predicted biology:stage-transition-carrier\_\_ordered-recurrent-stage-relation\_\_development-reproduction-and-death-held\_\_taxon-environment-bounded; source-derived biology:stage-transition-carrier\_\_ordered-recurrent-stage-relation\_\_development-reproduction-and-death-held\_\_taxon-environment-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:67a4e86b3ed951530efd79d96e84ba9bb8d48a8129dfa9f7521a2bb6e90fbff9; derivation seal sha256:526d27175bd3b92c4064f5bae7170362d6231926bb27a407c0f61ee83ae98de9; engine receipt sha256:796de364657e92a20c6bc7717a9a9beb334122f2aa559261e9b5492651762c87 at receipts/engine/model\_admitted/SFT-BIO-LIFE-CYCLE-001-796de364657e92a2.json; empirical validation sha256:49018d6efe54b6aea56350fc28a7dd7b1f0897531d59ae382bd9c06532f45fbf; measurement receipt sha256:78f8d7dd4fe9e8dbf34d96de7d621234cb6fa82da61f2b7867ab01b2edc44957; isolation sha256:6883888192e4c59b06e7a811f44239e73312bd646a9d9ac17024f3b7401de123; custody sha256:5a02bf5eb05149e9b1190632a60f179326de002388bcd6f82da4b4ae03389078.

## 61. Senescence and longevity

Claim identity: SFT-BIO-SENESCENCE-001

**Question and exact theorem.** Senescence is an age-linked change in maintenance or continuation measured on a declared lineage and population; longevity requires survival and censoring records and is not forced by denominator parity alone.

lineage-maintenance-history-carrier\_\_age-linked-maintenance-loss-relation\_\_survival-repair-reproduction-and-censoring-held\_\_population-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001



- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-LIFE-CYCLE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                     | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | lineage-maintenance-history-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | age-linked-maintenance-loss-relation            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | survival-repair-reproduction-and-censoring-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | population-condition-bounded                    | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record        | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                      | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure   | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                   | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: lineage-maintenance-history-carrier\_\_age-linked-maintenance-loss-relation\_\_survival-repair-reproduction-and-censoring-held\_\_population-condition-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** Ageing is not denominator parity. Survival, repair, reproduction, population, condition and censoring records are required; the law distinguishes senescence from cancer and from one cellular Hayflick observation.

Organismal identity composes living parts across a life cycle. Physiology, homeostatic control, reproduction and senescence keep environment, stage, population, survival and censoring boundaries explicit.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 05766741ecdfc8240c79043c47975ec952c42cd8cac271334a3924c575e3635b.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : b89ec47eb1ab5e86a17003c1ac76e660c0cc44e64f7feb1aaa9a56299147ddb1.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 51eb080ffd91634322c4b5bd0b22883032e3dfac398e611ca45b8e63c0fdda66.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 2bba67aae451b61957ecd49659a8a7e7f8a840ab51b8703c7c0374a6ca66a243.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : 04a6878bd24eb3b184c6a172e041811a617796ae9f53e035df35a36cec24ec1d; certificate  
sha256 : cee890f01fcb82c4b60869968361239a3a23c7279e596d3f04bf5b88296ac313; external-validation  
sha256 : d566570e517435d3feaff775e99aa3a33ad49947b9b29102bc886f4c6eaa5ae2.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : d513a6769b190c94d10fc8577e6b7aabcacf69bcd7baeafc7258a3f5c17f223a. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT - BIO - SRC - NCBI - AGING - SENESCENCE - V1** - NCBI Bookshelf, Developmental Biology;  
<https://www.ncbi.nlm.nih.gov/books/NBK10041/?report=printable>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-ncbi-aging-senescence-v1.html;  
sha256 : 920f58581ed5a30448598733f38a1b2eb85682fb9901cf5c1a5606513f6adb4c; scope: aging, lifespan, regeneration and cancer distinction.

Comparison record:

- **sft-bio-senescence-001-authority-correspondence:** predicted biology:lineage-maintenance-history-carrier\_\_age-linked-maintenance-loss-relation\_\_survival-repair-reproduction-and-censoring-held\_\_population-condition-bounded; source-derived biology:lineage-maintenance-history-carrier\_\_age-linked-maintenance-loss-relation\_\_survival-repair-reproduction-and-censoring-held\_\_population-condition-bounded; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal  
experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256:3184ec75401e962e259ff2c0dfb2430ae7c374b349ccfcb751027dd5c1ed3aa6; derivation seal  
sha256:c7de98a2227fc95cd824d5f1e5bfff2c8caf344d6fd4e6ce1cd5ce5463a5e783e; engine receipt  
sha256:b24bee7ff30267308f7f6aee985f931e54b45b545774914a628bd79f4801acbc at  
receipts/engine/model\_admitted/SFT-BIO-SENESCENCE-001-b24bee7ff3026730.json; empirical  
validation sha256:d62b0e1879707202b3116e33463e90e9d1f1dd78b6e5702ad28adf0d619ccd75;  
measurement receipt sha256:d92e5b8fa07be2de887626d6279abdf81220d8b6c2d4cf807932a37e0ea174c0;  
isolation sha256:c9c8e8f630ed03b27aae167667a4f21455e17639a41cc4bc62532527d54aa3ca; custody  
sha256:2972ad2c5bfa52d7bd634eecbdaad9055dc0496a57b09a79e2f67af0bd0d08e2.

20. Population Ecology

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

62. Biological population

Claim identity: SFT-BIO-POPULATION-001

**Question and exact theorem.** A population is a declared set of organismal individuals connected by place, time and relevant biological interaction, with membership and sampling rules explicit.

individual-set-space-time-carrier\_membership-and-interaction-relation\_identities-lineages-and-sampling-held\_taxon-place-time-bounded\_complete-state-transition-control-record\_root-bound-forward-forcing\_positive-finite-successor-and-adverse-closure\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-SENESCENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | individual-set-space-time-carrier             | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | membership-and-interaction-relation           | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | identities-lineages-and-sampling-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | taxon-place-time-bounded                      | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: individual-set-space-time-carrier\_membership-and-interaction-relation\_identities-lineages-and-sampling-held\_taxon-place-time-bounded

`d__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule.`

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a population is a declared set of organismal individuals connected by place, time and relevant biological interaction, with membership and sampling rules explicit. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 4451a848379ebef63ec97c2daef83f7a0b9c64f42fa0b8f4ceed696bd36876db.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7bd47375b27a6045b94797c2e9ffc7f15e3ec3e2b5aef3a6c2848bbf42eba707.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 077eb310477b8c9bc29c54f6c14f96af3149ee56cd3a2c9241dd58b2bfb8a892.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 085b42aa1947c769f992316ef975cfcab135af58ce04e00f5d09e85390316285.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256 : b580d736812e87d0b23fda9938854aa6ced587cd3c91022f75908bf719c723d5; certificate  
sha256 : ecfdcfcad61dad9abb150fdabe481adca40ee582e7a91b37a779d04f34fbf6f07; external-validation  
sha256 : 131894257055aeacae781c7eed8402fb80b61c7df4053dbb34e98c1cf482ace.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256 : 70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshot  
s/sft-bio-src-gbif-darwin-core-archive.html;  
sha256 : 89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon,  
occurrence and sampling-event data structures.
- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 - Global Biodiversity Information Facility;  
<https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html;

sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-population-001-authority-correspondence: predicted biology:individual-set-space-time-carrier\_\_membership-and-interaction-relation\_\_identities-lineages-and-sampling-held\_\_taxon-place-time-bounded; source-derived biology:individual-set-space-time-carrier\_\_membership-and-interaction-relation\_\_identities-lineages-and-sampling-held\_\_taxon-place-time-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:6862f95ea74a8e76f3e1e75ac2be7184d713a61f1a0b2503355b906f51fe936b; derivation seal sha256:d602e2c4e1ac1bef70cd2543c8d488e08578942c7c4aa8585d19e8445251f526; engine receipt sha256:d92a8470023c0f5ae5f93f0fb88655af2986bd5bf5d09e164adb53124d11a52d at receipts/engine/model\_admitted/SFT-BIO-POPULATION-001-d92a8470023c0f5a.json; empirical validation sha256:b85182c14245679d90ef7661c785ca56fbb6fc02ede2818f2982597fae277ca8; measurement receipt sha256:850b4f77eca7e7b9b664fea6fb7c403cc02ab7a415aa234dadfca3c07cbe4599; isolation sha256:cbfc70a1e9ae49ea0a68069c334209820aa8b9f3abc76d02d0a64b3ec2614a8a; custody sha256:70e146649932394fad5bb4643019a20fb5bf3051530073c28435363082218571.

## 63. Species and reproductive boundary

Claim identity: SFT-BIO-SPECIES-BOUNDARY-001

**Question and exact theorem.** A species claim is an explicitly chosen, evidence-bound equivalence class over populations; reproductive, phylogenetic and ecological criteria may not be silently merged.

population-lineage-carrier\_\_within-between-continuation-comparison\_\_gene-flow-history-and-method-held\_\_species-concept-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-POPULATION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                              | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | population-lineage-carrier               | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | within-between-continuation-comparison   | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | gene-flow-history-and-method-held        | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | species-concept-declared                 | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |
| record       | result-without-transition-record   | complete-state-transition-control-record | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.    |



| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: population-lineage-carrier\_\_within-between-continuation-comparison\_\_gene-flow-history-and-method-held\_\_species-concept-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a species claim is an explicitly chosen, evidence-bound equivalence class over populations; reproductive, phylogenetic and ecological criteria may not be silently merged. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:c329dd9096e8cd31248433efa65ac56beb52870665627445bec1d2dd612bd57e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ede378074543aa548a719db790635a7d49406d490aaebf1a74ca043fc74f8277.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:33ecb3a2d75849b19a74a7e03f2153802d689a5d813607e0a4b262bb8d359d47.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:dbad48d4d1dddac29994e13820480284419f4ca693b9ed25d5932af340241d98.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:2ca65d546b14a70159f9249625a9bdde98895926ea77521e893d53899c5c5e1e; certificate  
sha256:0156ba8c4e255875fa7e21005b87e540b8885fd88e12279a2e2aba7dd12bb4d6; external-validation  
sha256:cffdbd3822560644f4829a4fb08f1fbcc72100e5bd25b79947493100cd06311b.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshot/s/sft-bio-src-gbif-darwin-core-archive.html;  
sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.
- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 - Global Biodiversity Information Facility;  
<https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html;  
sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-species-boundary-001-authority-correspondence: predicted biology:population-lineage-carrier\_\_within-between-continuation-comparison\_\_gene-flow-history-and-method-held\_\_species-concept-declared; source-derived biology:population-lineage-carrier\_\_within-between-continuation-comparison\_\_gene-flow-history-and-method-held\_\_species-concept-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:17a48e947bade89b624a9e797f09989d554df2f16d4b92bad88febc8014b7142; derivation seal sha256:513ebbd80de2c3d3aa2134974ae9db71d5865c52cc0096ba087c3bf61035878e; engine receipt sha256:5594e8750cc5de11e2d96a74f5cc72a126e81b7e66efeffc35088866a24f35a9 at receipts/engine/model\_admitted/SFT-BIO-SPECIES-BOUNDARY-001-5594e8750cc5de11.json; empirical validation sha256:27cba82aa46c4d571a8681176a1574955b76c0ff52b703418e32ac16591d7499; measurement receipt sha256:677c504e36ddb79551f56b91589e87e8e2d94bf5f72c5f91702ad1462e217854; isolation sha256:fcf5cfde6863387fd7148999316bab89e2e3f581de360f3684d666e353c8f335; custody sha256:5abc164a225afa449f306bbdf1a3c3e99a73bc4861dd72decc3e34c1925c245f.

## 64. Ecological community

Claim identity: SFT-BIO-COMMUNITY-001

**Question and exact theorem.** A community is a place- and time-bounded network of distinguishable populations and their observed interactions, including absent and unresolved links.

co-located-population-carrier\_\_interpopulation-interaction-network\_\_species-identities-and-observation-effort-held\_\_place-time-scale-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-SPECIES-BOUNDARY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis    | Eliminated form               | Forced form                   | Exact elimination / retention basis                                                                                                              |
|---------|-------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | co-located-population-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained. |

| Axis         | Eliminated form                        | Forced form                                    | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | interpopulation-interaction-network            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | species-identities-and-observation-effort-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | place-time-scale-bounded                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record       | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                     | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure  | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                  | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `co-located-population-carrier__interpopulation-interaction-network__species-identities-and-observation-effort-held__place-time-scale-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a community is a place- and time-bounded network of distinguishable populations and their observed interactions, including absent and unresolved links. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256 : f0d4d796a40d001288b6c3e702439b404174468bfa032bbdc5b7c1b92f92d5f`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256 : c0d5757dc63dcc187e8c97ae183200f62df17a0c2c7dd3d60ad2904215d6bbbed`.

- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: eba0f8db3b33d992037a642f71f22db8929f2e515e8b38a324bdc8ade7326884.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 2efe8f2062c6f6b5ea8087d3c0a2efd835e5b62e9d1577783df704ffac0ad4db.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 6831462293c165f4a54d5ac19b857160db5f549bfb96dbd3ad652ddb0cdd0e63; certificate  
sha256: ee3b4a998cddbdcbb5aec8b19350e39d74907464005f69b6874b7a9aa273e03; external-validation  
sha256: 228468af1ffbafe9e3e5767d7da91001c5171776f880becc301d73d304df53c78.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256: 70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshot  
s/sft-bio-src-gbif-darwin-core-archive.html;  
sha256: 89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon,  
occurrence and sampling-event data structures.
- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 - Global Biodiversity Information Facility;  
<https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html;  
sha256: 58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological  
survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-community-001-authority-correspondence: predicted biology:co-located-population-carrier\_\_interpopulation-interaction-network\_\_species-identities-and-observation-effort-held\_\_place-time-scale-bounded; source-derived biology:co-located-population-carrier\_\_interpopulation-interaction-network\_\_species-identities-and-observation-effort-held\_\_place-time-scale-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition

- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:bad644c140ae64abc04b6375c82e4e6f8f4cb8337d6be6dc034db2d642449653; derivation seal sha256:5eda91184c2c797d07f5304f1508c50f86c59e891ec8e5988c4db3c031a2da35; engine receipt sha256:ce593cef58bc0880814d28ed4efccb0bfedd325d83511767a36b752c8bf68bbe at receipts/engine/model\_admitted/SFT-BIO-COMMUNITY-001-ce593cef58bc0880.json; empirical validation sha256:45a90ef79864a04a95b91f0436dcb61a697b336ecb4429da7ea992582634df4f; measurement receipt sha256:391a8fcb9147f271e644d569246425741461e81e3f9b46fe59dc5943be765698; isolation sha256:0c130295c31770a43dedada6ca98ae5b7d819ac44ae546a6f05fec9d9196fb9f6; custody sha256:168a2b11a53a4ffe9edd6ba4dfb72bc929d25b6f8dc0a1a88807975f4ac17fd3.

## 65. Trophic and resource network

Claim identity: SFT-BIO-TROPHIC-NETWORK-001

**Question and exact theorem.** A trophic network is a directed resource-transfer graph retaining source, consumer, amount basis, direction, loss and observation boundary for every edge.

population-resource-directed-network\_\_consumption-and-transfer-edge\_\_amount-direction-loss-and-provenance-held\_\_ecosystem-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-COMMUNITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | population-resource-directed-network          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | consumption-and-transfer-edge                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | amount-direction-loss-and-provenance-held     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | ecosystem-window-declared                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: population-resource-directed-network\_\_consumption-and-transfer-edge\_\_amount-direction-loss-and-provenance-held\_\_ecosystem-window-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a trophic network is a directed resource-transfer graph retaining source, consumer, amount basis, direction, loss and observation boundary for every edge. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never



promoted into universal constants without a separately generated and externally tested law.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:035048bc126fdea554f728b0967d07f600a91ee41bfaeb32dd79b89fa7276e9d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b8b1a3b4da4f641c8d604ef608d740d8820b076004c3d606cada6cfd5c35b0ee.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:c4586dbb655ad991ddabe26c7d042dcf4fa3532dca6dcc1d85a0a1977a06bb18.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:c15b7400d83b0960d2193c7c3bf6ca305a4cc31213f969b471b83268e594aa29.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:d56cdb6c3408d4f9a937fb3b949cba5bf6e3be45c6ffc25e666b0e814f44da61; certificate  
sha256:d3369cf4ddc173341d0507ec3ac3635cd37f5f445f687e0142fe13ebb2412b87; external-validation  
sha256:c13a30298a052352c5f78ce6c70ba3932f17e6bf230340859f74ec4301e18487.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE** - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-darwin-core-archive.html;  
sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.
- **SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1** - Global Biodiversity Information Facility;  
<https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html;  
sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-trophic-network-001-authority-correspondence: predicted biology:population-resource-directed-network\_\_consumption-and-transfer-edge\_\_amount-direction-loss-and-provenance-held\_\_ecosystem-window-declared; source-derived biology:population-resource-directed-network\_\_consumption-and-transfer-edge\_\_amount-direction-loss-and-provenance-held\_\_ecosystem-window-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:1c85c481b2d065023efdf30f93e29113d782535c780ee7b61f02fb8a6c0cd048; derivation seal sha256:47d29f3f90f4b5e3864ef6dbd14a1bed08816fbc1f37f6b4ba82f3c7e05c2006; engine receipt sha256:0bca610fd19ea03b8711a6622ecef413729d9e7cf890e336184baaea08396a0 at receipts/engine/model\_admitted/SFT-BIO-TROPHIC-NETWORK-001-0bca610fd19ea03b.json; empirical validation sha256:785755d3e0fb90bf91a22ead792a280ca349185e755ba1a57e148022965c69a2; measurement receipt sha256:72a472871b91c1d50feb7273633a552271d910aece0af4eea08795e2d41d6f4; isolation sha256:56dbc2452563d3e887d6e913a8995a980ab138ab3552e98abfbd7d90c45a7cf6; custody sha256:1099c9590901dc77f324ec3aac491dcb6dd067b64434f94db07f5afafbfdf6a1a.

## 66. Ecosystem organization

Claim identity: SFT-BIO-ECOSYSTEM-001

**Question and exact theorem.** An ecosystem is a bounded joint organization of communities and abiotic supports with complete registered stocks, transfers, losses and observation history.

community-environment-joint-carrier\_\_biotic-abiotic-transfer-network\_\_stocks-flows-boundaries-and-history-held\_\_spatial-temporal-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-TROPHIC-NETWORK-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | community-environment-joint-carrier           | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | biotic-abiotic-transfer-network               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | stocks-flows-boundaries-and-history-held      | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | spatial-temporal-boundary                     | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: community-environment-joint-carrier\_\_biotic-abiotic-transfer-network\_\_stocks-flows-boundaries-and-history-held\_\_spatial-temporal-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces an ecosystem is a bounded joint organization of communities and abiotic supports with complete registered stocks, transfers, losses and observation history. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:035f91987da03ae0597255cd0134c916db8636420418f375669ac3ebac364c2b.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6a1aa75d9f4cead9e891cd07185bf17e834735dcf75d827fd3cb45d0051da07a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:c4c5490830dc845f3ae781e10a5026da47059d9153fac971f0ebe934301ff4d5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:d972e376f9955abe0e750c8c642a497687ad1b37ea40bdaa4aa0a07363328d54.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:92096e2b1842297bd2d328502d1584e129c2e6a8f3c3875eb6bb70bd82974905; certificate  
sha256:9f7381ebaf2003739540212ca1097b91d92f5c88fc0cda31f62025c83a0a1c79; external-validation  
sha256:5229c9848122c7cb78a3658d5a1b2ca87ba72266f0e56baddf243aeaeb1360ff.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshot  
s/sft-bio-src-gbif-darwin-core-archive.html;  
sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon,  
occurrence and sampling-event data structures.
- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 - Global Biodiversity Information Facility;  
<https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot  
experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html;  
sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological  
survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-ecosystem-001-authority-correspondence: predicted biology:community-environment-joint-carrier\_\_biotic-abiotic-transfer-network\_\_stocks-flows-boundaries-and-history-held\_\_spatial-temporal-boundary; source-derived biology:community-environment-joint-carrier\_\_biotic-abiotic-transfer-network\_\_stocks-flows-boundaries-and-history-held\_\_spatial-temporal-boundary;

exact match True

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:ba49d6faec0b0474347b6ba77af8e7f92f85ca26d80512e3a134481c330d4d52; derivation seal sha256:0cc265cf12f6ccdc1f4512f1b765eca75721b86e0eb1215ea4772ff113447508; engine receipt sha256:7f75dd42da58bc7246f8bd78837d2b6562a655a5b6be15622c0c06e1826ef006 at receipts/engine/model\_admitted/SFT-BIO-ECOSYSTEM-001-7f75dd42da58bc72.json; empirical validation sha256:d550ec68b9b7f38c7475adf49fd641b54c34aab43a03f0f9c95d67cde067b9de; measurement receipt sha256:63145dd729dd1fe5afc95a553aff893a81d73885faa79a65ddef9d5b03108c33; isolation sha256:cf4e9a062cb438756430c58692772cfeefae9d08014e06a319ac126e4be3ea8; custody sha256:585bd8f10dc417847c0928d8e9c6db040c2b5820ef8ba3ada945cd3e2e55bdc0.

## 67. Ecological recurrence and stability

Claim identity: SFT-BIO-ECOLOGICAL-RECURRENCE-001

**Question and exact theorem.** Ecological recurrence is a return of a complete declared state class; stability is persistence or recovery under registered perturbations and cannot be inferred from one bounded-denominator orbit alone.

time-indexed-ecosystem-state-carrier\_\_state-return-and-perturbation-relation\_\_species-resources-disturbance-and-effort-held\_\_finite-series-boundary\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ECOSYSTEM-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | time-indexed-ecosystem-state-carrier          | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | state-return-and-perturbation-relation        | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | species-resources-disturbance-and-effort-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | finite-series-boundary                        | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |



| Axis      | Eliminated form                      | Forced form   | Exact elimination / retention basis                                                                                                                                                               |
|-----------|--------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extension | free-fit-exception<br>-or-extra-rule | no-extra-rule | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor and depth independence.** Sole survivor: `time-indexed-ecosystem-state-carrier__state-return-and-perturbation-relation__species-resources-disturbance-and-effort-held__finite-series-boundary__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** A bounded periodic orbit alone is not ecosystem stability. Species, resources, perturbations and observation effort remain in the state; stability means persistence or recovery under registered perturbation at a declared boundary.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256:1e89bba08d5e0b48c52159eb5a7f13e571d4c800a39c740d3fea8abd7a8a6cd6`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:18e68b73880f76d5dc9e248f06effc9104e2951065d85eac3639f43281f816a`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
`sha256:5889f86530f44787e55d3cae165bfd0114adb9cbc0e6509ed541b5560eb8d247`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
`sha256:9e7cf064c2194e42c4d751afb9680c32ab22ac7664342296a19c35d73601ac81`.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
`sha256:e417726ab1d7584afe1304a47d82fe0410096a3e7498087f3ea725cec1fb876a`; certificate  
`sha256:57f650b0078821cc4dbc8112996f66a33ec0ccca03101db46b2e430ae6903409`; external-validation  
`sha256:4853973f3ba00d02ae6afbcdd61e85c1f6664ecce2cc6dc5a04b6ae62801e46`.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
`sha256:70ac6481691357c34731332b0e30bcee903ab6bf1fbf3fbc3127e81feaa7c92d`. Target opened after derivation seal: `true`. All rows preserved: `true`. Exact comparison: `true`. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility;  
<https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-darwin-core-archive.html;



sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.

- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 - Global Biodiversity Information Facility; <https://docs.gbif.org/guide-publishing-survey-data/en/>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-survey-guide-v1.html; sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; scope: ecological survey protocol, effort, occurrence, absence and monitoring.

Comparison record:

- sft-bio-ecological-recurrence-001-authority-correspondence: predicted biology:time-indexed-ecosystem-state-carrier\_\_state-return-and-perturbation-relation\_\_species-resources-disturbance-and-effort-held\_\_finite-series-boundary; source-derived biology:time-indexed-ecosystem-state-carrier\_\_state-return-and-perturbation-relation\_\_species-resources-disturbance-and-effort-held\_\_finite-series-boundary; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:4e4145c27d07ff2be7b0f162351e0c7bcb38bc6b8c6b96c18ae6ffd24c686555; derivation seal sha256:f720deefa2637ba006436c815a69c95f2185ec05d19a192e517ad1e879d7ac62; engine receipt sha256:b670971a03f2f2d088debfbd791529419052470c1f86eb22ba94c009f44461b4 at receipts/engine/model\_admitted/SFT-BIO-ECOLOGICAL-RECURRENCE-001-b670971a03f2f2d0.json; empirical validation sha256:4219b5f443bf62164b274740adb2f298e1bfa23953de31cbfe514876717a9239; measurement receipt sha256:8f45bbb2cae7e170e02d1ca6c3b13c1b847ae065d966e192c4ab9aea7565d8f5; isolation sha256:c250db6700fb37e4d8b7a12fc87572788b3146f065956cc5c0040334acc42df0; custody sha256:4f84b41f99000126c52248ebbdce17d268a2abe4f5c40c499fd0a1e7de66440d.

## 68. Ecological network topology

Claim identity: SFT-BIO-ECOLOGICAL-NETWORK-001

**Question and exact theorem.** Ecological network topology is the exact degree, path and clustering census of a declared interaction graph; scale-free or small-world labels require whole-graph and unfavorable-model comparison.

complete-population-interaction-graph\_\_degree-path-and-clustering-census\_\_nodes-edges-effort-and-missing-links-held\_\_sampling-scale-declared\_\_complete-state-tra

nsition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ECOLOGICAL-RECURRENCE-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                               | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | complete-population-interaction-graph     | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | degree-path-and-clustering-census         | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | nodes-edges-effort-and-missing-links-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | sampling-scale-declared                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-population-interaction-graph\_\_degree-path-and-clustering-census\_\_nodes-edges-effort-and-missing-links-held\_\_sampling-scale-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The external severe test reports that strongly scale-free networks are rare and that log-normal alternatives often fit as well or better. The admitted law is the complete graph and unfavorable-model comparison protocol, not the overclaim that biological networks are universally scale-free or small-world.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 97f52dd252514c914e85f9e49d2336f5b61e2600a7da68bd72c2e15b08fa311d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 0c1a909e152d807bcab387cd1805147ecc3d936eeb804f9d3f5e60d7a94f2336.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 62864b1f39ff575c4d3bd64ffb6580729d70f45901678d243ab5225301cf4886.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: b3423929e482c5a75257ba84b96d929d4435c23f52f34c16ca6b07c79207515d.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: fdc68b6bd34b7323176b72c15b09315fd754b9604d2b0ea94429987e9dc401c3; certificate

sha256:31e0be3bb11b21f4ce62f2e1faca7d4cfb05d9e9fca65d62315c75e69c07e52c; external-validation sha256:d457dd1650a3556631d7ca2ce3e9f2437e80f351ba22d5ade01398de3d749610.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:30535c78690aaf9cc0088ad4346ef4397a9d371f0e1ce670871d99fc89867893. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-EUROPEPMC-SCALE-FREE-RARE-6399239-XML-V1 - Europe PMC;  
<https://www.ebi.ac.uk/europepmc/webservices/rest/PMC6399239/fullTextXML>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-europepmc-scale-free-rare-6399239-xml-v1.xml; sha256:3896db77ef007b8da965fb21a7f87d63790222cb566889ec89d978392ded21a8; scope: severe empirical scale-free test and alternative distributions.

Comparison record:

- sft-bio-ecological-network-001-authority-correspondence: predicted biology:complete-population-interaction-graph\_\_degree-path-and-clustering-census\_\_nodes-edges-effort-and-missing-links-held\_\_sampling-scale-declared; source-derived biology:complete-population-interaction-graph\_\_degree-path-and-clustering-census\_\_nodes-edges-effort-and-missing-links-held\_\_sampling-scale-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:0de24592b30ddddd00f26319b59649ca36669b3714d53b59f52cb0c0f9cf66ac2; derivation seal sha256:2da7060e422f059cc78c14179647023ef8b343daf84aae0243a1ea484dac978d; engine receipt sha256:f0671939caa9ed6d2fe121453e60df8452e681f31d5d37d871452b620522773b at receipts/engine/model\_admitted/SFT-BIO-ECOLOGICAL-NETWORK-001-f0671939caa9ed6d.json; empirical validation sha256:77f683f52a2cd0c9d6f9b3a5744b698ccfb92c8b4f69381fe4c637e0a61f52e8; measurement receipt sha256:2653519e069949e1c0e6cf1e32e304ac82b42e751f3b71e31269a61b17374cf0; isolation sha256:4f077d6b9d556fe131cf3ad7bca94086d4557ecba837bb3f0e2f12ca9dea4fc2; custody sha256:be5482b8139f865c02c89c9f721ec82eedc0588178a31b08f8aa6856930e6122.

## 69. Biological allometric relation

Claim identity: SFT-BIO-BIOLOGICAL-ALLOMETRY-001

**Question and exact theorem.** A biological allometric claim is a population-bounded comparison of trait and body-size recurrence with taxa, stages, conditions, residuals and alternative exponents retained; no universal exponent is imported.

organism-trait-size-carrier\_\_trait-to-size-scaling-comparison\_\_taxon-stage-condition-and-residuals-held\_\_range-and-sample-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-ECOLOGICAL-NETWORK-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                      | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | organism-trait-size-carrier      | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | trait-to-size-scaling-comparison | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | taxon-stage-condition-and-residuals-held      | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | range-and-sample-bounded                      | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `organism-trait-size-carrier__trait-to-size-scaling-comparison__taxon-stage-condition-and-residuals-held__range-and-sample-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- `carrier`: the biological carrier is explicit and nonempty; passed `true`.
- `relation`: relation and organization remain separately reconstructible; passed `true`.
- `boundary`: observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The source record retains 3/4, 2/3, unity, size-range dependence and evidence against one universal exponent. The Fold theorem requires a source-, taxon-, range- and condition-bounded allometric comparison; it does not fit or select a preferred exponent.

Populations, species criteria, communities, trophic graphs and ecosystems are bounded data structures. Network and allometric universality claims receive explicit hostile alternatives rather than preferred-law status.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:784677be38d812ca8cc4cc92af8ce33fe5868278cbe699e3ecdb495b5b957179`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:72440f1e0a1c6e3fa0aac4c0cfae6289db5dbd463b5701c6e22f317252a29af3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4ffc02d7b0dfdfa82b62990c0373b299cf3350d83f801fc3f85ac8c6eae69962f`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 6174e624458ca246ad8d07c4fd3e9a5e0549da806e72c5d2c76eb0841be9da61.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256: 1cd8e26f30b6b2f6cc655bd2c34e87e917c03d29b193298fc18571111230b28c; certificate  
sha256: 7e8e920e30a536e3be88933bbf1a3e079a89a6a1cd5c178c8ac61a2f306448b7; external-validation  
sha256: 26e8ac7c35fe89a1bd0deff2a83f693300cf7340fa0968831e03b6f9dab1f3d3.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 3 purpose-matched discriminators with ordered identity  
sha256: e1b9cbafa9ce6fb2ed3b90399162b5958138b06f492ec88eae91aa21e944f169. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-EUROPEPMC-ALLOMETRY-11782306-XML-V1 - Europe PMC;  
<https://www.ebi.ac.uk/europepmc/webservices/rest/PMC11782306/fullTextXML>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-europepmc-allometry-11782306-xml-v1.xml;  
sha256: 214f06e309b9505e2ec0852d9b30cb02b6c4863c7e17f16d16ceb17378462e99; scope: three-quarter, two-thirds, unity and nonuniversal allometric evidence.

Comparison record:

- sft-bio-biological-allometry-001-authority-correspondence: predicted biology:organism-trait-size-carrier\_\_trait-to-size-scaling-comparison\_\_taxon-stage-condition-and-residuals-held\_\_range-and-sample-bounded; source-derived biology:organism-trait-size-carrier\_\_trait-to-size-scaling-comparison\_\_taxon-stage-condition-and-residuals-held\_\_range-and-sample-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal  
experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256: 1261eabf83adaf1d23b950d64745d830e9f9dcc9c7d97b6055387e91dd905701; derivation seal  
sha256: 41cd2a111514f99e287eb7fa53c828f302cb1b08d60f3ae928f8b4e7057fcdeb; engine receipt  
sha256: d5ce1f4e35ed88884352ee720bbd29ca4dfdc2513e90bf70cdf210f2e723c596 at  
receipts/engine/model\_admitted/SFT-BIO-BIOLOGICAL-ALLOMETRY-001-d5ce1f4e35ed8888.json;



empirical validation sha256:92f70b58bee23e841f4a64f8d38ff7afb4c67a7bd312e7cb8d38790794a339a6;  
 measurement receipt sha256:ff777763c0d3697eb5f15a7f88bb9b906faa9f42711ec939237f635b8234c4c4;  
 isolation sha256:e2b5d16a4a13c5662b8b9573d597351cfd19f8cd0e6e1668c934ed84961ad885; custody  
 sha256:7ca0acb6383a1c26d9a11d0a2b373161d01dc0fe2fa51518fffc90041434c7cb.

## 21. Observation Handoffs

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

### 70. Biological specimen and sampling

Claim identity: SFT-BIO-BIO-SPECIMEN-001

**Question and exact theorem.** A biological specimen is a finite source-bound part with collection, preservation, preparation, subsampling and representativeness boundaries explicit.

source-bound-biological-part\_\_specimen-to-source-provenance\_\_collection-preservation-and-subselection-held\_\_representativeness-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIOLOGICAL-ALLOMETRY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-bound-biological-part                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | specimen-to-source-provenance                 | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | collection-preservation-and-subselection-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | representativeness-declared                   | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `source-bound-biological-part__specimen-to-source-provenance__collection-preservation-and-subselection-held__representativeness-declared__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological specimen is a finite source-bound part with collection, preservation, preparation, subsampling and representativeness boundaries explicit. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 3d27156399308c45f1a240c4ba54d72cebae6b5edf61acdf224f73f3179931ce.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: ffc0241b8af32e90076d8efe9aa6192d91cc2fd35809d9c11c985104683fa2f8.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 39a13818c4d348a5e248f8b482b40240fef859275a578dc9ad06e249340f8223.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: d033ac4b38c920cca742a2106142b9cf05eecb28bcf7fea231c76f06530134c5.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256: c8ce42de0811f900fe0d429491bf67ee9c282599cbe9abe14a3010493d695b15; certificate  
sha256: 5f76e083df654d005da343a8a76acf71ea34fd55fcbb920ac53641e528f47de2; external-validation  
sha256: 8c418402433dbb2704c1a8c80a9324dfe82450dc12175b75585880b3970dc28b.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256: b595f4309c9f88fc5c7b556785b4c252382b77a71178717089db70fbb27e2d0. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot-experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot-experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256: b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.
- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility; <https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-darwin-core-archive.html; sha256: 89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.

Comparison record:

- sft-bio-bio-specimen-001-authority-correspondence: predicted biology:source-bound-biological-part\_\_specimen-to-source-provenance\_\_collection-preservation-and-subselection-held\_\_representativeness-declared; source-derived biology:source-bound-biological-part\_\_specimen-to-source-provenance\_\_collection-preservation-and-subselection-held\_\_representativeness-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology

- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:334a3a64a5c8f8e98a410b22730574dca33b23fed37ec9d21daa7e9db5db05b3; derivation seal sha256:5b08efce103cc645bd5826afd5e3452981dbaa1d6d74605756c22349c5276206; engine receipt sha256:e8adfad0f9b420176de99c2d31f622fb178b6ef766494cb9d9df2adb333c14e1 at receipts/engine/model\_admitted/SFT-BIO-BIO-SPECIMEN-001-e8adfad0f9b42017.json; empirical validation sha256:c3b57685834720d760247ca330c88953363b2b5c3ba29294fe37769e6ac5845f; measurement receipt sha256:0edc200c019d99d401a6bf5569004522aa43fec099a4716e7db5e12b067f3a70; isolation sha256:28dd59b9d99409ac695f9b043f2b2ae3df5c28ee9f292b80e4712008e0ab55b7; custody sha256:1d0598e6f518c8aec75d7d64d1897122d8da67bd398d3cdbe034c416af3e9a78.

## 71. Biological condition and control

Claim identity: SFT-BIO-BIO-CONDITION-001

**Question and exact theorem.** A biological comparison requires experimental units, conditions, controls, timing and context to be registered before the outcome is opened.

experimental-unit-condition-carrier\_\_condition-to-outcome-comparison\_\_controls-exposure-time-and-context-held\_\_design-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIO-SPECIMEN-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | experimental-unit-condition-carrier           | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | condition-to-outcome-comparison               | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | controls-exposure-time-and-context-held       | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | design-boundary-declared                      | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: experimental-unit-condition-carrier\_\_condition-to-outcome-comparison\_\_controls-exposure-time-and-context-held\_\_design-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

**Operational witnesses.**

- **carrier**: the biological carrier is explicit and nonempty; passed **true**.
- **relation**: relation and organization remain separately reconstructible; passed **true**.
- **boundary**: observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological comparison requires experimental units, conditions, controls, timing and context to be registered before the outcome is opened. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

**Adverse controls.**

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:0e2af1cb5310e8e0f464f8ea29f839cf971c2c2c0c58e2c2183f29aa014c699e.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:acb8fcc6f8e79ae9d096f563a8046de12bad059dfaaf2bf16b7c3bec38950543.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:655d2f877578d7b9f006945454cd52cd11e228707af39c72ce06df551919129d.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:4ef9f6bdc4b213d9ed84b63a2a785c4366a0c197af0aa0c4748497c5cd2a94e4.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256:58a59445cd08d4925af025b5ed9e8f00cf7c719f1db3c7023289131d14ebecfd; certificate  
sha256:297c5e0636bd72b075ca4d031eb422ecb2671218ed947992567ccb35a85da1b0; external-validation  
sha256:5603d5745b7801631c97ea3377bc8932ec0d1a38f4c2c447c4af64f0b23b63b1.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256:b595f4309c9f88fcf5c7b556785b4c252382b77a71178717089db70fbb27e2d0. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE** - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot/experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot/experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256:b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.
- **SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE** - Global Biodiversity Information Facility; <https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-darwin-core-archive.html; sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.

Comparison record:

- **sft-bio-bio-condition-001-authority-correspondence**: predicted biology:experimental-unit-condition-carrier\_\_condition-to-outcome-comparison\_\_controls-exposure-time-and-context-held\_\_design-boundary-declared; source-derived biology:experimental-unit-condition-carrier\_\_condition-to-outcome-comparison\_\_controls-exposure-time-and-context-held\_\_design-boundary-declared; exact match **True**

- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:f30f215faff8252bbf6453f9221561b3d8a8c2b3f1843ec72064f6d08f43f378; derivation seal sha256:085891333c799525589e3799aca96f6f2d8a44df48937a73f49b0fea323be2e0; engine receipt sha256:fc6ea229bc38e9de68a4d0504d386a2b90bc33b950bcd408c24c49c27e238700 at receipts/engine/model\_admitted/SFT-BIO-BIO-CONDITION-001-fc6ea229bc38e9de.json; empirical validation sha256:20e57fd6cf6f66b45fc9fdd1d26c731fb0c3415e0153ff32b797a791f7e14676; measurement receipt sha256:bb559b6af10ac78000e0b401e223b8addcad040246d230ec7918ebd69b17b6af; isolation sha256:601d01b83be0f7653f4e0551c52f092830b084b9fb36b1c59086c8e3e848a13b; custody sha256:41c89703bb761a8139ce051ad24aebc7125fa2e2d58dbc5530691bb296a7f223.

## 72. Biological assay

Claim identity: SFT-BIO-BIO-ASSAY-001

**Question and exact theorem.** A biological assay is a specimen-bound observation process with calibration, detection, processing, exclusions and method-validity boundaries retained.

```
specimen-method-result-carrier__method-to-observation-relation__calibration-detection-and-processing-held__method-validity-bounded__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule
```

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001



- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIO-CONDITION-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | specimen-method-result-carrier                | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | method-to-observation-relation                | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | calibration-detection-and-processing-held     | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | method-validity-bounded                       | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: specimen-method-result-carrier\_\_method-to-observation-relation\_\_calibration-detection-and-processing-held\_\_method-validity-bounded\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological assay is a specimen-bound observation process with calibration, detection, processing, exclusions and method-validity boundaries retained. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 6171e7c61a2490e9400a173c29a4e881cb85c295d18e268fab6536ad2ee9d70c.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 7db406cd9e29392837134c44674898575fa7d8585abb78e2affff88b01533c53e.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 0d74ddbabc54ed1b3eaa4831436fe1b446c5267c66384b28872caac7ade15799.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : c638ab631ee09ce30b97df71947f7e96c20cfbadc912ea971dad3019765a8fd9.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : e1af41c6cf0795f4ab53f85f050ae73d840a8cd47a6f64ee47551c01f78d9012; certificate  
sha256 : e59c809972497cd704aff22217f2d2d32a1b05a09d142d7c76583ca1611f470a; external-validation  
sha256 : 2879dd09e320d3c950446ddbbae8379b555eaa1b49d3b831996668ffa6a6135a4.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : b595f4309c9f88fcf5c7b556785b4c252382b77a71178717089db70fbb27e2d0. Target opened after derivation seal: **true**. All rows preserved: **true**. Exact comparison: **true**. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE** - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot-experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot-experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256 : b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.

- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE - Global Biodiversity Information Facility; <https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshot/s/sft-bio-src-gbif-darwin-core-archive.html; sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.

Comparison record:

- sft-bio-bio-assay-001-authority-correspondence: predicted biology:specimen-method-result-carrier\_\_method-to-observation-relation\_\_calibration-detection-and-processing-held\_\_method-validity-bounded; source-derived biology:specimen-method-result-carrier\_\_method-to-observation-relation\_\_calibration-detection-and-processing-held\_\_method-validity-bounded; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

#### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

#### Immutable evidence identities. Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:1c3d638ad8372bea86d8e2fd5226f12aac7cce7dab1d20cae3fe69e53cc36335; derivation seal sha256:9bb5f47d73993aa1c970b33491d791e91413b7f4e8809f7970c410ad4e465924; engine receipt sha256:0f51212a550b8a30526dc82853d945b037c359b272dd8a66f96279ae3071df5c at receipts/engine/model\_admitted/SFT-BIO-BIO-ASSAY-001-0f51212a550b8a30.json; empirical validation sha256:68530af6aecc969286cc3c89e2932d1306dc14d6fcc98d5d04fea0d93d2dc33d; measurement receipt sha256:1f2708c45c9ae840bf8a65f64ad6b07eab9a955e56a399a04a9763b812f0d153; isolation sha256:e920194e7675908b0a5f63f2ba58dcd57ce8ee0f547c0884c4955f55bc03a48e; custody sha256:12b9a770f77b4d6ca5d824179d8078dea9631dc680817f081f20f48f21ea5ee2.

## 73. Biological uncertainty and missingness

Claim identity: SFT-BIO-BIO-UNCERTAINTY-001

**Question and exact theorem.** Biological evidence must retain biological variation, measurement error, missingness, censoring and unresolved states rather than collapsing them into one favorable estimate.

complete-observation-record\_\_result-to-uncertainty-partition\_\_variation-error-missing-and-censored-held\_\_inference-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIO-ASSAY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                    | Forced form                               | Exact elimination / retention basis                                                                                                                                                             |
|--------------|------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | complete-observation-record               | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                |
| relation     | relation-imported-fitted-or-erased | result-to-uncertainty-partition           | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                 |
| organization | organization-collapsed             | variation-error-missing-and-censored-held | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.        |
| observation  | observation-boundary-unrecorded    | inference-boundary-declared               | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law. |

| Axis       | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record     | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension  | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: complete-observation-record\_\_result-to-uncertainty-partition\_\_variation-error-missing-and-censored-held\_\_inference-boundary-declared\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: depth\_independent; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed **true**.
- **relation:** relation and organization remain separately reconstructible; passed **true**.
- **boundary:** observation boundary cannot silently become the biological law; passed **true**.

**Scientific meaning.** Gene Ontology's evidence rules explicitly state that missing annotations do not imply absence and preserve NOT assertions separately. The Fold record likewise distinguishes variation, error, missingness, censoring and demonstrated negation.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 44ba3f15856ea93f77e889dd25979b68f8bca505bdf3ec1b6360da28eb3afd0d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ea04da1ed8ab8ddd5cd14de8ed72ebb1d58b21115e32abb5f6efa43effaaa87a.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 21f1d678d6f1b1dac0da0bd6469ff47f6353382993d3ca3431fc4656578bd4a3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 440eb6924e7e371245912f3bd16f042144af0b7371822e5fe11ffe537629d630.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:f1f9e9a57b31e6c5283fbc32b092e2efa8c8b4ea18f722b0c796f62042a2eb4c; certificate  
 sha256:b876c2bd57409b7fbc3cd1df05d7a6ba74b4891b7223460ee219b6a50e52fbb9; external-validation  
 sha256:edd8f6ef94c32d1bd17d8e06a8ab02d9aa3361df30cc5ce9caa7bca9cf08b486.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:6b9bd6569dddef5b7c3b084ded976fc81ba05910859fb505199b16643886e638. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot/experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot/experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256:b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.

Comparison record:

- sft-bio-bio-uncertainty-001-authority-correspondence: predicted biology:complete-observation-record\_\_result-to-uncertainty-partition\_\_variation-error-missing-and-censored-held\_\_inference-boundary-declared; source-derived biology:complete-observation-record\_\_result-to-uncertainty-partition\_\_variation-error-missing-and-censored-held\_\_inference-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:dc71487ae792084dc6bb7dc418ee18e8ec9a0b61529a452d020e37b4d8910318; derivation seal sha256:fd1468f2f20370975408e00aeded59300b12e4b4763151ad113b3ffade5b2c28; engine receipt sha256:fe2987400977553ecfa40a7d3b5816a5a4e960b84ea807c1b5d255f2453bc37c at receipts/engine/model\_admitted/SFT-BIO-BIO-UNCERTAINTY-001-fe2987400977553e.json; empirical validation sha256:8412cb610b7b19891ed872a572e2a08113b1812214b15a139038e275a35e50bd; measurement receipt sha256:b94c2e3be8914e43e38dd492210699c2dbc4adbfe8b12baf9e09f0f25529792f; isolation sha256:9daabf1fd96edf14c09ee1c69ef0176ac5b11c9c9f2c04c9bc3ce07b1f9a5c6e; custody sha256:1b2af5f734690467e35154f398531b3869aa542bef0cbaf29817d6f6b36cf06f.

## 74. Biological causal and associative boundary

Claim identity: SFT-BIO-BIO-CAUSALITY-001

**Question and exact theorem.** A biological causal claim requires a registered intervention or adequate control structure that eliminates generated alternatives; association alone remains explicitly noncausal.

intervention-exposure-outcome-carrier\_\_counterfactual-control-comparison\_\_confounders-timing-and-selection-held\_\_design-specific-causal-scope\_\_complete-state-transition-control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adverse-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIO-UNCERTAINTY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis     | Eliminated form                    | Forced form                           | Exact elimination / retention basis                                                                                                                                             |
|----------|------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | intervention-exposure-outcome-carrier | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                |
| relation | relation-imported-fitted-or-erased | counterfactual-control-comparison     | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted. |



| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | confounders-timing-and-selection-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | design-specific-causal-scope                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `intervention-exposure-outcome-carrier__counterfactual-control-comparison__confounders-timing-and-selection-held__design-specific-causal-scope__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biological causal claim requires a registered intervention or adequate control structure that eliminates generated alternatives; association alone remains explicitly noncausal. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b685881b29ddf20901d2566fa923ed6d0239887268e24ff4e886a76bf734494b`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:addc22f4a36bebbba1a2f99e43c7a12423c4f7cd9986ac9519ea46257b52ac032`.

- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:1a69e94e409bb1d51a611ef199557189c84c7d93725814d25949c9f029fea8d1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:3e3e0626209153b5faa9774c5fd535d4c90714a9d9dfa250edf339a969f81f96.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation

sha256:b658684e8e4c5718fc4bc9472a918a3a6b79b4b22305de15251c447442072647; certificate  
sha256:e538760d2bc7c4631b4954ea6dc57090129618689f2402f19c58557ba6d14f3; external-validation  
sha256:5782a6613a3f969f819efd07b95738fdd0aa4e4a6c145bd6447ae10df048bb5a.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity

sha256:ec2f3af291be8147cc0b7c3cdb4a1877b6b45229f6a6477ef73740e465f2bef2. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE** - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot-experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot-experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256:b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.

Comparison record:

- sft-bio-bio-causality-001-authority-correspondence: predicted biology:intervention-exposure-outcome-carrier\_\_counterfactual-control-comparison\_\_confounders-timing-and-selection-held\_\_design-specific-causal-scope; source-derived biology:intervention-exposure-outcome-carrier\_\_counterfactual-control-comparison\_\_confounders-timing-and-selection-held\_\_design-specific-causal-scope; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

Falsification boundary: Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

### Explicit exclusions.

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value
- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest  
sha256:56947f42672d8374f9cf66c12acb11b3e54a1f069ffbed12b9a2c8b2e7c3f7c0; derivation seal  
sha256:563b0de131cd5274411816be3d450e7bdae8a10d1dbafceb2bd4309f398f5b92; engine receipt

sha256:89b1826edcf95ef8571cf058e28a46253279e75125df8cc8a0f4e62ca162c883 at  
 receipts/engine/model\_admitted/SFT-BIO-BIO-CAUSALITY-001-89b1826edcf95ef8.json; empirical  
 validation sha256:927b3076760be37de8abaabfe60e6028eb48dabd79a869a3b4519e85cb5dddb2;  
 measurement receipt sha256:2d961a7dd203c22769a54036c2623b62056771c48eb12aa14fd22d5ca013cfd3;  
 isolation sha256:eb57de061dc4deb654523ec1c1a652f83cd2c96179be826d2fd04183c836c92c; custody  
 sha256:04f9fd176b85307ea76d780b8b7cfcf8595e13e3cbecc51c4eb876550df4ed72.

## 75. Cross-branch biological handoff

Claim identity: SFT-BIO-BIO-HANDOFF-001

**Question and exact theorem.** A Biology claim cites upstream mathematical, informational, physical, chemical and material laws without reowning them and exports organismal consequences without claiming medical efficacy, consciousness or engineering implementation.

source-law-interface-carrier\_\_upstream-to-biological-composition\_\_ownership-assu  
 mptions-and-losses-held\_\_discipline-boundary-declared\_\_complete-state-transition  
 -control-record\_\_root-bound-forward-forcing\_\_positive-finite-successor-and-adver  
 se-closure\_\_no-extra-rule

**Rooted dependency chain.** The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-MAT-MICRO-INTERFACE-001
- SFT-MAT-FUNC-BIOMATERIAL-001
- SFT-BIO-BIO-CAUSALITY-001

The dependency graph independently reaches the premise-free root theorem; a branch label never substitutes for a receipt.

**Generated grammar.** Generate the Cartesian product of every registered binary preservation dimension; do not inspect external targets or prior answers.

Boundary: Exactly eight binary dimensions and therefore 256 forms; the statement is conditional only on this explicit complete grammar.

The exact product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

| Axis         | Eliminated form                        | Forced form                                   | Exact elimination / retention basis                                                                                                                                                                                         |
|--------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-law-interface-carrier                  | Erasing the biological carrier prevents reconstruction of the claimed living distinction. The complete generated biological carrier is retained.                                                                            |
| relation     | relation-imported-fitted-or-erased     | upstream-to-biological-composition            | An imported, fitted or absent relation can select a familiar biological answer without forcing it. Only the generated biological transition or comparison relation is admitted.                                             |
| organization | organization-collapsed                 | ownership-assumptions-and-losses-held         | Collapsing organization merges living states the question requires to remain distinct. Every boundary, lineage, context and organization distinction required by the claim remains held.                                    |
| observation  | observation-boundary-unrecorded        | discipline-boundary-declared                  | An unrecorded organism, condition, assay, scale or interval cannot identify the biological observation class. The complete declared observation boundary is retained and cannot select the law.                             |
| record       | result-without-transition-record       | complete-state-transition-control-record      | A biological label without initial state, transitions, controls and outcomes is not auditable. Initial state, transitions, controls, results and retained or lost distinctions are recorded.                                |
| provenance   | authority-target-or-prior-selected-law | root-bound-forward-forcing                    | Authority, consensus, a prior SFT answer or target data may test but cannot select a Fold law. Every decision traces through admitted dependencies to the premise-free root theorem.                                        |
| generality   | single-specimen-favorable-instance     | positive-finite-successor-and-adverse-closure | One favorable organism, specimen or experiment cannot close the generated biological class. Base carrier, every positive finite successor and all registered adverse controls preserve the relation at the stated boundary. |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                                 | A free coefficient, fitted threshold, exception or opaque predictor can manufacture a desired answer. No rule beyond admitted Fold dependencies and generated preservation conditions is present.                           |

**Unique survivor and depth independence.** Sole survivor: `source-law-interface-carrier__upstream-to-biological-composition__ownership-assumptions-and-losses-held__discipline-boundary-declared__complete-state-transition-control-record__root-bound-forward-forcing__positive-finite-successor-and-adverse-closure__no-extra-rule`.

Base: The least positive finite carrier retains every required biological distinction and its complete control record.

Successor: Adding one lawful finite successor preserves all existing identities, relations, boundaries and provenance and appends its new transition and adverse alternatives.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

#### Operational witnesses.

- **carrier:** the biological carrier is explicit and nonempty; passed `true`.
- **relation:** relation and organization remain separately reconstructible; passed `true`.
- **boundary:** observation boundary cannot silently become the biological law; passed `true`.

**Scientific meaning.** The result is an exact relational classification at the declared biological boundary. It forces a biology claim cites upstream mathematical, informational, physical, chemical and material laws without reowning them and exports organismal consequences without claiming medical efficacy, consciousness or engineering implementation. Specimen-, organism-, population-, method- and condition-dependent magnitudes remain explicit records and are never promoted into universal constants without a separately generated and externally tested law.

Biological observation is part of the science: specimen, condition, assay, uncertainty, missingness, controls and causal scope are retained. Upstream laws are cited without reownership and downstream medical or consciousness claims are not smuggled into Biology.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 366b2cbda7cc250cb4a9f632713c2b3ac16b166d9a26f0c9d63bfe6f94d5cd5e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : d2d8eb982f3a7b3b35bba2f1de68b20f21bf1e8b9d84b87619588217168c3702.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : c3a35e0b9f3c778ea83e420b242822106bb96ebd5864faec0330dc6525a2c028.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 3c12b0e4c309302505bb215951d93f67bedcada7e1ab5fca4cae2dba99b51443.

**Independent reconstruction.** A separately executed implementation regenerated the literal product, candidate order, every decision, the one survivor, depth-independent closure and all four control classes. Implementation  
sha256 : b57885f3ff7313888b5ab4c0d9906d96c46e365695ea3f4dae59d1147179eca9; certificate  
sha256 : 84a4561e102a9360dd8f9ba7ee65e181bd965e1806391518272816a549a5712b; external-validation  
sha256 : 9f1764112a91d4819f44db6d1697b73ac0b7d597da65cedba0d5d463e01045fe.

**Post-seal external comparison.** The entire 75-law prediction family was sealed before any source identity was chosen. This claim requires 2 purpose-matched discriminators with ordered identity  
sha256 : b595f4309c9f88fcf5c7b556785b4c252382b77a71178717089db70fbb27e2d0. Target opened after derivation seal: true. All rows preserved: true. Exact comparison: true. A deliberately changed observation was rejected.

Sources:

- **SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE** - Gene Ontology Consortium; [https://geneontology.org/docs/go-annotations/snapshot-experiments/external\\_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html](https://geneontology.org/docs/go-annotations/snapshot-experiments/external_sources/biology/snapshots/sft-bio-src-go-annotation-evidence.html); sha256 : b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; scope: evidence codes; context; negation; missingness; causal activity models.
- **SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE** - Global Biodiversity Information Facility; <https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>; snapshot experiments/external\_sources/biology/snapshots/sft-bio-src-gbif-darwin-core-archive.html; sha256 : 89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; scope: taxon, occurrence and sampling-event data structures.

Comparison record:

- sft-bio-bio-handoff-001-authority-correspondence: predicted biology:source-law-interface-carrier\_\_upstream-to-biological-composition\_\_ownership-assumptions-and-losses-held\_\_discipline-boundary-declared; source-derived biology:source-law-interface-carrier\_\_upstream-to-biological-composition\_\_ownership-assumptions-and-losses-held\_\_discipline-boundary-declared; exact match True
- every required authoritative fragment and snapshot hash reproduced
- transport/content failures and adverse scientific rows retained
- deliberately tampered unfavorable control rejected

**Falsification boundary:** Reject if any registered authoritative biological source contradicts a required content relation, if a generated alternative also preserves every dimension, or if any specimen, control, adverse row, condition or provenance distinction is missing.

**Explicit exclusions.**

- numerical-zero ontology
- negative proof quantity
- irrational or imaginary proof value

- fitted coefficient or threshold
- external target leakage
- opaque predictor
- unrecorded specimen or condition
- erased adverse result
- prior answer as premise

**Immutable evidence identities.** Pre-source seal

experiments/sealed\_predictions/biology\_foundation\_complete\_pre\_source.json; source manifest sha256:f6207ed89fec52578462bfe7ffd6031f5cf22cd3e0ef04559a6c137e211b67c0; derivation seal sha256:cb9cc40456235dafad99bc7030b04f2f650a964fe71ba849ab44f892b6776592; engine receipt sha256:d36e9dbd732ed6331adbad9f25c6dca5ddb0b6d56511c1671e47920c35147e5f at receipts/engine/model\_admitted/SFT-BIO-BIO-HANDOFF-001-d36e9dbd732ed633.json; empirical validation sha256:562c36879e42e31a3838cc961365f91e9baceb52455736c04c4354b675e41f29; measurement receipt sha256:8bacbc6350632bfc58ef058367a188d94a55e5997049a0c56fe23de05ae6790e; isolation sha256:19d893aba43bf4b799d645b8c673b62e71c84dc13025b07c2c81e86c534fe7f7; custody sha256:7019f7f049f35ee2041e1d482dcd488cc7c166f19c728c2264ac7633c6451eb2.

## 22. Integrated audit result

- Prior entries reviewed: 763.
- Biology-relevant prior rows: 18.
- Biology-owned atoms: 30.
- Same-strength V3-closed atoms: 30.
- Open atoms: 0.
- Corrected prior atoms: 21.
- Audit identity: sha256:3d76ea2913c6cd798d9fdd6a53b0246fd88e7deed89cb008102230b882d44449.

## 23. External-source ledger

- SFT-BIO-SRC-NASEM-UNITY-BIOLOGY-2010 — <https://www.ncbi.nlm.nih.gov/books/NBK32506/?report=printable>; sha256:62b53a959de8aa5b377997063a3a2719a87ff029a99f36225a383c7a4cfad9e6; DNA inheritance; cell; multicellularity; organism; community; ecosystem; biological energy.
- SFT-BIO-SRC-NCBI-CELL-ORIGIN-METABOLISM — <https://www.ncbi.nlm.nih.gov/books/NBK9841/?report=printable>; sha256:ce64ef3e815f9a4f14dc6938e708f449bfd3a39de108c317aac8a941bff68b6b; cells; membranes; compartments; metabolism; replication.
- SFT-BIO-SRC-GO-ONTOLOGY-DOCUMENTATION — <https://geneontology.org/docs/ontology-documentation/>; sha256:a55c5b5fc204dae486fa1bdfa85d50035f70bf2286fc089783d1a961fa8fcf26; molecular function; cellular component; biological process and relations.
- SFT-BIO-SRC-GO-ANNOTATION-EVIDENCE — <https://geneontology.org/docs/go-annotations/>; sha256:b16235ab261f6b02afa71826546aad87823680ff267e0bd1c433d54c838b7ebd; evidence codes; context; negation; missingness; causal activity models.
- SFT-BIO-SRC-NCBI-GENETIC-CODE — <https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi>; sha256:c329091f012ae379949dc3a39f1961d2338a7b38ddad60245f9c12d8a31db514; complete genetic-code tables and ordered base positions.
- SFT-BIO-SRC-RCSB-1COP-CIF — <https://files.rcsb.org/download/1COP.cif>; sha256:6eddefaa66f6191a3080ca3b0dd333c9e7859448d6aaf17b79b60a3e32ad9a3c; primary Lambda Cro NMR ensemble and homodimer assembly.



- SFT-BIO-SRC-GBIF-DARWIN-CORE-ARCHIVE — <https://ipt.gbif.org/manual/en/ipt/latest/dwca-guide>;  
sha256:89adbe8ba230fd51495154e86843e624d2805d632bc17e55136e88f85c7cad1f; taxon, occurrence and sampling-event data structures.
- SFT-BIO-SRC-SEQUENCE-ONTOLOGY-OBO-RELEASE-V1 —  
[https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology\\_Files/so.obo](https://raw.githubusercontent.com/The-Sequence-Ontology/SO-Ontologies/master/Ontology_Files/so.obo);  
sha256:dde032d4c7cfb89a7013f2f8ab7420a8ef7dc469fbc2b0ffb38bef2a064a1d1f; sequence features, gene, genome, transcript, polypeptide and mutation classes.
- SFT-BIO-SRC-GBIF-SURVEY-GUIDE-V1 — <https://docs.gbif.org/guide-publishing-survey-data/en/>;  
sha256:58b745acf0df0e8235a5c50e9ff86a128593f26fb6982fd92834ca544bf31be8; ecological survey protocol, effort, occurrence, absence and monitoring.
- SFT-BIO-SRC-NCBI-EVOLUTION-GENETICS-V1 — <https://www.ncbi.nlm.nih.gov/books/NBK595925/?report=printable>;  
sha256:49fd44aa874381d681845301eb337b361ed763fb4272118befc61af4a07eaceb; heritable variation, mutation, selection, population size and drift.
- SFT-BIO-SRC-NCBI-PROTEIN-FOLDING-V1 — <https://www.ncbi.nlm.nih.gov/books/NBK9843/?report=printable>;  
sha256:ba95b96559c7a21af68f3dc044011bb842b1d641ef86beeb2713a8b360e14313; sequence, folding, intermediates, chaperones, function and assembly.
- SFT-BIO-SRC-NCBI-GENOME-REGULATION-DEVELOPMENT-V1 —  
<https://www.ncbi.nlm.nih.gov/books/NBK21127/?report=printable>;  
sha256:e1d430fae9b5cee3fdd0778c89ccb4a578be472db6a81f5ba802191d94e909ce; genome expression, signalling, differentiation and development.
- SFT-BIO-SRC-NCBI-AGING-SENESCENCE-V1 — <https://www.ncbi.nlm.nih.gov/books/NBK10041/?report=printable>;  
sha256:920f58581ed5a30448598733f38a1b2eb85682fb9901cf5c1a5606513f6adb4c; aging, lifespan, regeneration and cancer distinction.
- SFT-BIO-SRC-NCBI-HOMEOSTASIS-V1 — <https://www.ncbi.nlm.nih.gov/books/NBK559138/?report=printable>;  
sha256:687e91d0923fa79516112a8c3bd99df2a4b911d3fc79d0c488954caae4c7c24; physiology and homeostatic control loops.
- SFT-BIO-SRC-PUBMED-HOMOCHIRALITY-35969306-V1 —  
<https://pubmed.ncbi.nlm.nih.gov/35969306/?format=pubmed>;  
sha256:8980303d43e13f638eccaf22e24edc141fc51b26037c5e9336facdea4f5c9bca; observed L-amino-acid and D-sugar predominance with mechanism unresolved.
- SFT-BIO-SRC-EUROPEPMC-SCALE-FREE-RARE-6399239-XML-V1 —  
<https://www.ebi.ac.uk/europepmc/webservices/rest/PMC6399239/fullTextXML>;  
sha256:3896db77ef007b8da965fb21a7f87d63790222cb566889ec89d978392ded21a8; severe empirical scale-free test and alternative distributions.
- SFT-BIO-SRC-EUROPEPMC-ALLOMETRY-11782306-XML-V1 —  
<https://www.ebi.ac.uk/europepmc/webservices/rest/PMC11782306/fullTextXML>;  
sha256:214f06e309b9505e2ec0852d9b30cb02b6c4863c7e17f16d16ceb17378462e99; three-quarter, two-thirds, unity and nonuniversal allometric evidence.

Preserved unsuccessful source rows are recorded in `experiments/external_sources/biology/biology_foundation_source_manifest.json` and `experiments/external_sources/biology/biology_foundation_family_source_manifest_v1.json`; their official transport/content replacements are separately registered, not substituted invisibly.

## 24. Reproducibility and falsification

The repository's one-command verification route checks engine and verification-authority seals, schemas, dependency resolution, source hashes, inventory cardinality, execution manifests, receipts and branch gates. The Biology-focused suite independently checks 75 inventory/spec alignments, all 19,200 candidates, 75 unique survivors, dependency order, all 153 registered source fragments, the exact codon certificate and prohibited blueprint fields.



The branch is falsified at its declared boundary if any source hash changes without a new version, any required external fragment is absent, any target becomes accessible before sealing, any adverse or failed row is omitted, any tampered target is accepted, any dependency loses its root path, any candidate census is incomplete, any claim has other than one survivor, or any implementation-distinct validator fails.

## 25. Full-field Biology roadmap

This paper completes Layer One. Later versions extend, in dependency order, through biochemistry at the living boundary; structural and quantitative biology; genetics, genomics and epigenetics; RNA and cellular machinery; microbiology and virology; taxonomy and phylogeny; development and regeneration; comparative plant, fungal, animal, marine and extremophile biology; neuroscience at the substrate boundary; immunology; behaviour and sensory biology; biodiversity and conservation; systems and synthetic biology; biological data inference; origins-of-life hypotheses; ageing; biosphere processes; and astrobiology handoffs. Fold Protein remains a downstream validation application and cannot select these laws.

## 26. Limitations

- The 75-law foundation is complete at its frozen current-evidence boundary, not a permanent declaration that no new biological law can exist.
- Most foundational results are exact relational and classification laws. Only the code counts are universal numerical results in this edition.
- External authority and primary data test correspondence; they are not premises of the derivation.
- The paper does not claim one universal life definition outside the registered finite operational grammar.
- Clinical efficacy belongs to Medicine; consciousness and qualia belong to their own branch; engineering applications cannot select Biology laws.
- Allometry, network topology, protein ensembles, senescence, ecological stability and similar quantities remain population-, condition-, scale- and method-bounded.

## 27. Conclusion

Foundational Biology is current-evidence closed and extension-open: seventy-five required laws, 19,200 exact candidates and decisions, seventy-five unique survivors, three hundred passing adverse controls, seventy-five independent reconstructions, seventy-five post-seal external comparisons, seventy-five root traces, and thirty of thirty legacy Biology atoms reconciled. The result is not a list of biological vocabulary. It is a public chain from the premise-free root theorem through exact finite alternatives to falsifiable, source-bound statements about life, heredity, evolution, cells, organisms and ecosystems.

The corrections matter as much as the correspondences. The engine did not reward familiar slogans: it rejected inevitable fixation, rigid one-state proteins, parity-selected homochirality, universal three-quarter allometry, ubiquitous scale-free networks and recurrence-only biological diagnoses. That is the empirical standard Ernos Labs is meant to make reproducible: open inquiry without open admission-by-assertion, and no credential, institution, model or author exempt from the same public gate.

## 28. Repository and publication status

- Canonical repository prepared for later authorized push: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Zenodo DOI: reserved only after explicit publication authorization
- Author: Maria Smith, Ernos Labs
- Contact: [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com)
- Submissions: <https://discord.gg/ucwGryVxGr>
- Current state: local prepublication; no remote action performed

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